

Solid Mechanics (ME 302)

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Books:

1. Strength of Materials: Part I, II, S. Timoshenko, CBS Publishers, 1985.
2. Engineering Mechanics of Solids, E. P. Popov, PHI, 1993.
3. Introduction to Solid Mechanics, I. H. Shames and J. M. Pittarresi, PHI, 2003.
4. Strength of Materials, F. L. Singer and A. Pytel, HarperCollins Publishers, 1991

Syllabus

- ❖ Stress, strain and elasticity; elastic limit, ultimate strength, working stress; factor of safety; Hooke's law; strain energy for axial loads
- ❖ Two dimensional stresses, strains and Mohr's circle
- ❖ Simple theory of bending; shearing forces and bending moments of beams; moment of resistance; section modulus; shear flow and shear centre
- ❖ Deflection of beams by Double-integration method and area-moment method; propped cantilever and fixed beams
- ❖ Columns, short and long columns; Euler's formula for columns
- ❖ Torsion of circular shafts
- ❖ Thin pressure vessels; thick walled cylinders and spheres, rotating discs
- ❖ Analysis of shafts and beams under combined loadings
- ❖ Helical springs
- ❖ Theories of failures

LECTURE 1: INTRODUCTION AND REVIEW

Preamble: Engineering science is usually subdivided into number of topics such as

- Solid Mechanics
- Fluid Mechanics
- Heat Transfer
- Properties of materials and soon Although there are close links between them in terms of the physical principles involved and methods of analysis employed.

The solid mechanics as a subject may be defined as a branch of applied mechanics that deals with behaviors of solid bodies subjected to various types of loadings. This is usually subdivided into further two streams i.e, Mechanics of rigid bodies or simply Mechanics and Mechanics of deformable solids.

The mechanics of deformable solids which is branch of applied mechanics is known by several names i.e. strength of materials, mechanics of materials etc.

Mechanics of rigid bodies: The mechanics of rigid bodies is primarily concerned with the static and dynamic behavior under external forces of engineering components and systems which are treated as infinitely strong and undeformable. Primarily we deal here with the forces and motions associated with particles and rigid bodies.

Mechanics of deformable solids:

Mechanics of solids: The mechanics of deformable solids is more concerned with the internal forces and associated changes in the geometry of the components involved.

Of particular importance are the properties of the materials used, the strength of which will determine whether the components fail by breaking in service, and the stiffness of which will determine whether the amount of deformation they suffer is acceptable.

Therefore, the subject of mechanics of materials or strength of materials is central to the whole activity of engineering design.

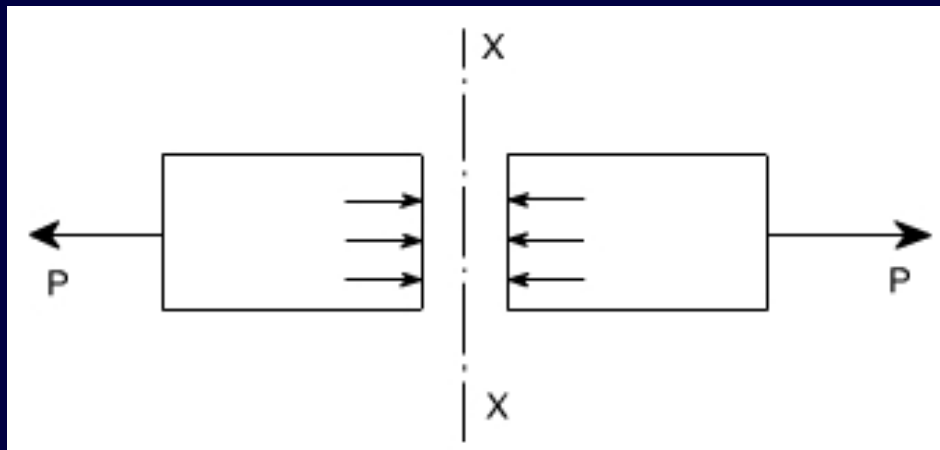
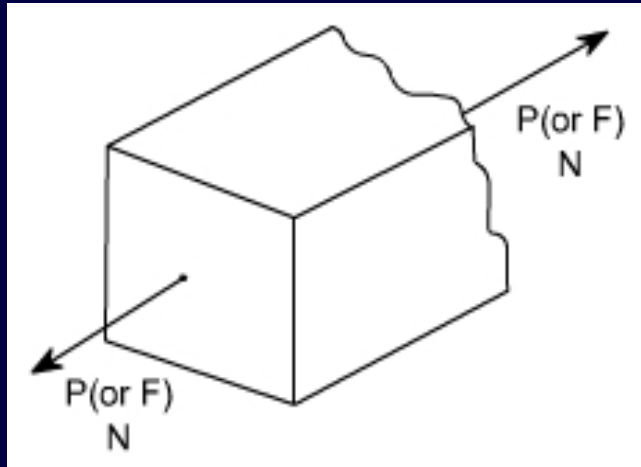
Usually the objectives in analysis here will be the determination of the stresses, strains, and deflections produced by loads. Theoretical analyses and experimental results have an equal roles in this field.

Analysis of stress and strain

Concept of stress

- Main problem of engineering mechanics of material is the investigation of the **internal resistance of the body**, i.e., the nature of forces set up within a body to balance the effect of the externally applied forces.
- The **externally applied forces are termed as loads**. These externally applied forces may be due to any one of the reason.
 - ❖ due to service conditions,
 - ❖ due to environment in which the component works,
 - ❖ through contact with other members,
 - ❖ due to fluid pressures,
 - ❖ due to gravity or inertia forces.
- In mechanics of deformable solids, externally applied forces acts on a body and body suffers a deformation. From equilibrium point of view, this action should be opposed or reacted by internal forces which are set up within the particles of material due to cohesion.
- These internal forces give rise to a concept of stress. Therefore, let us define a term stress

Stress



- Let us consider a rectangular bar of some cross sectional area and subjected to some load or force (in Newtons)
- Let us imagine that the same rectangular bar is assumed to be cut into two halves at section XX. The each portion of this rectangular bar is in equilibrium under the action of load P and the internal forces acting at the section XX has been shown
- Now stress is defined as the force intensity or force per unit area. Here we use a symbol sigma to represent the stress.

$$\sigma = \frac{P}{A}$$

- Where A is the area of the XX section

- ❖ Here we are using an assumption that the total force or total load carried by the rectangular bar is uniformly distributed over its cross section.
- ❖ But the stress distributions may be far from uniform, with local regions of high stress known as stress concentrations.
- ❖ If the force carried by a component is not uniformly distributed over its cross sectional area, A , we must consider a small area, dA which carries a small load dP , of the total force P , Then definition of stress is

$$\sigma = \frac{\delta F}{\delta A}$$

Units : The basic units of stress in S.I units are N / m^2 (or Pa)

$MPa = 10^6 \text{ Pa}$; $GPa = 10^9 \text{ Pa}$; $KPa = 10^3 \text{ Pa}$

Some times N / mm^2 units are also used, because this is an equivalent to MPa. While US customary unit is pound per square inch psi.

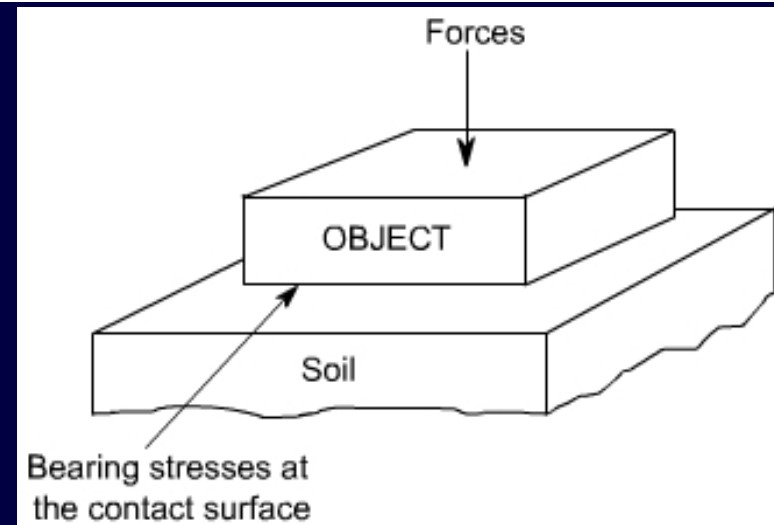
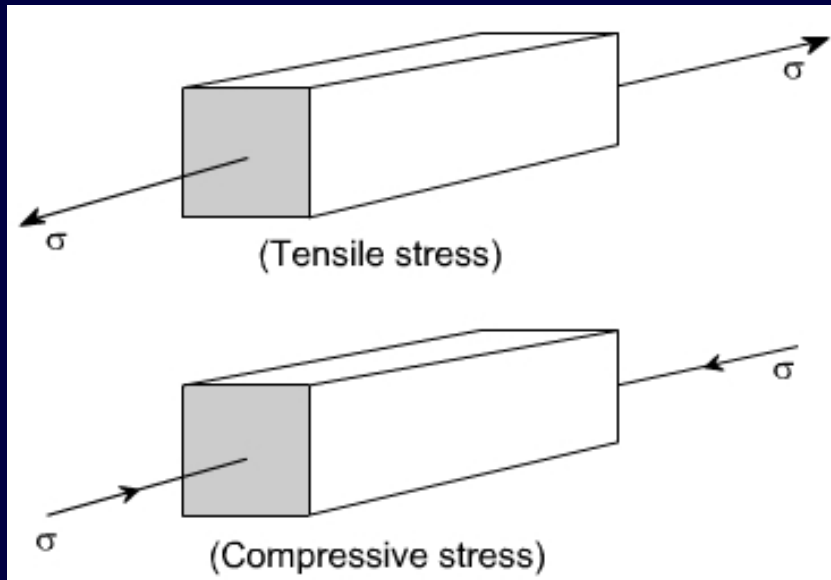
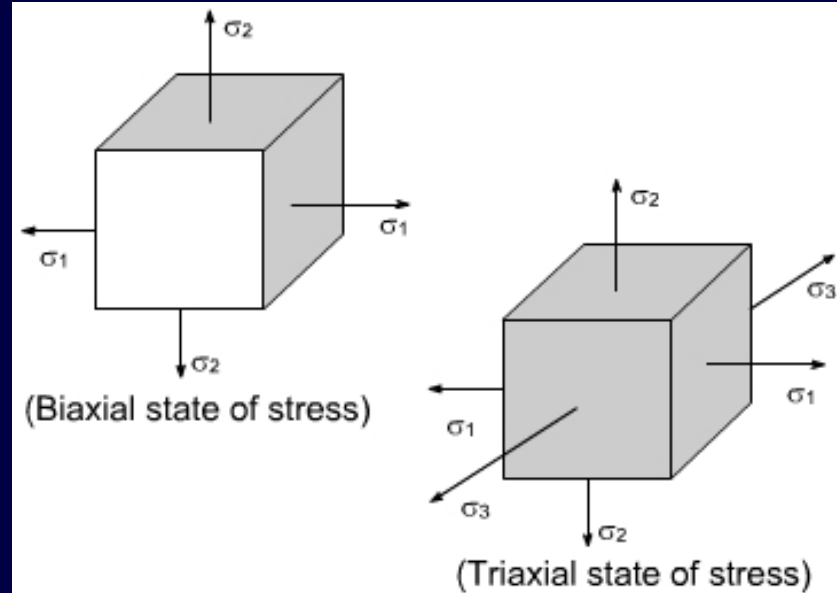
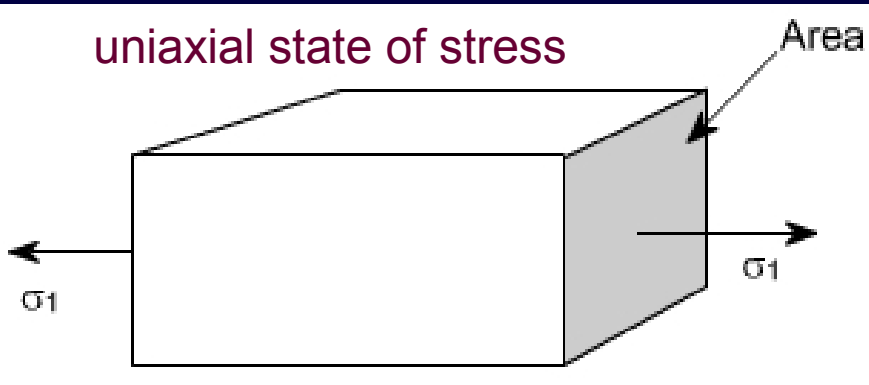
TYPES OF STRESSES: only two basic stresses exists :

- (1) normal stress
- (2) shear stress.

Other stresses either are similar to these basic stresses or are a combination of these e.g. bending stress is a combination of tensile, compressive and shear stresses. Torsional stress, as encountered in twisting of a shaft is a shearing stress.

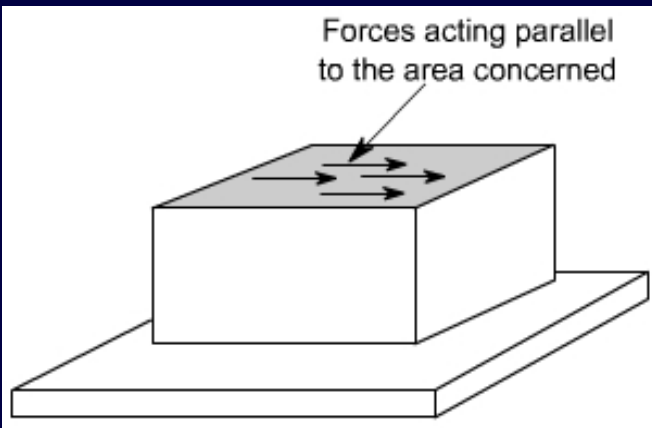
Normal stresses : If the stresses are normal to the areas concerned, then these are termed as normal stresses. The normal stresses are generally denoted by a Greek letter (σ)

uniaxial state of stress



Bearing Stress : When one object presses against another, it is referred to a bearing stress (They are in fact the compressive stresses).

Shear stresses : Where the cross sectional area of a block of material is subject to a distribution of forces which are parallel, rather than normal, to the area concerned. Such forces are associated with a shearing of the material, and are referred to as shear forces.

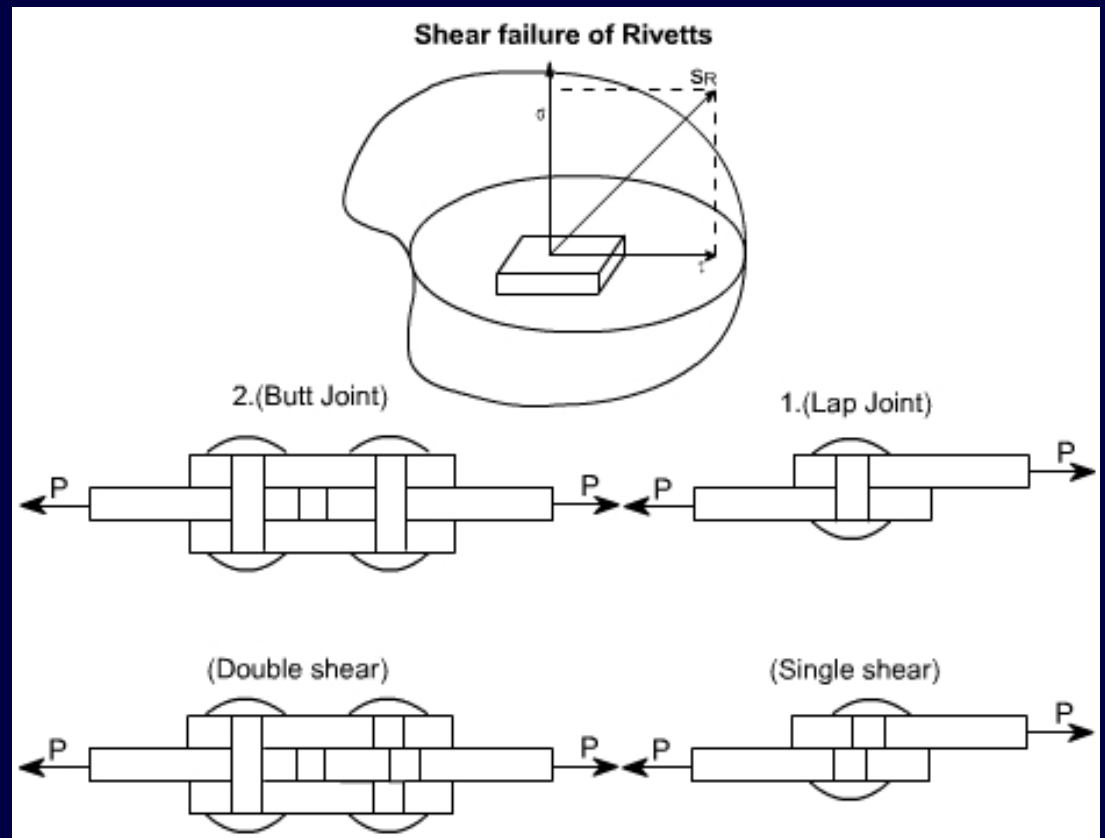


The resulting force intensities are known as shear stresses, the mean shear stress being equal to

$$\tau = \frac{P}{A}$$

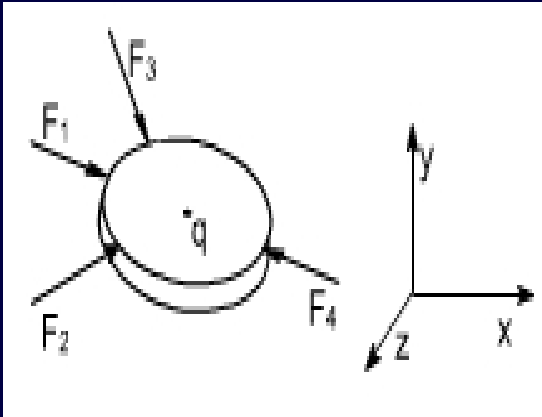
where P is the total force and A the area over which it acts.

The stress (resultant stress) at any point in a body is basically resolved into two components σ and τ one acts perpendicular and other parallel to the area concerned, as it is clearly defined in the following figure.

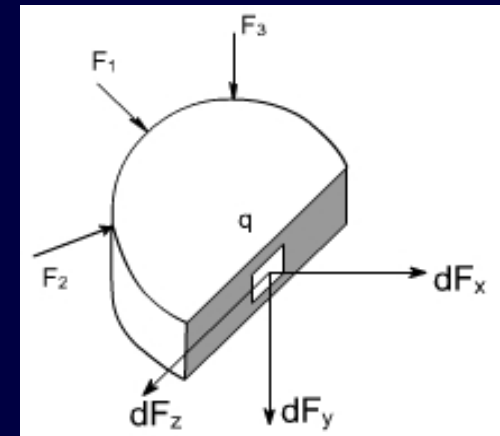


Lecture 2: ANALYSIS OF STRESSES

General State of stress at a point : Stress at a point in a material body has been defined as a force per unit area. But this definition is somewhat ambiguous since it depends upon what area we consider at that point. Let us, consider a point 'q' in the interior of the body



Let us pass a cutting plane through a point 'q' perpendicular to the x - axis as shown below



The corresponding force components can be shown like this

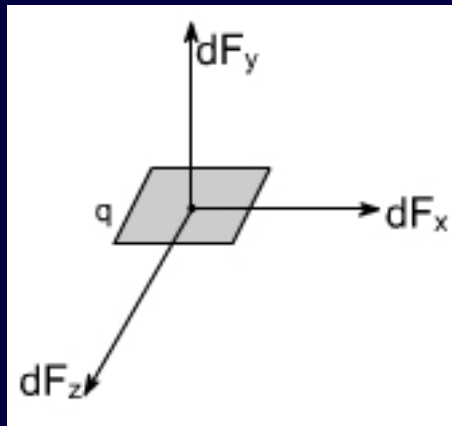
$$dF_x = \sigma_{xx} \cdot da_x$$

$$dF_y = \tau_{xy} \cdot da_x$$

$$dF_z = \tau_{xz} \cdot da_x$$

where da_x is the area surrounding the point 'q' when the cutting plane perpendicular is to x - axis.

In a similar way it can be assumed that the cutting plane is passed through the point 'q' perpendicular to the y - axis. The corresponding force components are shown below



The corresponding force components may be written as

$$dF_x = \tau_{yx} \cdot da_y$$

$$dF_y = \sigma_{yy} \cdot da_y$$

$$dF_z = \tau_{yz} \cdot da_y$$

where da_y is the area surrounding the point 'q' when the cutting plane perpendicular to y - axis.

In the last it can be considered that the cutting plane is passed through the point 'q' perpendicular to the z - axis.

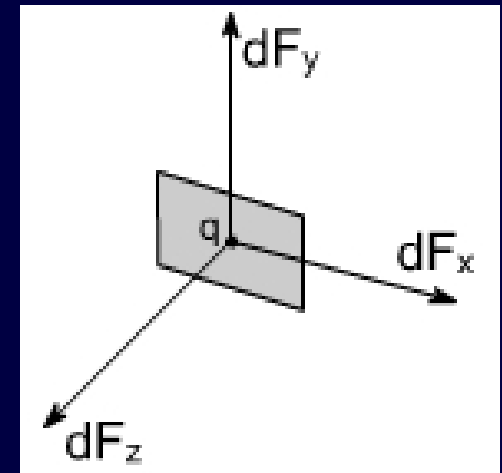
The corresponding force components may be written as

$$dF_x = \tau_{zx} \cdot da_z$$

$$dF_y = \sigma_{zy} \cdot da_z$$

$$dF_z = \tau_{zz} \cdot da_z$$

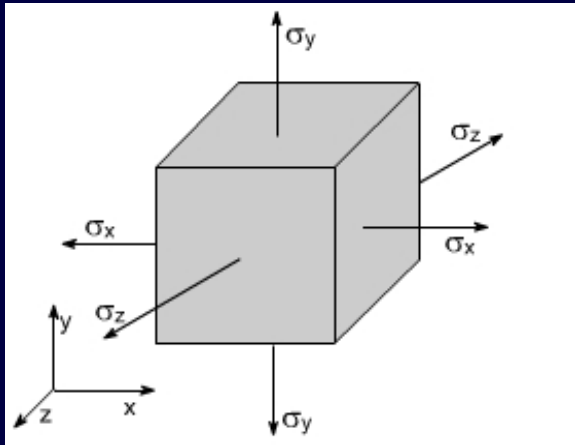
where da_z is the area surrounding the point 'q' when the cutting plane perpendicular to z - axis.



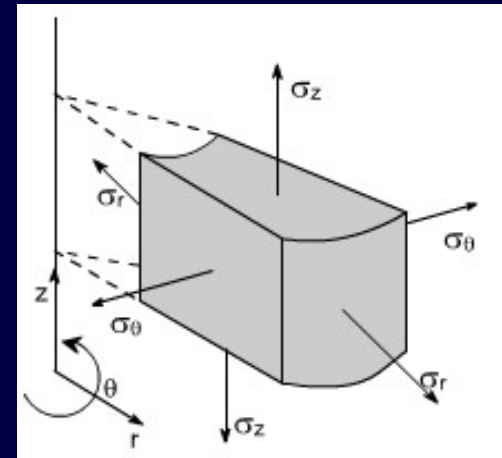
Salient Features

- It is amply clear that there is nothing like stress at a point 'q'
- There exist a situation where it is a combination of state of stress at a point q.
- Thus, it becomes imperative to understand the term state of stress at a point 'q'.
- Therefore, it becomes easy to express a state of stress by the scheme as discussed earlier, where the stresses on the three mutually perpendicular planes are labeled in the manner as shown earlier. the state of stress as depicted earlier is called the general or a tri-axial state of stress that can exist at any interior point of a loaded body.
- Before defining the general state of stress at a point. Let us make ourselves conversant with the notations for the stresses.
- We have already chosen to distinguish between normal and shear stress with the help of symbols σ and τ .

Cartesian - co-ordinate system:



Cylindrical - co-ordinate system:



Sign convention : The tensile forces are termed as (+ve) while the compressive forces are termed as negative (-ve).

Normal Stress:

- tensile +ve
- compressive -ve

First subscript : it indicates the direction of the normal to the surface.

Second subscript : it indicates the direction of the stress.

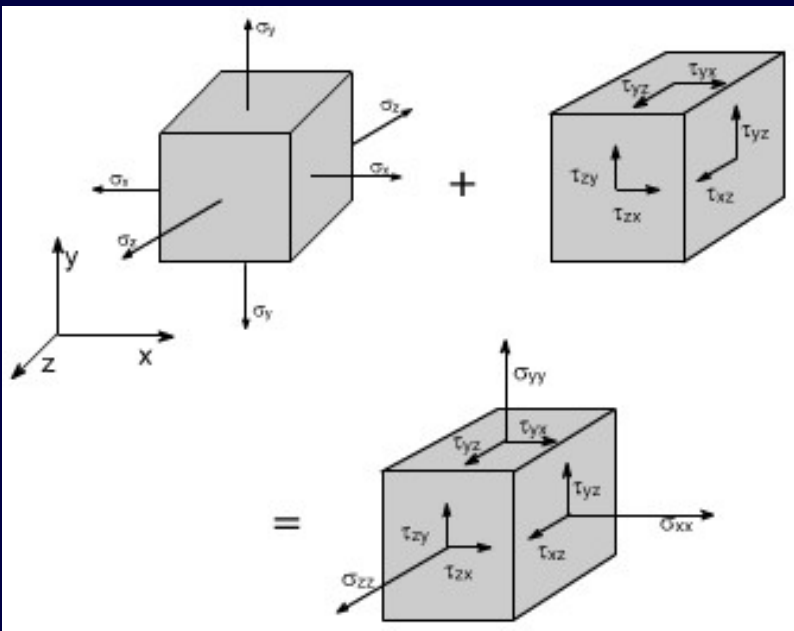
It may be noted that in the case of normal stresses the double script notation may be dispensed with as the direction of the normal stress and the direction of normal to the surface of the element on which it acts is the same. Therefore, a single subscript notation as used is sufficient to define the normal stresses.

Shear Stresses

Single subscript notation is not practical, because such stresses are in direction parallel to the surfaces on which they act. We therefore have two directions to specify, that of normal to the surface and the stress itself.

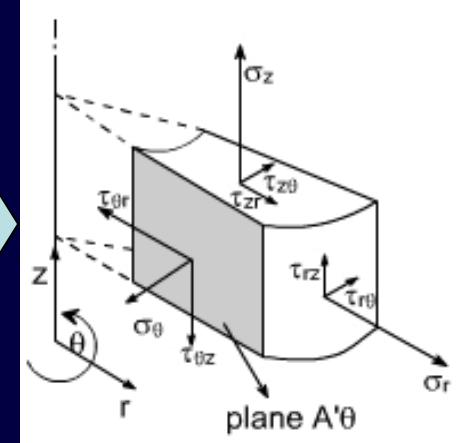
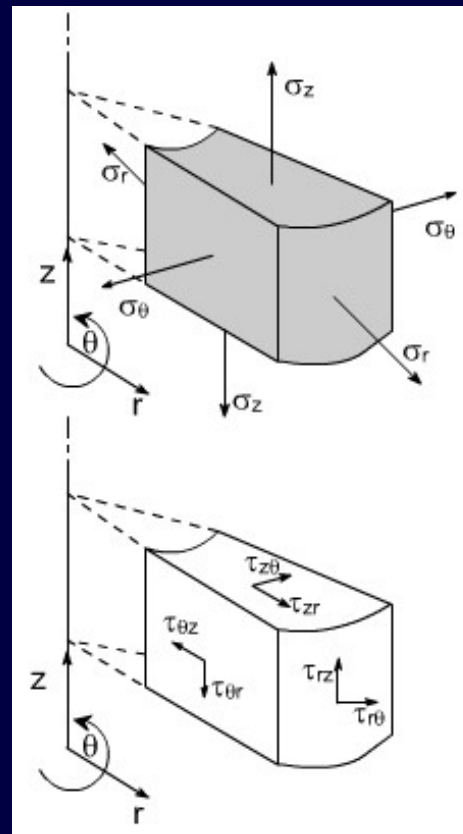
Cartesian - co-ordinate system:

$$\tau_{xy}, \tau_{yx}, \tau_{yz}, \tau_{zy}, \tau_{zx}, \tau_{xz}$$



Cylindrical - co-ordinate system:

$$\tau_{rq}, \tau_{qr}, \tau_{qz}, \tau_{zq}, \tau_{zr}, \tau_{rz}$$



State of stress at a point

By state of stress at a point, we mean an information which is required at that point such that it remains under equilibrium. or simply a general state of stress at a point involves all the normal stress components, together with all the shear stress components as shown in earlier figures.

Therefore, we need nine components, to define the state of stress at a point


$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \sigma_y & \tau_{yx} & \tau_{yz} \\ \sigma_z & \tau_{zx} & \tau_{zy} \end{bmatrix}$$

we apply the conditions of equilibrium which are as follows:

$$\begin{aligned} \sum F_x &= 0 ; \quad \sum M_x = 0 \\ \sum F_y &= 0 ; \quad \sum M_y = 0 \\ \sum F_z &= 0 ; \quad \sum M_z = 0 \end{aligned}$$

Then we get $\tau_{xy} = \tau_{yx}$; $\tau_{yz} = \tau_{zy}$; $\tau_{zx} = \tau_{xz}$

Then we will need only six components to specify the state of stress at a point i.e,

$$\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{yz}, \tau_{zx}$$

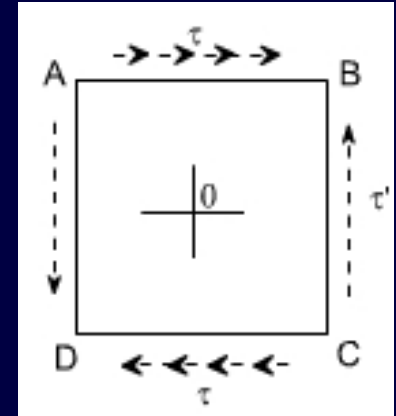
Shear stresses:

- tending to turn the element C.W +ve.
- tending to turn the element C.C.W -ve.

Complementary shear stresses

The existence of shear stresses on any two sides of the element induces complementary shear stresses on the other two sides of the element to maintain equilibrium.

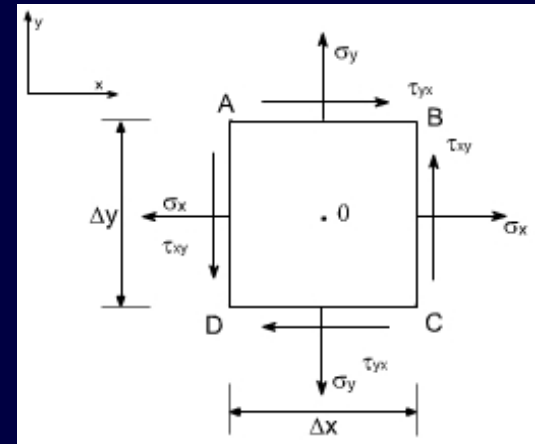
- ❖ On planes AB and CD, the shear stress τ acts.
- ❖ To maintain the static equilibrium of this element, on planes AD and BC, τ' should act,
- ❖ we shall see that τ' which is known as the complementary shear stress would come out to equal and opposite to the τ .



Let us prove this thing for a general case as discussed below:

The figure shows a small rectangular element with sides of length Δx , Δy parallel to x and y directions. Its thickness normal to the plane of paper is Δz in z direction. All nine normal and shear stress components may act on the element, only those in x and y directions are shown.

Now forces on top and bottom surfaces produce a couple which must be balanced by the forces on left and right hand faces



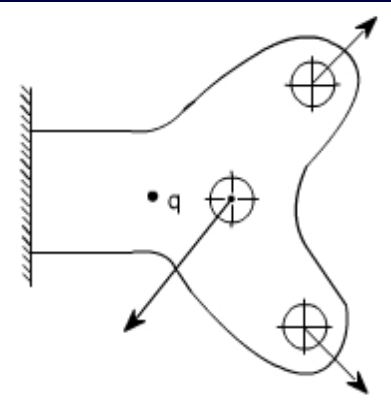
$$\tau_{yx} \cdot \Delta x \cdot \Delta z \cdot \Delta y = \tau_{xy} \cdot \Delta x \cdot \Delta z \cdot \Delta y$$

$$\Rightarrow \tau_{yx} = \tau_{xy}$$

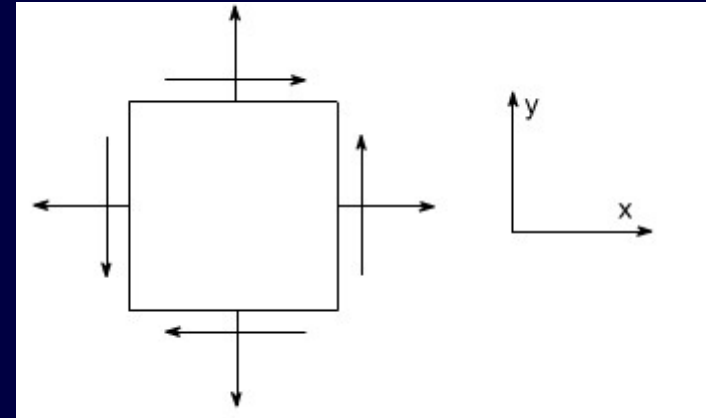
$$\text{Similarly, } \tau_{yz} = \tau_{zy} \text{ and } \tau_{xz} = \tau_{zx}$$

The complementary shear stresses are equal in magnitude

LECTURE 3: ANALYSIS OF STRESSES

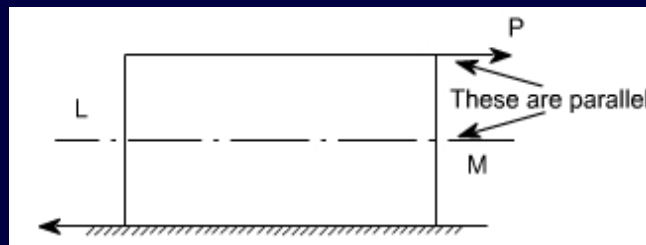


Consider a point 'q' in some sort of structural member. Assuming that at point 'q', a plane state of stress exist. i.e., the state of state stress is to describe by a parameters σ_x , σ_y and τ_{xy} . These stresses could be indicated on the two dimensional diagram as shown here:



This is a common way of representing the stresses. It must be realized that the material is unaware of what we have called the x and y axes. i.e. the material has to resist the loads irrespective less of how we wish to name them or whether they are horizontal, vertical or otherwise further more, the material will fail when the stresses exceed beyond a permissible value.

Thus, a fundamental problem in engineering design is to determine the **maximum normal stress or maximum shear stress at any particular point in a body**. There is no reason to believe a priori that σ_x , σ_y and τ_{xy} are the maximum value. Rather the maximum stresses may associates themselves with some other planes located at 'q'. Thus, it becomes imperative to determine the values of σ_q and τ_q . In order to achieve this let us consider the following.

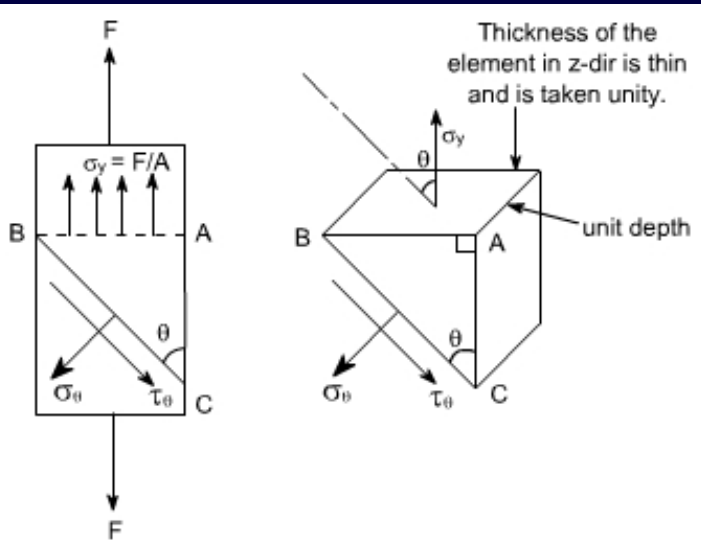


Stresses on oblique plane

Till now we have dealt with either pure normal direct stress or pure shear stress. In many instances, however both direct and shear stresses act and the resultant stress across any section will be neither normal nor tangential to the plane.

A plane state of stress is a 2-D state of stress in a sense that the stress components in one direction are all zero i.e

$$\sigma_z = \tau_{yz} = \tau_{zx} = 0$$



Let the block be of unit depth now considering the equilibrium of forces on the triangle portion ABC. Resolving forces perpendicular to BC, gives

$$\sigma_{\theta} \cdot BC \cdot 1 = \sigma_y \sin \theta \cdot AB \cdot 1$$

$$\text{but } AB/BC = \sin \theta \text{ or } AB = BC \sin \theta$$

$$\sigma_{\theta} \cdot BC \cdot 1 = \sigma_y \sin \theta \cdot BC \sin \theta \cdot 1$$

$$\Rightarrow \sigma_{\theta} = \sigma_y \sin^2 \theta = \frac{1}{2} \sigma_y (1 - \cos 2\theta)$$

Now resolving the forces parallel to BC

$$\tau_{\theta} \cdot BC \cdot 1 = \sigma_y \cos \theta \cdot AB \sin \theta \cdot 1$$

$$\text{again } AB = BC \cos \theta$$

$$\tau_{\theta} \cdot BC \cdot 1 = \sigma_y \cos \theta \cdot BC \sin \theta \cdot 1 \text{ or } \tau_{\theta} = \sigma_y \sin \theta \cos \theta$$

$$\Rightarrow \tau_{\theta} = \frac{1}{2} \sigma_y \sin 2\theta$$

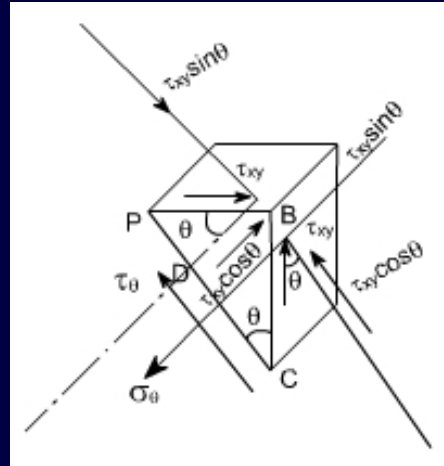
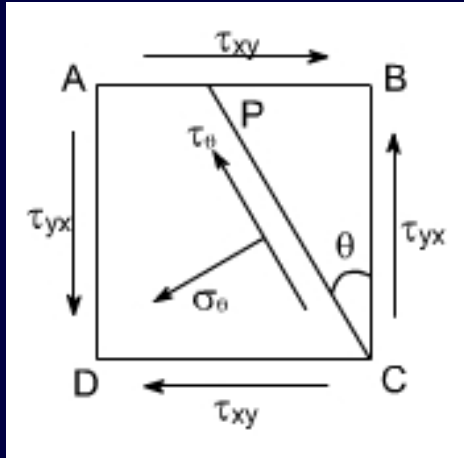
By examining the equations (1) and (2), the following conclusions may be drawn

- ❖ The value of direct stress σ_{θ} is maximum and is equal to σ_y when $\theta = 90^\circ$.
- ❖ The shear stress τ_{θ} has a maximum value of $0.5\sigma_y$ when $\theta = 45^\circ$
- ❖ The stresses σ_{θ} and τ_{θ} are not simply the resolution of σ_y

Material Subjected to Pure Shear

Shear stresses have been applied to the sides AB and DC

Consider the equilibrium of portion of PBC



Assuming unit depth and resolving normal to PC or in the direction of σ_θ

$$\begin{aligned}\sigma_\theta \cdot PC \cdot 1 &= \tau_{xy} \cdot PB \cdot \cos\theta \cdot 1 + \tau_{xy} \cdot BC \cdot \sin\theta \cdot 1 \\ &= \tau_{xy} \cdot PB \cdot \cos\theta + \tau_{xy} \cdot BC \cdot \sin\theta\end{aligned}$$

Now $PB/PC = \sin\theta$ $BC/PC = \cos\theta$

$$\sigma_\theta \cdot PC \cdot 1 = \tau_{xy} \cdot \cos\theta \sin\theta PC + \tau_{xy} \cdot \cos\theta \cdot \sin\theta PC$$

$$\sigma_\theta = 2 \tau_{xy} \sin\theta \cos\theta = \tau_{xy} \cdot \sin 2\theta \dots\dots\dots (1)$$

Now resolving forces parallel to PC or in the direction τ_θ .

$$\text{then } \tau_{xy} PC \cdot 1 = \tau_{xy} \cdot PB \sin\theta - \tau_{xy} \cdot BC \cos\theta$$

-ve sign has been put because this component is in the same direction as that of τ_θ .

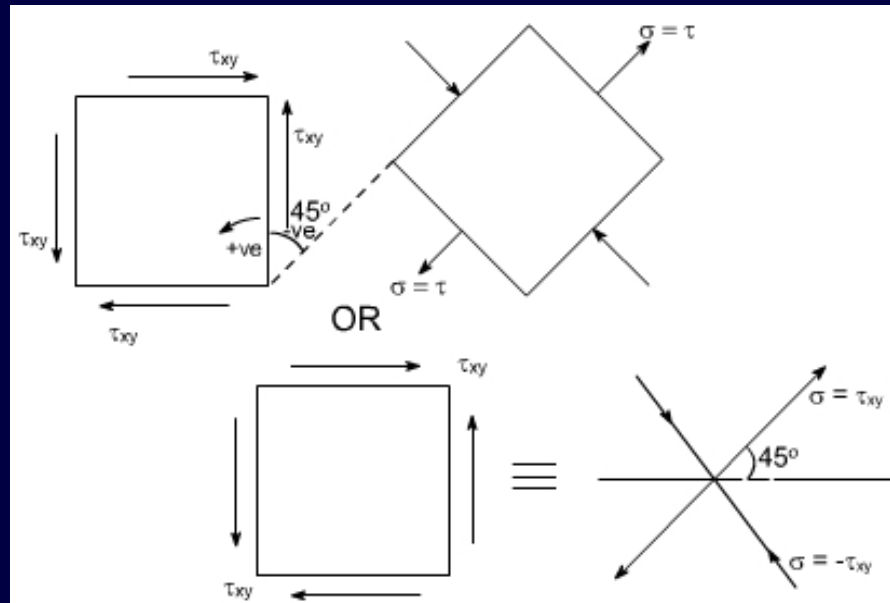
again converting the various quantities in terms of PC we have

$$\tau_{xy} PC \cdot 1 = \tau_{xy} \cdot PB \cdot \sin^2\theta - \tau_{xy} \cdot PC \cos^2\theta$$

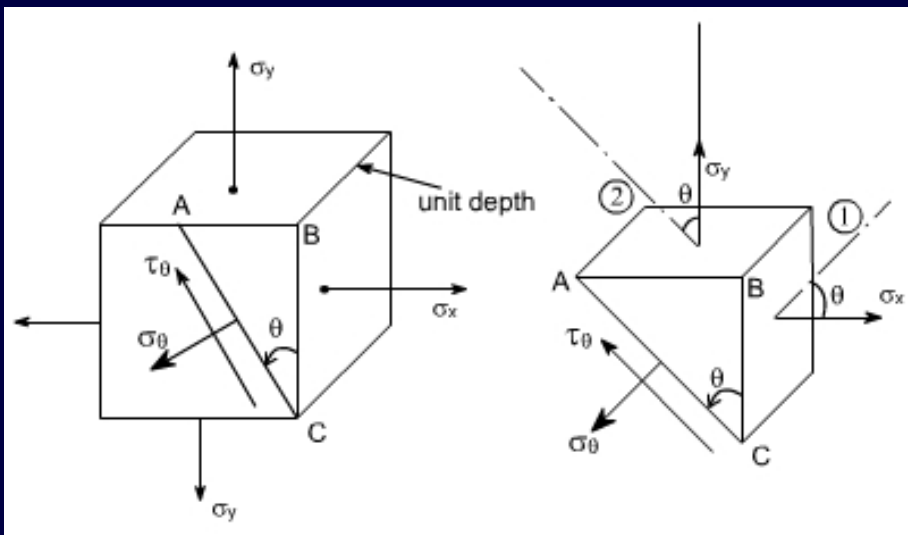
$$\tau_\theta = -[\tau_{xy} (\cos^2\theta - \sin^2\theta)] = -\tau_{xy} \cos 2\theta \dots\dots\dots (2)$$

Salient Features

- The negative sign means that the sense of τ_θ is opposite to that of assumed one. Let us examine the equations (1) and (2) respectively
- Equation (1) ($\sigma_\theta = \tau_{xy} \sin 2\theta$), represents that maximum value of $\sigma_\theta = \tau_{xy}$ when $\theta = 45^\circ$.
- Equation (2) ($\tau_\theta = -\tau_{xy} \cos 2\theta$), which indicates that the maximum value of τ_θ is τ_{xy} when $\theta = 0^\circ$ or 90° . it has a value zero when $\theta = 45^\circ$.
- From equation (1) it may be noticed that the normal component σ_θ has maximum and minimum values of $+\tau_{xy}$ (tension) and $-\tau_{xy}$ (compression) on plane at $\pm 45^\circ$ to the applied shear and on these planes the tangential component τ_θ is zero.
- Hence the system of pure shear stresses produces an equivalent direct stress system, one set compressive and one tensile each located at 45° to the original shear directions as depicted in the figure below:



Material subjected to two mutually perpendicular direct stresses



For equilibrium of the portion ABC, resolving perpendicular to AC

$$\sigma_{\theta} \cdot AC.1 = \sigma_y \sin\theta \cdot AB.1 + \sigma_x \cos\theta \cdot BC.1$$

Converting AB and BC in terms of AC so that AC cancels out from the sides On rearranging the various terms we get

$$\sigma_{\theta} = \left(\frac{\sigma_x + \sigma_y}{2} \right) + \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos 2\theta \dots\dots\dots (3)$$

Now resolving parallel to AC: $\tau_{\theta} \cdot AC.1 = -\sigma_y \cdot \cos\theta \cdot AB.1 + \sigma_x \cdot BC \cdot \sin\theta.1$

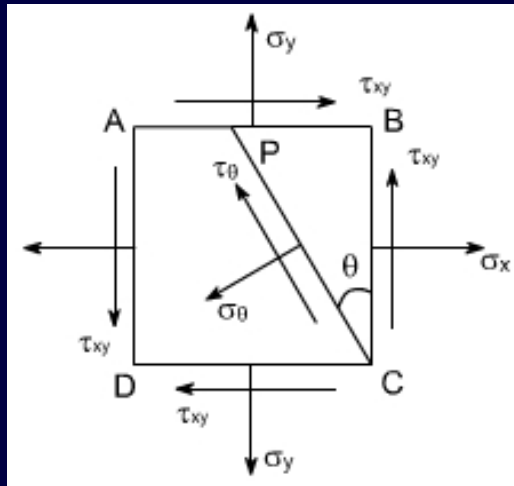
Again converting the various quantities in terms of AC so that the AC cancels out from the two sides.

$$\tau_{\theta} = \frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta \dots\dots\dots (4)$$

- ❖ The maximum direct stress would be equal to σ_x or σ_y which ever is the greater, when $\theta = 0^\circ$ or 90°
- ❖ The maximum shear stress in the plane of the applied stresses occurs when $\theta = 45^\circ$

$$\tau_{\max} = \frac{(\sigma_x - \sigma_y)}{2}$$

LECTURE 4: Material subjected to combined direct and shear stresses



By looking at this state of stress, it may be observed that this state of stress is combination of two different cases:

- Material subjected to pure state of stress shear. In this case the various formulas deserved are as follows

$$\sigma_{\theta} = \tau_{yx} \sin 2\theta$$

$$\tau_{\theta} = -\tau_{yx} \cos 2\theta$$

- Material subjected to two mutually perpendicular direct stresses. In this case the various formula's derived are as follows.

$$\sigma_{\theta} = \left(\frac{\sigma_x + \sigma_y}{2} \right) + \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos 2\theta$$

$$\tau_{\theta} = \frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta$$

To get the required equations for the case under consideration, let us add the respective equations for the above two cases such that

$$\sigma_{\theta} = \left(\frac{\sigma_x + \sigma_y}{2} \right) + \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos 2\theta + \tau_{yx} \sin 2\theta$$

$$\tau_{\theta} = \frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta - \tau_{yx} \cos 2\theta$$

- These are the equilibrium equations for stresses at a point. They do not depend on material proportions and are equally valid for elastic and inelastic behavior
- This eqn. gives two values of 2θ that differ by 180° . Hence the planes on which maximum and minimum normal stresses occur are 90° apart.

For σ_θ to be a maximum or minimum $\frac{d\sigma_\theta}{d\theta} = 0$

Now

$$\sigma_\theta = \frac{(\sigma_x + \sigma_y)}{2} + \frac{(\sigma_x - \sigma_y)}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

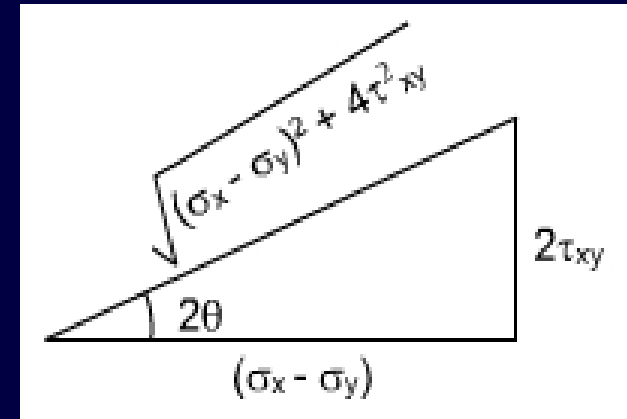
$$\begin{aligned} \frac{d\sigma_\theta}{d\theta} &= -\frac{1}{2}(\sigma_x - \sigma_y) \sin 2\theta \cdot 2 + \tau_{xy} \cos 2\theta \cdot 2 \\ &= 0 \end{aligned}$$

$$\text{i.e. } -(\sigma_x - \sigma_y) \sin 2\theta + \tau_{xy} \cos 2\theta \cdot 2 = 0$$

$$\tau_{xy} \cos 2\theta \cdot 2 = (\sigma_x - \sigma_y) \sin 2\theta$$

Thus,
$$\tan 2\theta = \frac{2\tau_{xy}}{(\sigma_x - \sigma_y)}$$

From the triangle it may be determined



$$\cos 2\theta = \frac{(\sigma_x - \sigma_y)}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

$$\sin 2\theta = \frac{2\tau_{xy}}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

Substituting the values of $\cos 2\theta$ and $\sin 2\theta$ in equation (5) we get

$$\sigma_{\theta} = \frac{(\sigma_x + \sigma_y)}{2} + \frac{(\sigma_x - \sigma_y)}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{\theta} = \frac{(\sigma_x + \sigma_y)}{2} + \frac{(\sigma_x - \sigma_y)}{2} \cdot \frac{(\sigma_x - \sigma_y)}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}} + \frac{\tau_{xy} \cdot 2\tau_{xy}}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

$$= \frac{(\sigma_x + \sigma_y)}{2} + \frac{1}{2} \cdot \frac{(\sigma_x - \sigma_y)^2}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}} + \frac{1}{2} \cdot \frac{4\tau_{xy}^2}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

or

$$= \frac{(\sigma_x + \sigma_y)}{2} + \frac{1}{2} \cdot \frac{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

$$= \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2} \cdot \frac{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2} \cdot \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

$$\sigma_{\theta} = \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2} \cdot \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

Hence we get the two values of σ_{θ} , which are designated σ_1 as σ_2 and respectively, therefore

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2} \cdot \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$\sigma_2 = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2} \cdot \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

The σ_1 and σ_2 are termed as the principle stresses of the system.

Substituting the values of $\cos 2\theta$ and $\sin 2\theta$ in equation (6) we see that

$$\tau_{\theta} = \frac{1}{2}(\sigma_x - \sigma_y) \sin 2\theta - \tau_{xy} \cos 2\theta$$

$$= \frac{1}{2}(\sigma_x - \sigma_y) \cdot \frac{2\tau_{xy}}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}} - \frac{\tau_{xy} \cdot (\sigma_x - \sigma_y)}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}}$$

$$\tau_{\theta} = 0$$

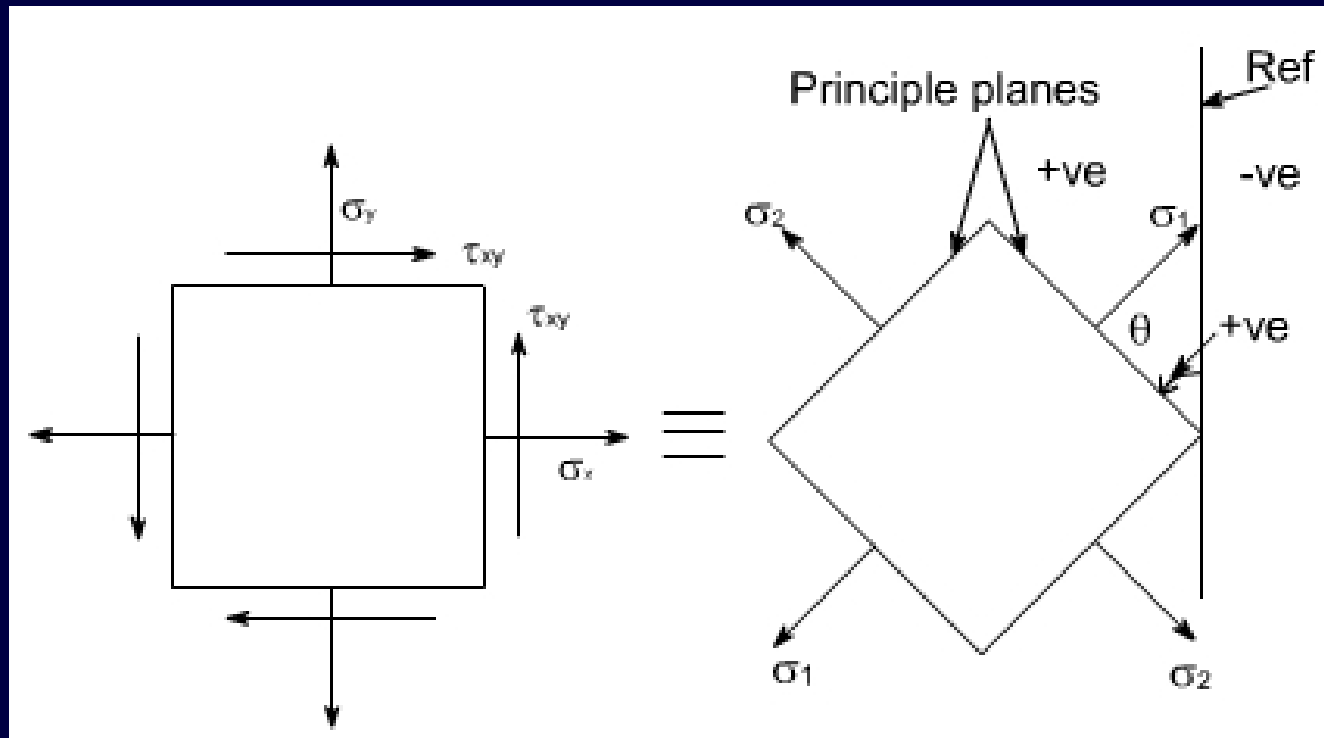
This shows that the values of shear stress is zero on the principal planes.

Maximum and minimum values of normal stresses occur on planes of zero shearing stress. The maximum and minimum normal stresses are called the principal stresses, and the planes on which they act are called principal plane the solution of equation

$$\tan 2\theta_p = \frac{2\tau_{xy}}{(\sigma_x - \sigma_y)}$$

will yield two values of 2θ separated by 180° i.e. two values of θ separated by 90° . Thus the two principal stresses occur on mutually perpendicular planes termed principal planes.

Therefore the two dimensional complex stress system can now be reduced to the equivalent system of principal stresses.



Let us recall that for the case of a material subjected to direct stresses the value of maximum shear stresses

$$\tau_{\max}^m = \frac{1}{2}(\sigma_x - \sigma_y) \text{ at } \theta = 45^\circ, \text{ Thus, for a 2-dimensional state of stress, subjected to principle stresses}$$

$$\tau_{\max}^m = \frac{1}{2}(\sigma_1 - \sigma_2), \text{ on substituting the values of } \sigma_1 \text{ and } \sigma_2, \text{ we get}$$

$$\tau_{\max}^m = \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

Alternatively this expression can also be obtained by differentiating the expression for τ_θ with respect to θ i.e.

$$\tau_\theta = \frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta - \tau_{xy} \cos 2\theta$$

$$\begin{aligned} \frac{d\tau_\theta}{d\theta} &= -\frac{1}{2}(\sigma_x - \sigma_y) \cos 2\theta \cdot 2 + \tau_{xy} \sin 2\theta \cdot 2 \\ &= 0 \end{aligned}$$

$$\text{or } (\sigma_x - \sigma_y) \cos 2\theta + 2\tau_{xy} \sin 2\theta = 0$$

$$\tan 2\theta_s = \frac{(\sigma_y - \sigma_x)}{2\tau_{xy}} = -\frac{(\sigma_x - \sigma_y)}{2\tau_{xy}}$$

$$\tan 2\theta_s = -\frac{(\sigma_x - \sigma_y)}{2\tau_{xy}}$$

Recalling that

$$\tan 2\theta_p = \frac{2\tau_{xy}}{(\sigma_x - \sigma_y)}$$

Thus,

$$\boxed{\tan 2\theta_p \cdot \tan 2\theta_s = 1}$$

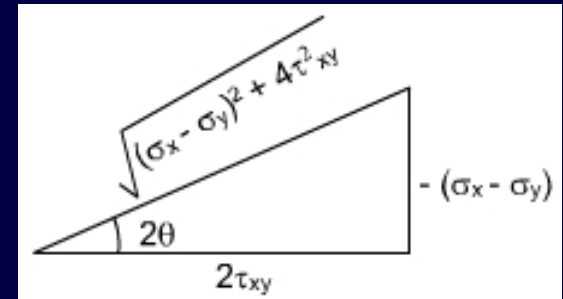
- Therefore, it can be concluded that the equation (2) is a negative reciprocal of equation (1) hence the roots for the double angle of equation (2) are 90° away from the corresponding angle of equation (1).
- This means that the angles that locate the plane of maximum or minimum shearing stresses form angles of 45° with the planes of principal stresses.
- Further, by making the triangle we get

$$\cos 2\theta = \frac{2\tau_{xy}}{\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}}$$

$$\sin 2\theta = \frac{-(\sigma_x - \sigma_y)}{\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}}$$

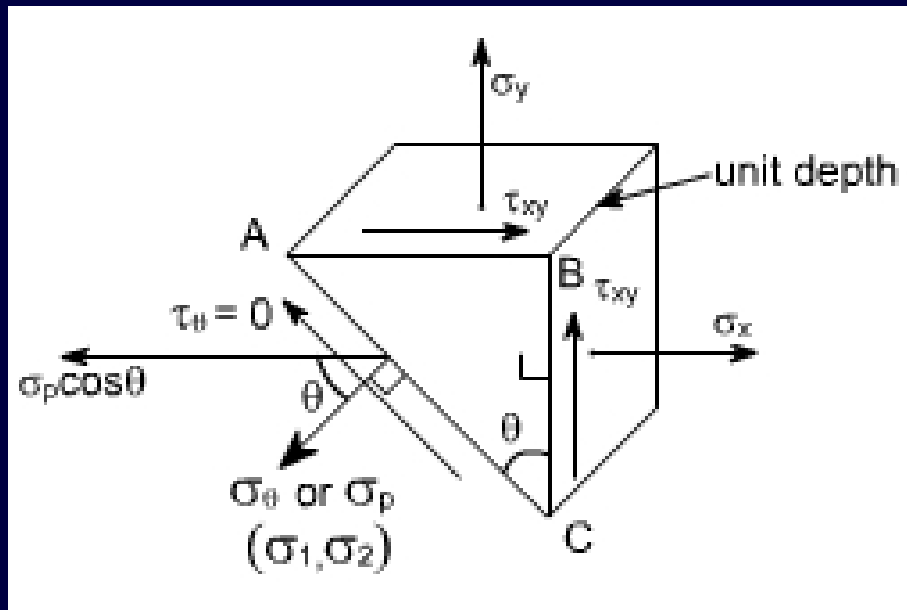
Therefore by substituting the values of $\cos 2\theta$ and $\sin 2\theta$ we have

$$\begin{aligned}\tau_\theta &= \frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta - \tau_{xy}\cos 2\theta \\ &= \frac{1}{2} \cdot \frac{(\sigma_x - \sigma_y) \cdot (\sigma_x - \sigma_y)}{\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}} - \frac{\tau_{xy} \cdot 2\tau_{xy}}{\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}} \\ &= -\frac{1}{2} \cdot \frac{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}{\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2}} \\ \tau_\theta &= \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}\end{aligned}$$



- ✓ Because of root the difference in sign convention arises from the point of view of locating the planes on which shear stress act. From physical point of view these sign have no meaning.
- ✓ The largest stress regard less of sign is always know as maximum shear stress.

Principal plane inclination in terms of associated principal stress



Consider once again the equilibrium of a triangular block of material of unit depth, Assuming AC to be a principal plane on which principal stresses σ_p acts, and the shear stress is zero.

$$\sigma_x \cdot + \tau_{xy} \frac{AB}{BC} = \sigma_p \cdot \cos \theta \cdot \frac{AC}{BC}$$

or

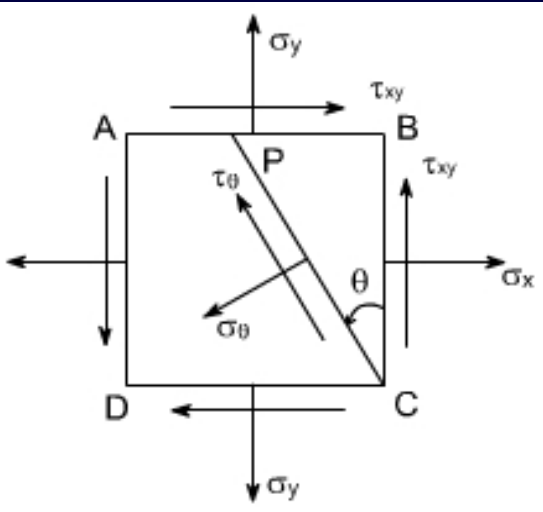
$$\sigma_x + \tau_{xy} \tan \theta = \sigma_p$$

Thus

$$\tan \theta = \frac{\sigma_p - \sigma_x}{\tau_{xy}}$$

LECTURE 5: GRAPHICAL SOLUTION / MOHR'S STRESS CIRCLE

The Mohr's stress circle is used to find out graphically the direct stress σ and shear stress τ on any plane inclined at θ to the plane on which σ_x acts. The direction of θ here is taken in anticlockwise direction from the BC.



STEPS:

1. Label the Block ABCD.
2. Set up axes for the direct stress (as abscissa) and shear stress (as ordinate)
3. Plot the stresses on two adjacent faces e.g. AB and BC, using the following sign convention.

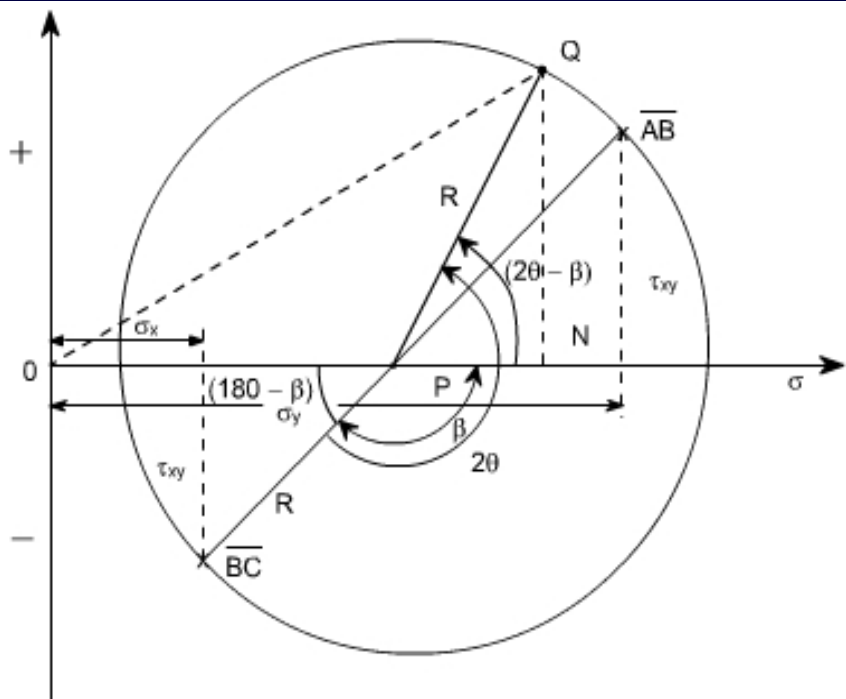
Direct stresses - tensile -- positive; compressive -- negative

Shear stresses - tending to turn block clockwise, positive
tending to turn block counter clockwise, negative

This gives two points on the graph which may then be labeled as respectively to denote stresses on these planes.

4. Join AB and BC
5. The point P where this line cuts the σ axis is then the center of Mohr's stress circle and the line joining AB and BC is diameter. Therefore the circle can now be drawn.

Now every point on the circle then represents a state of stress on some plane through C.



Proof: Consider any point Q on the circumference of the circle, such that PQ makes an angle 2θ with BC, and drop a perpendicular from Q to meet the σ axis at N. Then OQ represents the resultant stress on the plane an angle θ to BC. Here we have assumed that $\sigma_x > \sigma_y$. Now let us find out the coordinates of point Q.

These are ON and QN. From the figure drawn earlier $ON = OP + PN$ and $OP = OK + KP$

$$OP = \sigma_y + 1/2 (\sigma_x - \sigma_y) = (\sigma_x + \sigma_y)/2$$

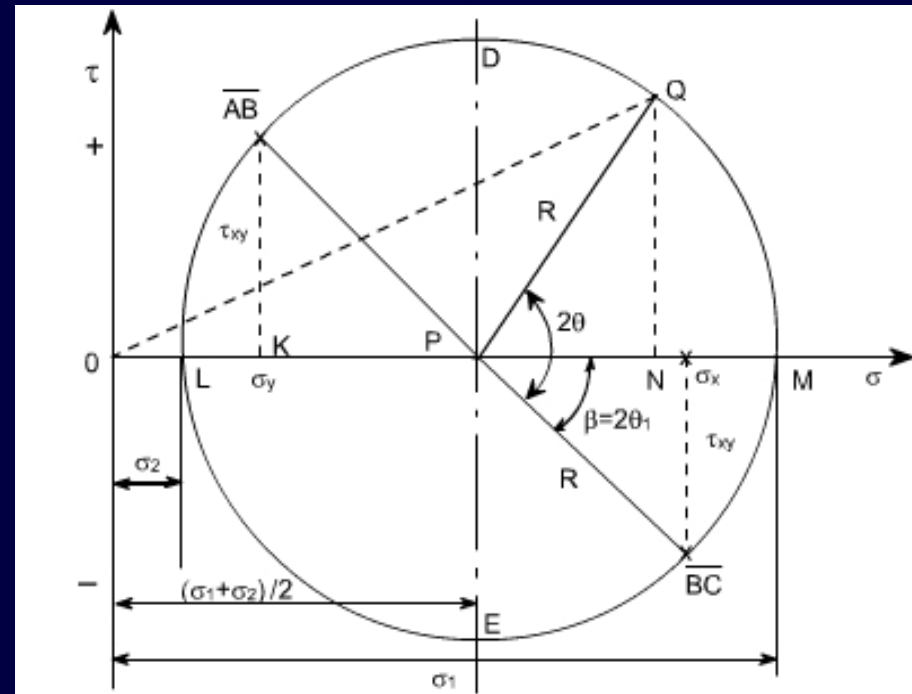
$$PN = R \cos(2\theta - \beta)$$

$$R \cos \beta = \frac{(\sigma_x - \sigma_y)}{2}; R \sin \beta = \tau_{xy}$$

$$ON = (\sigma_x + \sigma_y)/2 + 1/2 (\sigma_x + \sigma_y) \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\begin{aligned} QM &= R \sin(2\theta - \beta) = R \sin 2\theta \cos \beta - R \cos 2\theta \sin \beta \\ &= 1/2 (\sigma_x - \sigma_y) \sin 2\theta - \tau_{xy} \cos 2\theta \end{aligned}$$

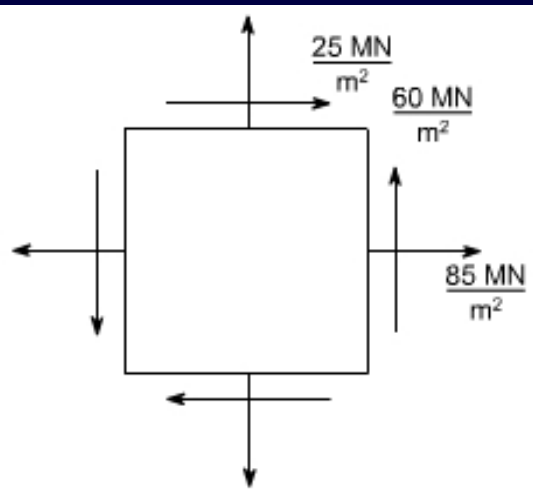
These are same as derived analytically



Salient Features

1. The direct stress is maximum when Q is at M and at this point obviously the shear stress is zero, hence by definition OM is the length representing the maximum principal stresses σ_1 and $2\theta_1$ gives the angle of the plane θ_1 from BC. Similar OL is the other principal stress and is represented by σ_2
2. The maximum shear stress is given by the highest point on the circle and is represented by the radius of the circle.
3. From the above point the maximum shear stress i.e. the Radius of the Mohr's stress circle would be $(\sigma_x - \sigma_y)/2$
4. As already defined the principal planes are the planes on which the shear components are zero. Therefore, we may conclude that on principal plane the shear stress is zero.
5. Since the resultant of two stress at 90° can be found from the parallelogram of vectors as shown in the diagram. Thus, the resultant stress on the plane at q to BC is given by OQ on Mohr's Circle.
6. The graphical method of solution for a complex stress problems using Mohr's circle is a very powerful technique, since all the information relating to any plane within the stressed element is contained in the single construction. It thus, provides a convenient and rapid means of solution. Which is less prone to arithmetical errors and is highly recommended.

ILLUSRATIVE PROBLEMS



σ_1 and σ_2

$$= \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$= \frac{1}{2}(85 + 25) \pm \frac{1}{2}\sqrt{(85 + 25)^2 + (4 \times 60^2)}$$

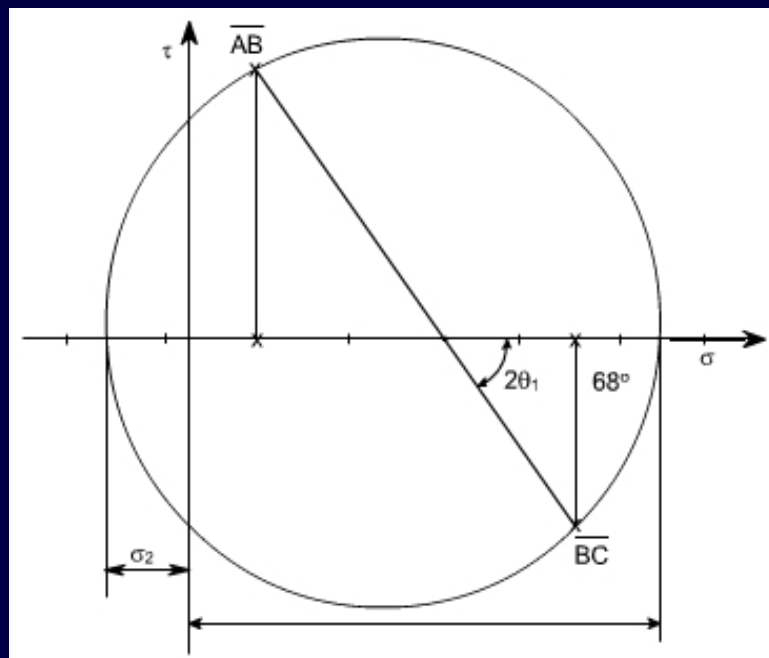
$$= 55 \pm \frac{1}{2} \cdot 60\sqrt{5} = 55 \pm 67$$

$$\Rightarrow \sigma_1 = 122 \text{ MN/m}^2$$

$$\sigma_2 = -12 \text{ MN/m}^2 \text{ (compressive)}$$

$$\tan 2\theta = \left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y} \right)$$

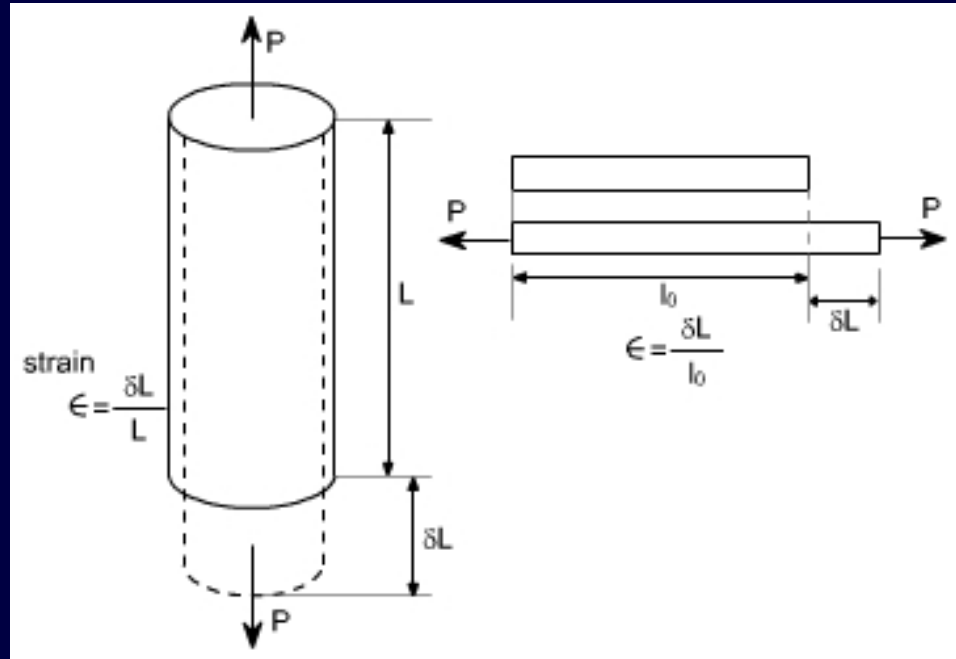
$$\Rightarrow \theta_1 = 31^\circ 71' \text{ and } \theta_2 = 121^\circ 71'$$



LECTURE 6: ANALYSIS OF STRAINS

Linear strain or normal strain or the longitudinal strain:

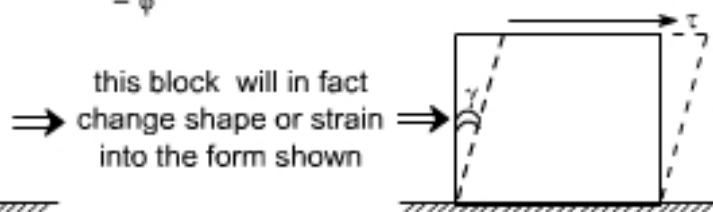
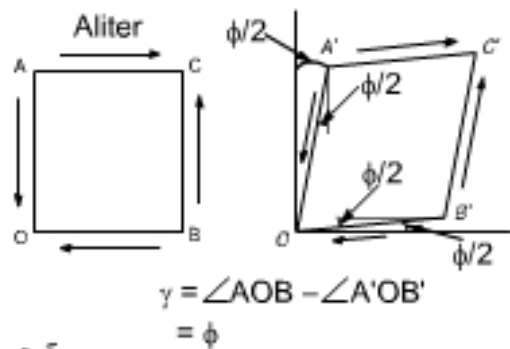
$$\text{strain}(\epsilon) = \frac{\text{change in length}}{\text{original length}} = \frac{\delta L}{L}$$



Sign convention for strain: Tensile strains are positive whereas compressive strains are negative.

NEXT SLIDE -- Shear Strain

- The tangent of the angle through which two adjacent sides rotate relative to their initial position is termed shear strain. In many cases the angle is very small and the angle itself is used, (in radians), instead of tangent, so that $\gamma = \angle AOB - \angle A'OB' = \phi$



Hook's Law: A material is said to be elastic if it returns to its original, unloaded dimensions when load is removed.

$$\frac{\text{stress}}{\text{strain}} = \text{constant}$$

This constant is given by the symbol E and is termed as the **modulus of elasticity** or **Young's modulus of elasticity**

Poisson's ratio: If a bar is subjected to a longitudinal stress there will be a strain in this direction equal to σ/E . There will also be a strain in all directions at right angles to σ . The final shape being shown by the dotted lines.

It has been observed that for an elastic materials, the lateral strain is proportional to the longitudinal strain. The ratio of the lateral strain to longitudinal strain is known as the poisson's ratio .

Poison's ratio (μ) = -lateral strain /longitudinal strain

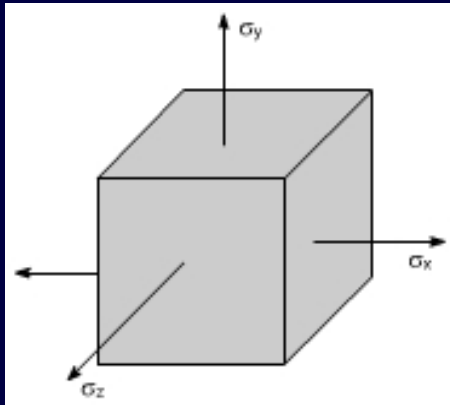
$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon} = \frac{P/A}{\delta L/L}$$

$$E = \frac{PL}{A\delta L}$$



For most engineering materials the value of μ is between 0.25 and 0.33.

Three dimensional state of strain: Consider an element subjected to three mutually perpendicular tensile stresses σ_x , σ_y and σ_z as shown in the figure below.



$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_z}{E}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E}$$

Principal strains in terms of stress: In the absence of shear stresses on the faces of the elements let us say that σ_x , σ_y and σ_z are in fact the principal stress. The resulting strain in the three directions would be the principal strains.

i.e. We will have the following

$$\epsilon_1 = \frac{1}{E} [\sigma_1 - \mu\sigma_2 - \mu\sigma_3]$$

$$\epsilon_2 = \frac{1}{E} [\sigma_2 - \mu\sigma_1 - \mu\sigma_3]$$

$$\epsilon_3 = \frac{1}{E} [\sigma_3 - \mu\sigma_1 - \mu\sigma_2]$$

The negative sign indicating that if σ_y and σ_z are positive i.e. tensile, these they tend to reduce the strain in x direction.

Two dimensional state of strain: stress in third direction is zero ($\sigma_z = 0$ or $\sigma_3 = 0$). Although the strains will be there in this direction due to the stresses σ_1 , σ_2

$$\epsilon_1 = \frac{1}{E} [\sigma_1 - \mu\sigma_2]$$

$$\epsilon_2 = \frac{1}{E} [\sigma_2 - \mu\sigma_1]$$

$$\epsilon_3 = \frac{1}{E} [-\mu\sigma_1 - \mu\sigma_2]$$

Hence a strain can exist without a stress in that direction

$$\text{i.e. if } \sigma_3 = 0; \epsilon_3 = \frac{1}{E} [-\mu\sigma_1 - \mu\sigma_2]$$

Also

$$\epsilon_1 \cdot E = \sigma_1 - \mu\sigma_2$$

$$\epsilon_2 \cdot E = \sigma_2 - \mu\sigma_1$$

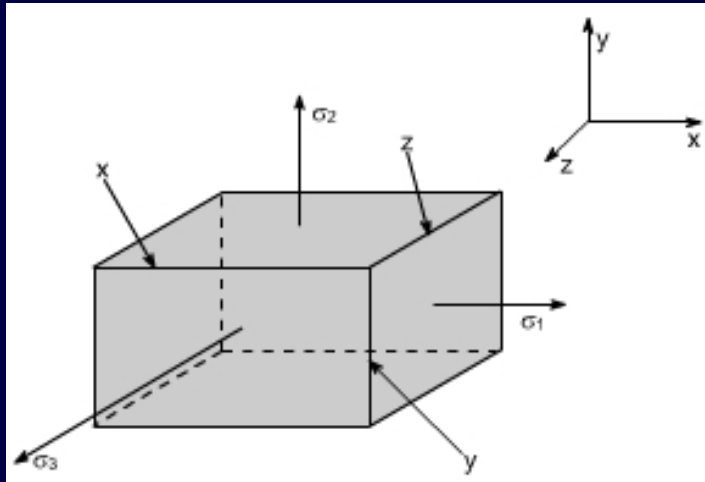
so the solution of above two equations yields

$$\sigma_1 = \frac{E}{(1 - \mu^2)} [\epsilon_1 + \mu\epsilon_2]$$

$$\sigma_2 = \frac{E}{(1 - \mu^2)} [\epsilon_2 + \mu\epsilon_1]$$

Volumetric Strain

$$\begin{aligned}
 \text{Volumetric strain} &= \frac{\text{Increase in volume}}{\text{Original volume}} \\
 &= \frac{x(1 + \epsilon_1)y(1 + \epsilon_2)(1 + \epsilon_3)z - xyz}{xyz} \\
 &= (1 + \epsilon_1)(1 + \epsilon_2)(1 + \epsilon_3) - 1 \approx \epsilon_1 + \epsilon_2 + \epsilon_3 \quad \left[\text{Neglecting the products of } \epsilon \text{'s} \right]
 \end{aligned}$$



$$\epsilon_1 = \frac{\sigma_1}{E} - \mu \frac{\sigma_2}{E} - \mu \frac{\sigma_3}{E}$$

$$\epsilon_2 = \frac{\sigma_2}{E} - \mu \frac{\sigma_1}{E} - \mu \frac{\sigma_3}{E}$$

$$\epsilon_3 = \frac{\sigma_3}{E} - \mu \frac{\sigma_1}{E} - \mu \frac{\sigma_2}{E}$$

$$\text{Further Volumetric strain} = \epsilon_1 + \epsilon_2 + \epsilon_3$$

$$= \frac{(\sigma_1 + \sigma_2 + \sigma_3)}{E} - \frac{2\mu(\sigma_1 + \sigma_2 + \sigma_3)}{E}$$

$$= \frac{(\sigma_1 + \sigma_2 + \sigma_3)(1 - 2\mu)}{E}$$

hence the

$\text{Volumetric strain} = \frac{(\sigma_1 + \sigma_2 + \sigma_3)(1 - 2\mu)}{E}$

Strains on an oblique plane ---- (a) Linear strain

Thus the movement of M parallel to OM, which since the strains are small is practically coincident with MM', and this would be the summation of portions (1) and (2) respectively and is equal to

$$= (\epsilon_y a \sin \theta) \sin \theta + (\epsilon_x a \cos \theta + \gamma_{xy} a \sin \theta) \cos \theta$$

$$= a [\epsilon_y \sin \theta \cdot \sin \theta + \epsilon_x \cos \theta \cdot \cos \theta + \gamma_{xy} \sin \theta \cdot \cos \theta]$$

hence the strain along OM

$$= \frac{\text{extension}}{\text{original length}}$$

$$\epsilon_\theta = \epsilon_x \cos^2 \theta + \gamma_{xy} \sin \theta \cdot \cos \theta + \epsilon_y \sin^2 \theta$$

$$\epsilon_\theta = \epsilon_x \cos^2 \theta + \epsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cdot \cos \theta$$

$$\text{Recalling } \cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\text{or } 2 \cos^2 \theta - 1 = \cos 2\theta$$

$$\cos^2 \theta = \left[\frac{1 + \cos 2\theta}{2} \right]$$

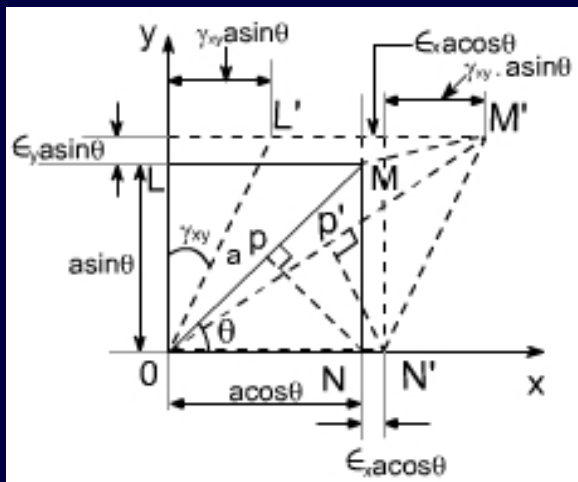
$$\sin^2 \theta = \left[\frac{1 - \sin 2\theta}{2} \right]$$

hence

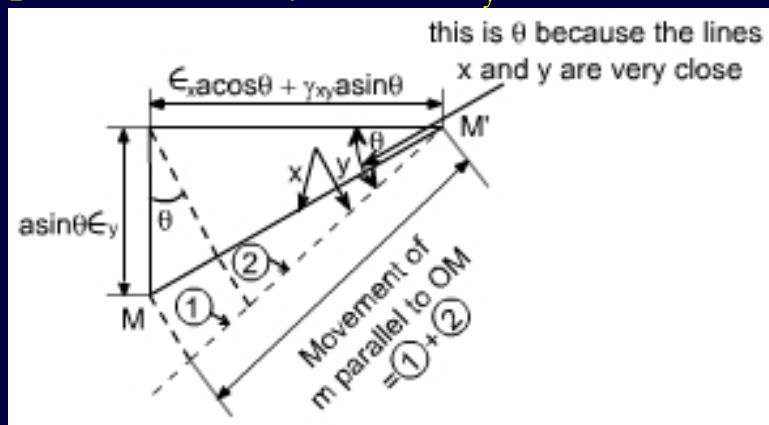
$$\epsilon_\theta = \epsilon_x \left[\frac{1 + \cos 2\theta}{2} \right] + \epsilon_y \left[\frac{1 - \sin 2\theta}{2} \right] + \gamma_{xy} a \sin \theta \cdot \cos \theta$$

$$= \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta + \frac{1}{2} \gamma_{xy} \sin 2\theta$$

$$\epsilon_\theta = \left\{ \frac{\epsilon_x + \epsilon_y}{2} \right\} + \left\{ \frac{\epsilon_x - \epsilon_y}{2} \right\} \cos 2\theta + \frac{1}{2} \gamma_{xy} \sin 2\theta$$



If M moves to M', then the movement of M parallel to x axis is $\epsilon_x a \cos \theta + \gamma_{xy} a \sin \theta$ and the movement parallel to the y axis is $\epsilon_y a \sin \theta$



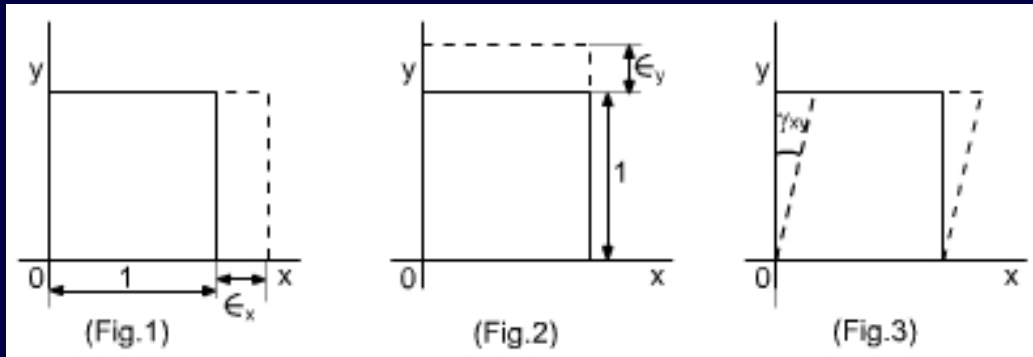
Shear strain: shear strain in the direction OM = displacement of point P at the foot of the perpendicular from N to OM; and the following expression can be derived as

$$\frac{1}{2} \gamma_{\theta} = - \left[\frac{1}{2} (\epsilon_x - \epsilon_y) \sin 2\theta - \frac{1}{2} \gamma_{xy} \cos 2\theta \right]$$

In the above expression $\frac{1}{2}$ is there so as to keep the consistency with the stress relations. Further -ve sign in the expression occurs so as to keep the consistency of sign convention, because OM' moves clockwise with respect to OM it is considered to be negative strain.

The other relevant expressions are the following :

Plane Strain: In xy plane three strain components may exist as can be seen from the following figures:



Principal planes :

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y}$$

Principal strains :

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2} \right)^2 + \left(\frac{\gamma_{xy}}{2} \right)^2}$$

Maximum shear strains :

$$\frac{\gamma_{\max}}{2} = \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2} \right)^2 + \left(\frac{\gamma_{xy}}{2} \right)^2}$$

Plane strain condition is defined only by the components , ϵ_y , γ_{xy} : $\epsilon_z = 0$; $\gamma_{xz} = 0$; $\gamma_{yz} = 0$

Plane stress is not the stress system associated with plane strain. The plane strain condition is associated with 3-D stress system and plane stress is associated with 3-D strain system.

Principal Strain and Mohr's Circle for Strain

$$\epsilon_{\theta} = \left\{ \frac{\epsilon_x + \epsilon_y}{2} \right\} + \left\{ \frac{\epsilon_x - \epsilon_y}{2} \right\} \cos 2\theta + \frac{1}{2} \gamma_{xy} \sin 2\theta \quad (1)$$

$$\frac{1}{2} \gamma_{\theta} = - \left[\frac{1}{2} (\epsilon_x - \epsilon_y) \sin 2\theta - \frac{1}{2} \gamma_{xy} \cos 2\theta \right] \quad (2)$$

Re writing the equation (1) as below :

$$\left[\epsilon_{\theta} - \left(\frac{\epsilon_x + \epsilon_y}{2} \right) \right] = \left\{ \frac{\epsilon_x - \epsilon_y}{2} \right\} \cos 2\theta + \frac{1}{2} \gamma_{xy} \sin 2\theta \quad (3)$$

squaring and adding equations (2) and (3)

$$\left[\epsilon_{\theta} - \left(\frac{\epsilon_x + \epsilon_y}{2} \right) \right]^2 + \left\{ \frac{1}{2} \gamma_{\theta} \right\}^2 = \left[\left\{ \frac{\epsilon_x - \epsilon_y}{2} \right\} \cos 2\theta + \frac{1}{2} \gamma_{xy} \sin 2\theta \right]^2 + \left[\frac{1}{2} (\epsilon_x - \epsilon_y) \sin 2\theta - \frac{1}{2} \gamma_{xy} \cos 2\theta \right]^2$$

$$\left[\epsilon_{\theta} - \left(\frac{\epsilon_x + \epsilon_y}{2} \right) \right]^2 + \left\{ \frac{1}{2} \gamma_{\theta} \right\}^2 = \left(\frac{\epsilon_x + \epsilon_y}{2} \right)^2 + \frac{\gamma_{xy}^2}{4}$$

Now as we know that

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2} \right)^2 + \left(\frac{\gamma_{xy}}{2} \right)^2}$$

$$\epsilon_1 + \epsilon_2 = \epsilon_x + \epsilon_y$$

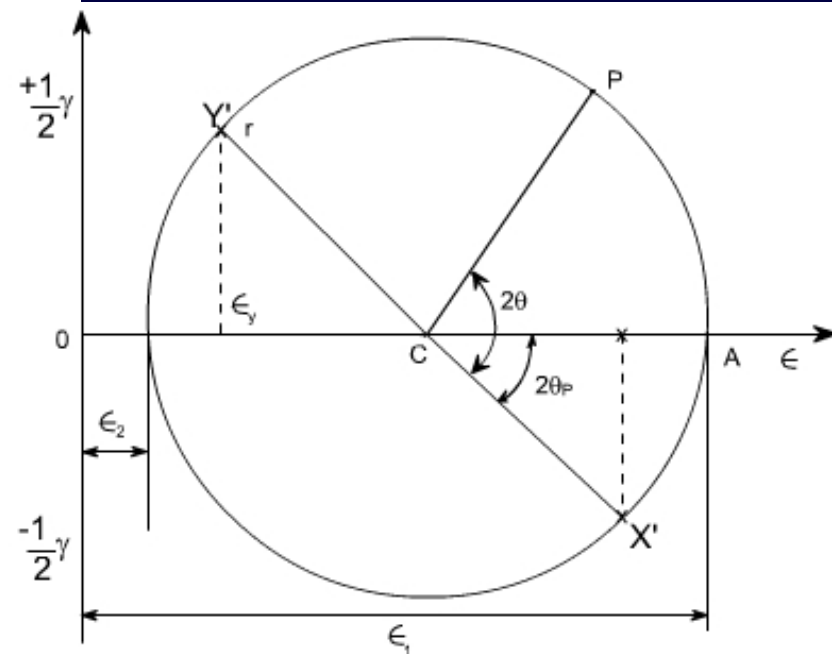
$$\left(\frac{\epsilon_1 - \epsilon_2}{2} \right)^2 = \left(\frac{\epsilon_x - \epsilon_y}{2} \right)^2 + \frac{\gamma_{xy}^2}{4}$$

Therefore the equation gets transformed to

$$\left[\epsilon_{\theta} - \left(\frac{\epsilon_1 + \epsilon_2}{2} \right) \right]^2 + \left[\frac{\gamma_{\theta}}{2} \right]^2 = \left(\frac{\epsilon_1 - \epsilon_2}{2} \right)^2 \quad (4)$$

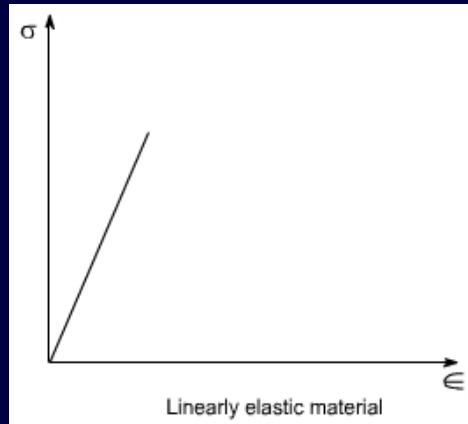
If we plot equation (4) we obtain a circle of radius $\left(\frac{\epsilon_1 - \epsilon_2}{2} \right)$ with center at $\left(\frac{\epsilon_1 + \epsilon_2}{2}, 0 \right)$

For the strains on an oblique plane we have two equations which are identical in form with the equation defining the direct stress on any inclined plane θ .

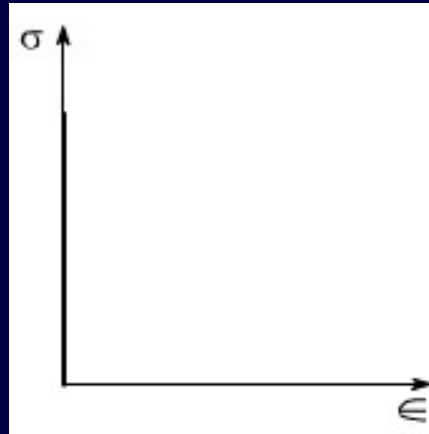


LECTURE 7: STRESS - STRAIN RELATIONS

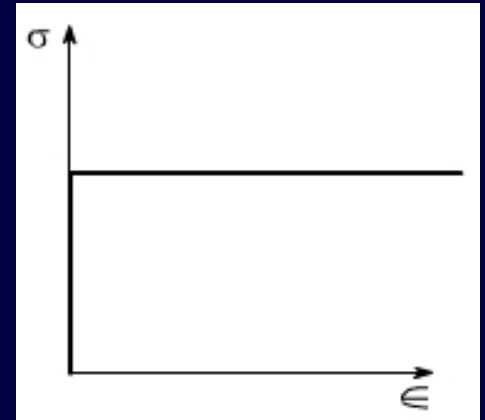
(i) Linear elastic material



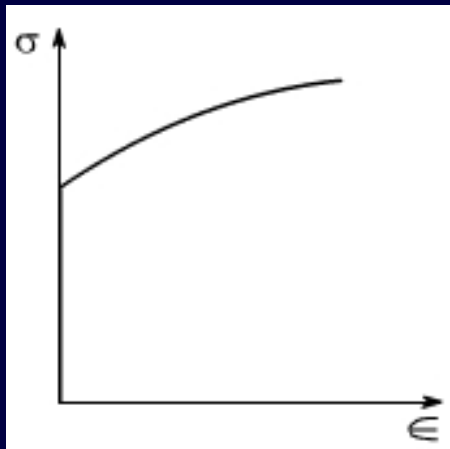
(ii) Rigid Materials



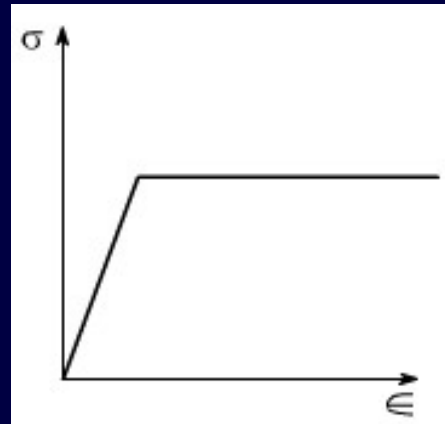
(iii) Perfectly plastic (non-strain hardening):



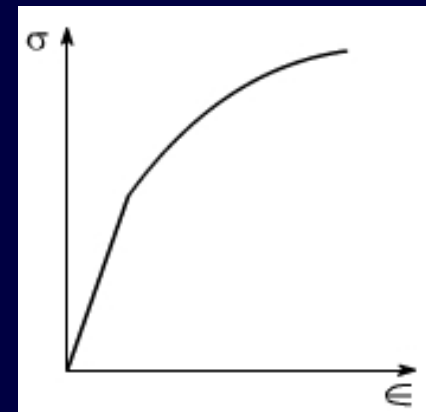
(iv) Rigid Plastic material (strain hardening)



(v) Elastic Perfectly Plastic material



(vi) Elastic Plastic material

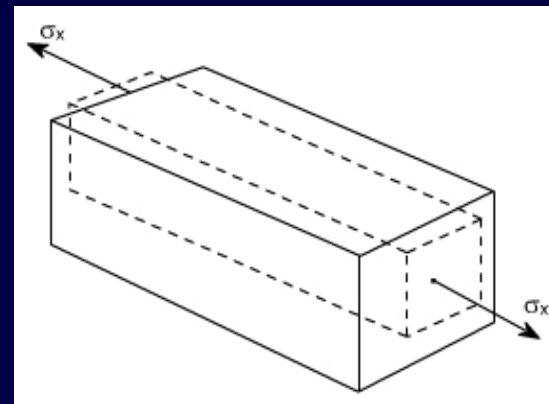
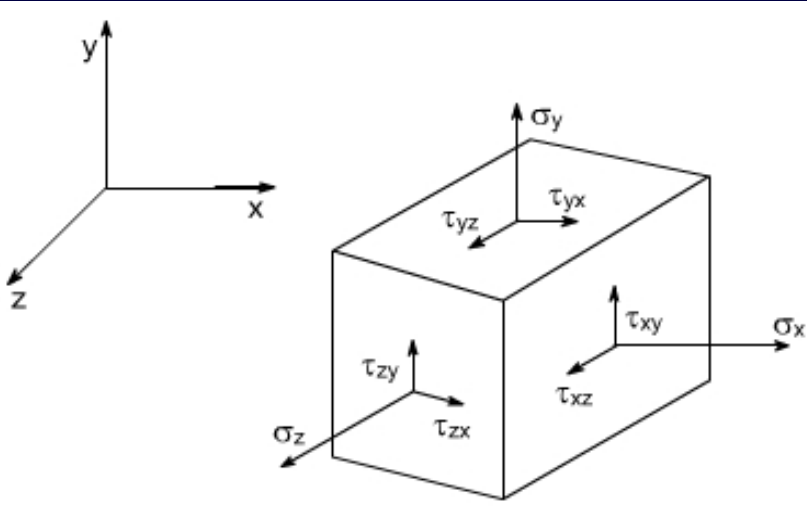


ISOTROPIC: If the response of the material is independent of the orientation of the load axis of the sample, then we say that the material is isotropic or in other words we can say that isotropy of a material in a characteristics, which gives us the information that the properties are the same in the three orthogonal directions x y z , on the other hand if the response is dependent on orientation it is known as an-isotropic.

Examples of an-isotropic materials, whose properties are different in different directions are

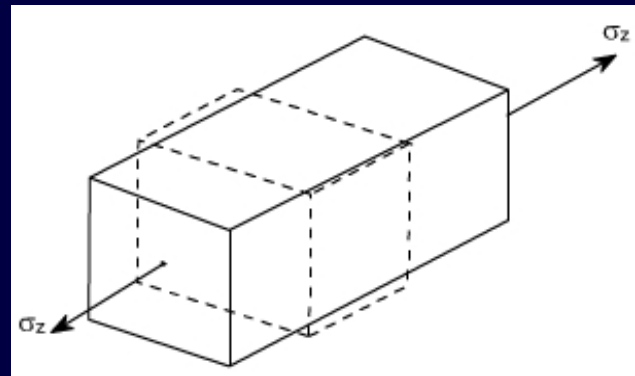
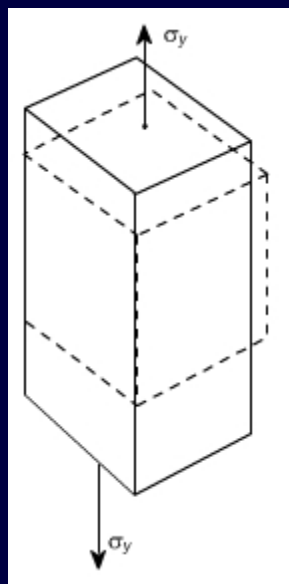
- ❖ Wood
- ❖ Fibre reinforced plastic
- ❖ Reinforced concrete

HOMOGENIUS: A material is homogenous if it has the same composition throughout body. Hence the elastic properties are the same at every point in the body. However, the properties need not to be the same in all the direction for the material to be homogenous. Isotropic materials have the same elastic properties in all the directions. Therefore, the material must be both homogenous and isotropic in order to have the lateral strains to be same at every point in a particular component.



$$\epsilon_x = \frac{\sigma_x}{E}, \epsilon_y = -\mu \epsilon_x; \epsilon_z = -\mu \epsilon_x$$

$$\epsilon_x = \frac{\sigma_x}{E}; \epsilon_y = -\mu \frac{\sigma_x}{E}; \epsilon_z = -\mu \frac{\sigma_x}{E}$$



$$\epsilon_z = \frac{\sigma_z}{E}, \epsilon_y = -\mu \epsilon_z; \epsilon_x = -\mu \epsilon_z$$

$$\epsilon_z = \frac{\sigma_z}{E}; \epsilon_y = -\mu \frac{\sigma_z}{E}; \epsilon_x = -\mu \frac{\sigma_z}{E}$$

Thus the total strain in any one direction is

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\mu}{E}(\sigma_y + \sigma_z) \quad (1)$$

In a similar manner, the total strain in the y and z directions become

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\mu}{E}(\sigma_x + \sigma_z) \quad (2)$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\mu}{E}(\sigma_x + \sigma_y) \quad (3)$$

$$\epsilon_y = \frac{\sigma_y}{E}, \epsilon_x = -\mu \epsilon_y; \epsilon_z = -\mu \epsilon_y$$

$$\epsilon_y = \frac{\sigma_y}{E}; \epsilon_x = -\mu \frac{\sigma_y}{E}; \epsilon_z = -\mu \frac{\sigma_y}{E}$$

In the following analysis shear stresses were not considered. It can be shown that for an isotropic material's a shear stress will produce only its corresponding shear strain and will not influence the axial strain. Thus, we can write Hook's law for the individual shear strains and shear stresses in the following manner.

$$\gamma_{xy} = \frac{\tau_{xy}}{G} \dots\dots\dots(4)$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G} \dots\dots\dots(5)$$

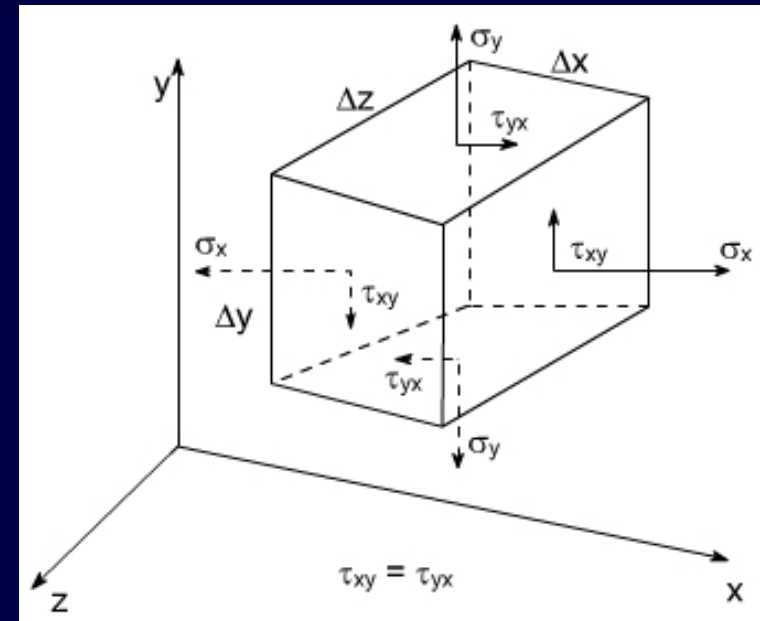
$$\gamma_{zx} = \frac{\tau_{zx}}{G} \dots\dots\dots(6)$$

$$\begin{aligned} \epsilon_x &= \frac{\sigma_x}{E} - \frac{\mu\sigma_y}{E} \\ \epsilon_y &= \frac{\sigma_y}{E} - \frac{\mu\sigma_x}{E} \\ \epsilon_z &= -\mu\frac{\sigma_x}{E} - \frac{\mu\sigma_y}{E} \text{ and } \tau_{xy} = \frac{\gamma_{xy}}{G} \end{aligned}$$

Their inverse relations can be also determined and are given as

$$\begin{aligned} \sigma_x &= \frac{E}{(1-\mu^2)}(\epsilon_x + \mu\epsilon_y) \\ \sigma_y &= \frac{E}{(1-\mu^2)}(\epsilon_y + \mu\epsilon_x) \\ \tau_{xy} &= G\gamma_{xy} \end{aligned}$$

PLANE STRESS



$$\sigma_z = \tau_{zx} = \tau_{zy} = 0$$

The Equations (1) through (6) are known as Generalized Hook's law and are the constitutive equations for the linear elastic isotropic materials. When these equations are used as written, the strains can be completely determined from known values of the stresses. To engineers the plane stress situation is of much relevance (i.e. $\sigma_z = \tau_{xz} = \tau_{yz} = 0$), Thus then the above set of equations reduces to

RELATION AMONG ELASTIC CONSTANTS

E = Young's Modulus of Rigidity = Stress / strain

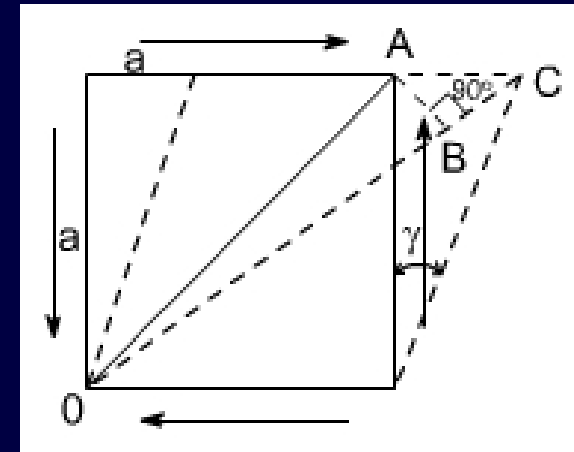
G = Shear Modulus or Modulus of rigidity = Shear stress / Shear strain

μ = Poisson's ratio = - lateral strain / longitudinal strain

K = Bulk Modulus of elasticity = Volumetric stress / Volumetric strain

where Volumetric strain = sum of linear strain in x, y and z direction.

Volumetric stress = stress which cause the change in volume.



Relation between E , G and μ :

Consider a cube of material of side 'a' subjected to the action of the shear and complementary shear stresses as shown in the figure and producing the strained shape as shown in the figure below.

Assuming that the strains are small and the angle ACB may be taken as 45° .

$$\begin{aligned} \text{strain on diagonal} &= \frac{\sigma_1}{E} - \mu \frac{\sigma_2}{E} \\ &= \frac{\tau}{E} - \mu \frac{(-\tau)}{E} \\ &= \frac{\tau}{E} (1 + \mu) \end{aligned}$$

equating the two strains one may get

$$\frac{\tau}{2G} = \frac{\tau}{E} (1 + \mu)$$

$$\text{or } \boxed{E = 2G(1 + \mu)}$$

$$\begin{aligned} \text{Thus, strain on diagonal OA} &= \frac{BC}{OA} \\ &= \frac{AC \cos 45^\circ}{OA} \end{aligned}$$

$$OA = \frac{a}{\sin 45^\circ} = a\sqrt{2}$$

hence

$$\begin{aligned} \text{strain} &= \frac{AC}{a\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{AC}{2a} \end{aligned}$$

but $AC = a\gamma$

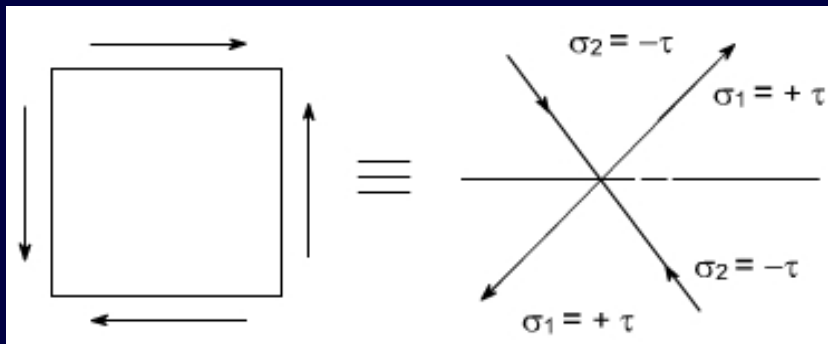
where γ = shear strain

$$\text{Thus, the strain on diagonal} = \frac{a\gamma}{2a} = \frac{\gamma}{2}$$

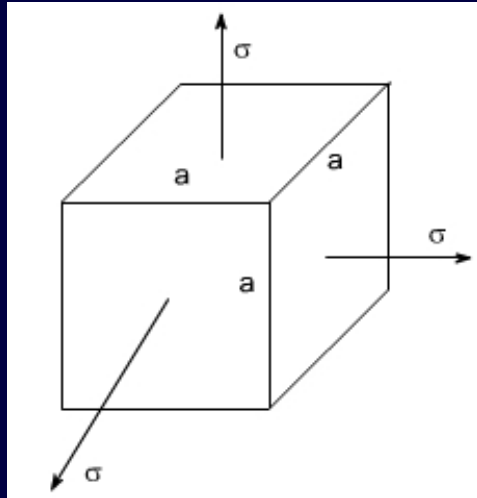
From the definition

$$G = \frac{\tau}{\gamma} \text{ or } \gamma = \frac{\tau}{G}$$

$$\text{thus, the strain on diagonal} = \frac{\gamma}{2} = \frac{\tau}{2G}$$



Relation between E, K and μ : Consider a cube subjected to three equal stresses s as shown in the figure below



The total strain in one direction or along one edge due to the application of hydrostatic stress or volumetric stress s is given as

$$= \frac{\sigma}{E} - \gamma \frac{\sigma}{E} - \gamma \frac{\sigma}{E}$$

$$= \frac{\sigma}{E} (1 - 2\gamma)$$

volumetre strain = 3.linear strain

$$\text{volumetre strain} = \epsilon_x + \epsilon_y + \epsilon_z$$

or thus, $\epsilon_x = \epsilon_y = \epsilon_z$

$$\text{volumetric strain} = 3 \frac{\sigma}{E} (1 - 2\gamma)$$

By definition

$$\text{Bulk Modulus of Elasticity (K)} = \frac{\text{Volumetric stress}(\sigma)}{\text{Volumetric strain}}$$

or

$$\text{Volumetric strain} = \frac{\sigma}{k}$$

Equating the two strains we get

$$\frac{\sigma}{k} = 3 \frac{\sigma}{E} (1 - 2\gamma)$$

$$\boxed{E = 3K(1 - 2\gamma)}$$

Relation between E, K and G

$$E = 2G(1 + \mu) \text{ and } E = 3K(1 - 2\mu)$$

$$\therefore E = \frac{9GK}{(3K + G)}$$

It may be noted that under condition of simple tension and simple shear, all real materials tend to experience displacements in the directions of the applied forces and Under hydrostatic loading they tend to increase in volume. In other words the value of the elastic constants E, G and K cannot be negative

$$E = 2G(1 + \mu) \text{ and } E = 3K(1 - 2\mu)$$

$$\text{Yields } -1 \leq \mu \leq 0.5$$

In actual practice no real material has value of Poisson's ratio negative . Thus, the value of μ cannot be greater than 0.5, if however $\mu > 0.5$ than $\epsilon_v = -ve$, which is physically unlikely because when the material is stretched its volume would always increase.

LECTURE 21: Members Subjected to Flexural Loads

In many engineering structures members are required to resist forces that are applied laterally or transversely to their axes. These type of members are termed as beam.

There are various ways to define the beams such as

Definition I: A beam is a laterally loaded member, whose cross-sectional dimensions are small as compared to its length.

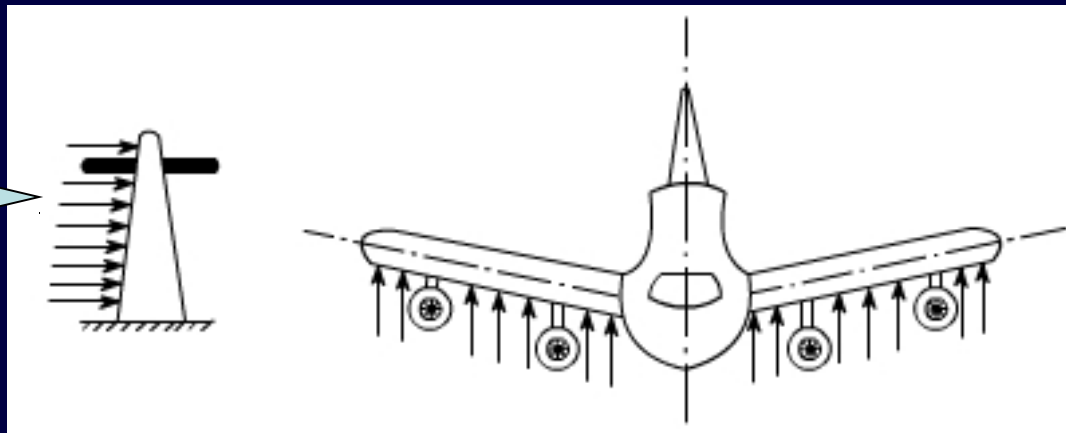
Definition II: A beam is nothing simply a bar which is subjected to forces or couples that lie in a plane containing the longitudinal axis of the bar. The forces are understood to act perpendicular to the longitudinal axis of the bar.

Definition III: A bar working under bending is generally termed as a beam.

Materials for Beam: The beams may be made from several usable engineering materials such commonly among them are Metal, Wood, Concrete, Plastic

Examples of Beams: Refer to the figures shown below that illustrates the beam

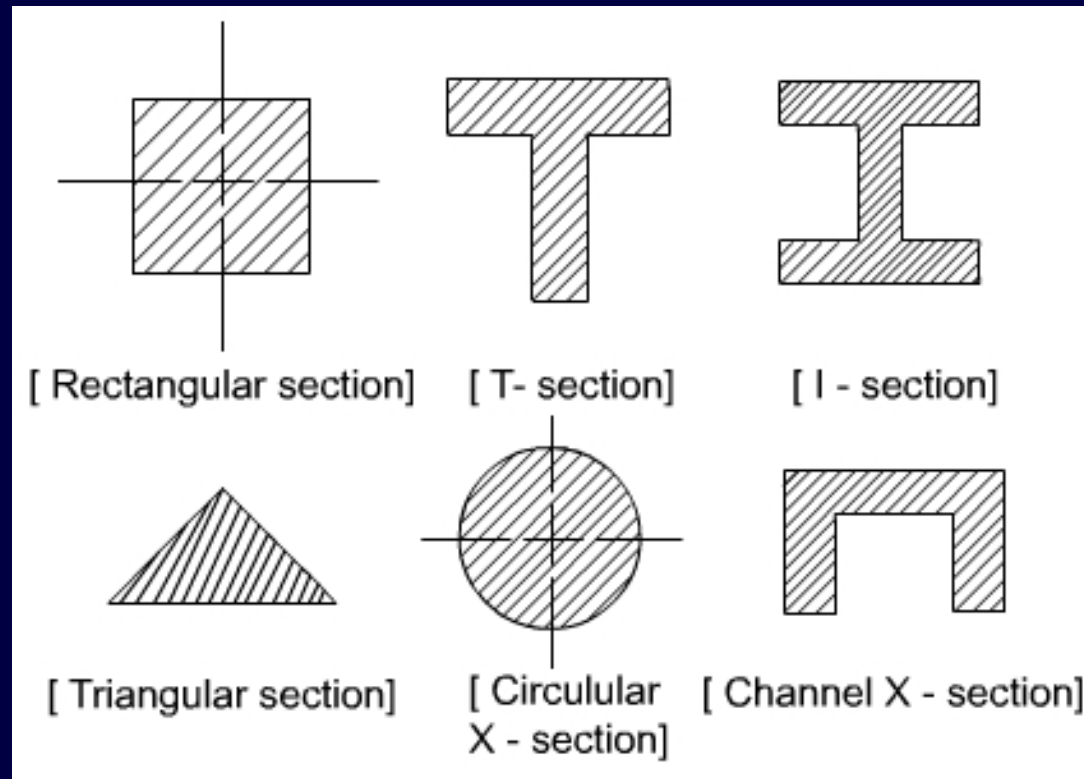
an electric pole



an aeroplane



Geometric forms of Beams: The Area of X-section of the beam may take several forms some of them have been shown below



Issues Regarding Beam: Designer would be interested to know the answers to following issues while dealing with beams in practical engineering application

- ❖ At what load will it fail
- ❖ How much deflection occurs under the application of loads.

Classification of Beams: Beams are classified on the basis of their geometry and the manner in which they are supported.

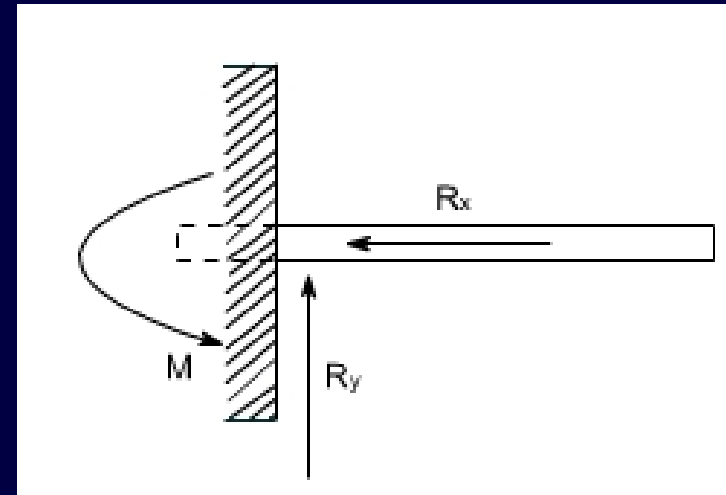
Classification I: The classification based on the basis of geometry normally includes features such as the shape of the X-section and whether the beam is straight or curved.

Classification II: Beams are classified into several groups, depending primarily on the kind of supports used. But it must be clearly understood why do we need supports.

The supports are required to provide constraint to the movement of the beams or simply the supports resist the movements either in particular direction or in rotational direction or both. As a consequence of this, the reaction comes into picture whereas to resist rotational movements the moment comes into picture.

On the basis of the support, the beams may be classified as follows:

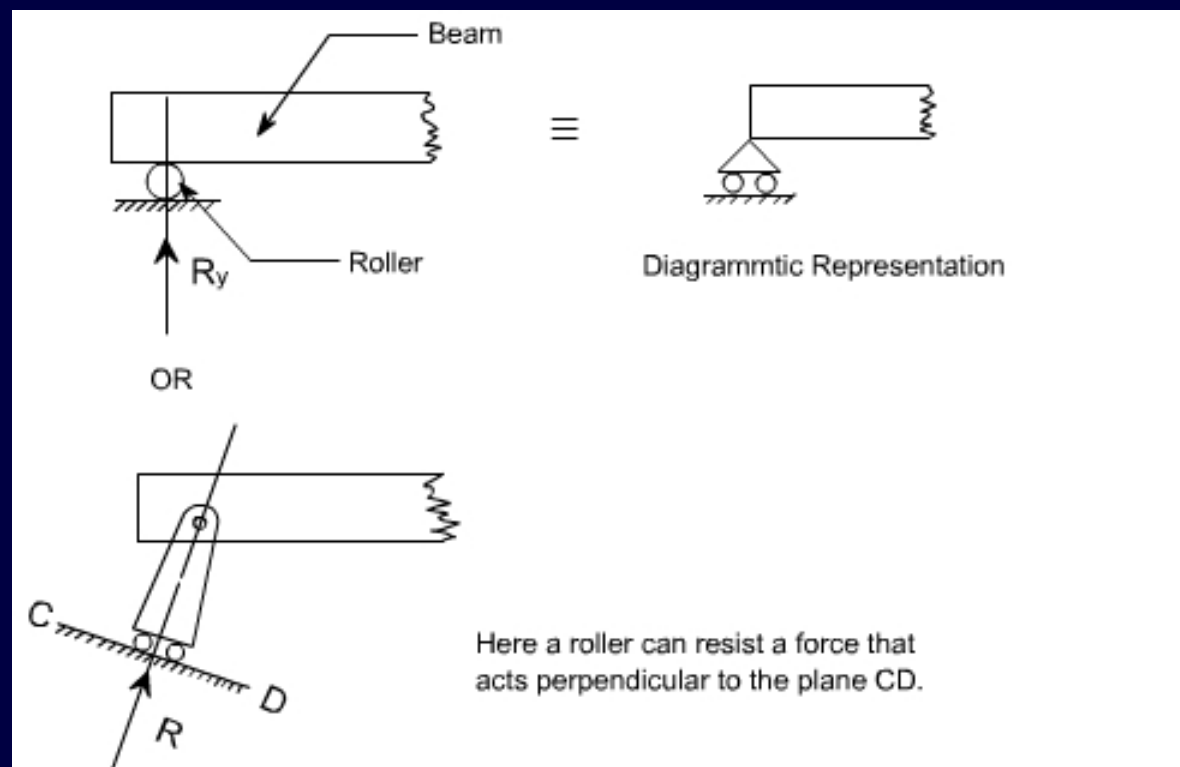
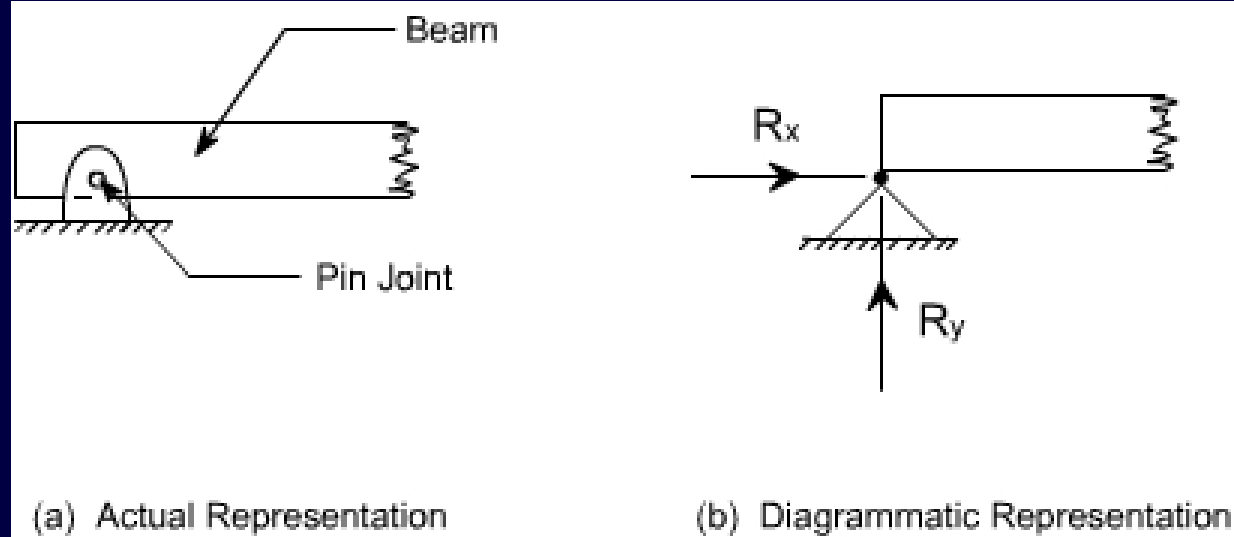
Cantilever Beam: A beam which is supported on the fixed support is termed as a cantilever beam: Now let us understand the meaning of a fixed support. Such a support is obtained by building a beam into a brick wall, casting it into concrete or welding the end of the beam. Such a support provides both the translational and rotational constraint to the beam, therefore the reaction as well as the moments appears.



Simply Supported Beam:

The beams are said to be simply supported if their supports creates only the translational constraints.

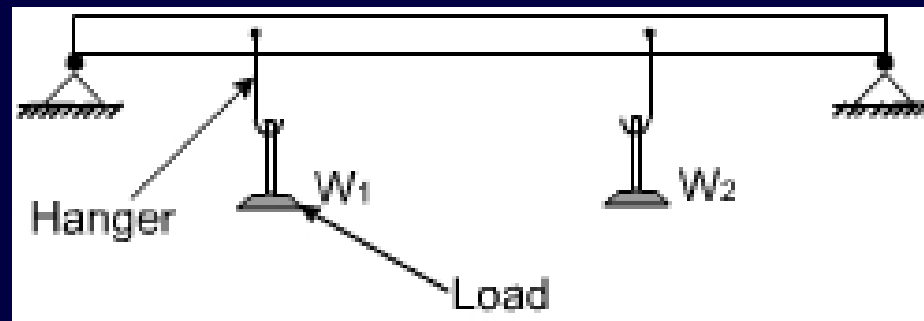
Some times the translational movement may be allowed in one direction with the help of rollers and can be represented like this



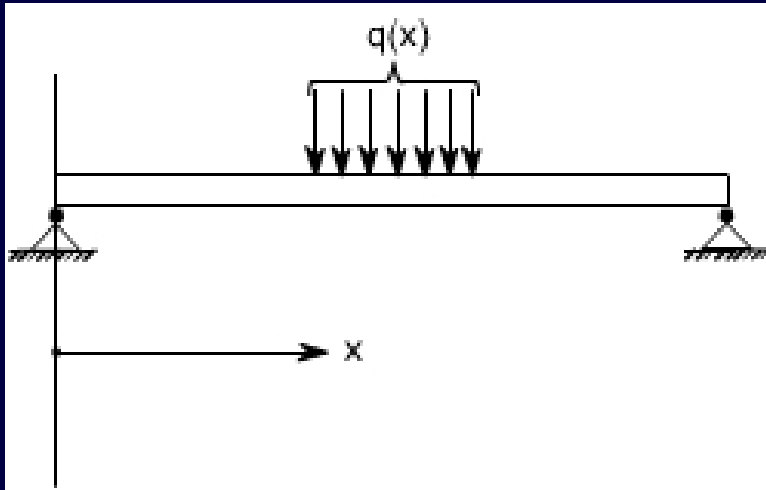
Statically Determinate or Statically Indeterminate Beams: If all the external forces and moments acting on it can be determined from the equilibrium conditions alone then. It would be referred as a statically determinate beam, whereas in the statically indeterminate beams one has to consider deformation i.e. deflections to solve the problem.

Types of loads acting on beams: A beam is normally horizontal where as the external loads acting on the beams is generally in the vertical directions. In order to study the behaviors of beams under flexural loads. It becomes pertinent that one must be familiar with the various types of loads acting on the beams as well as their physical manifestations.

A. Concentrated Load: It is a kind of load which is considered to act at a point. By this we mean that the length of beam over which the force acts is so small in comparison to its total length that one can model the force as though applied at a point in two dimensional view of beam. Here in this case, force or load may be made to act on a beam by a hanger or through other means

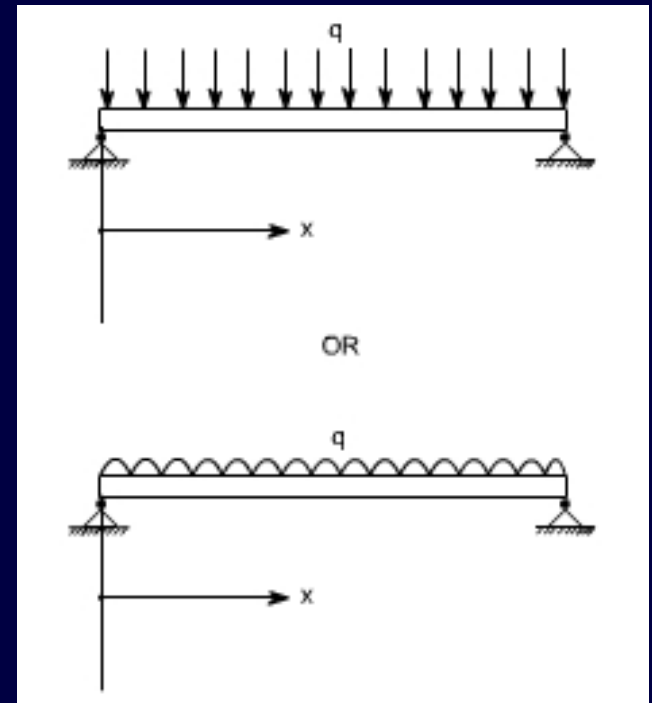
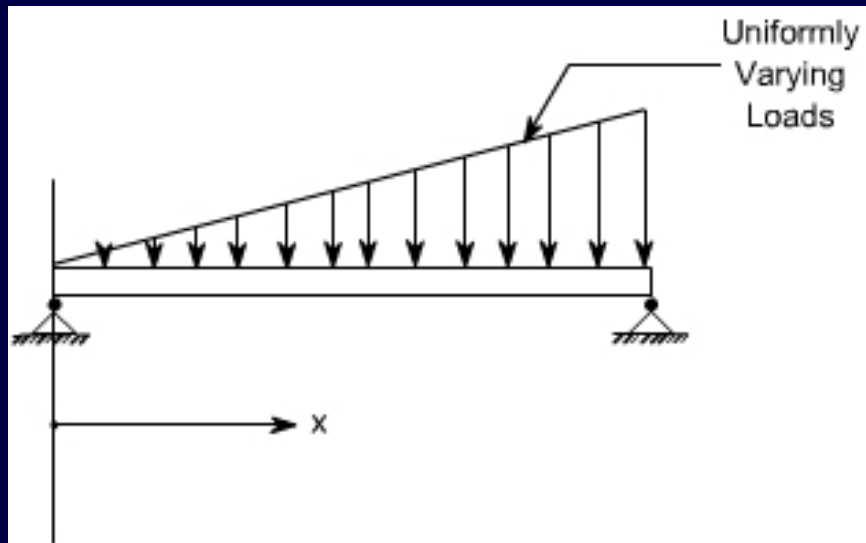


B. Distributed Load: The distributed load is a kind of load which is made to spread over a entire span of beam or over a particular portion of the beam in some specific manner

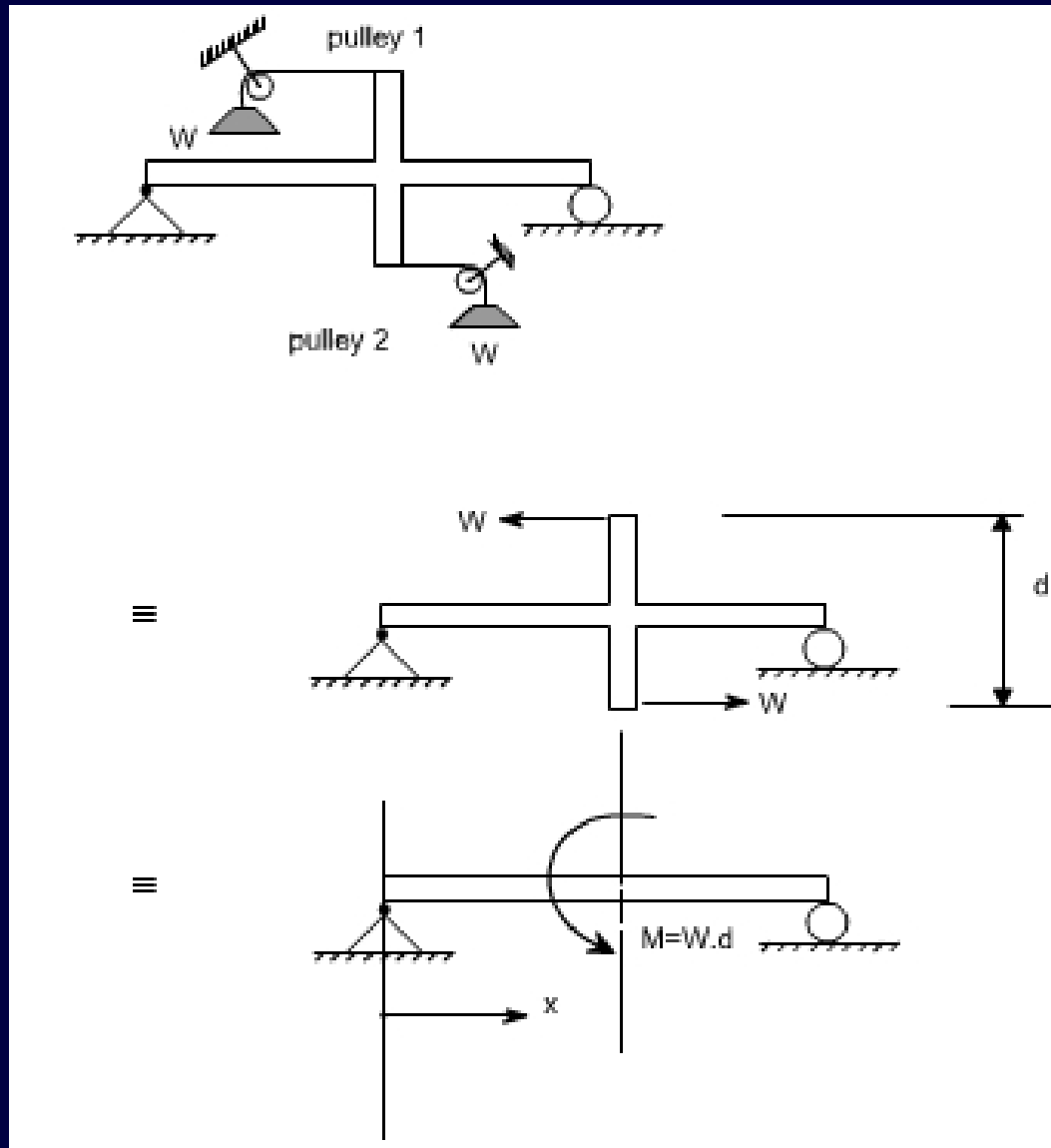


Here, the rate of loading ' q ' is a function of x i.e. span of the beam, hence this is a non uniformly distributed load.

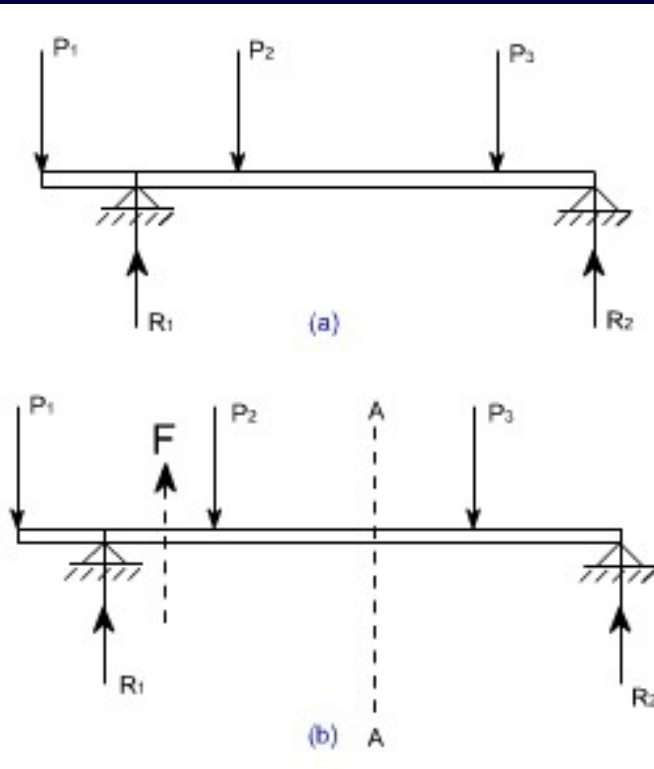
The rate of loading ' q ' over the length of the beam may be uniform over the entire span of beam, then we call this as a uniformly distributed load (U.D.L). The U.D.L may be represented in either of the way on the beams



Concentrated Moment: The beam may be subjected to a concentrated moment essentially at a point. One of the possible arrangements for applying the moment is being shown in the figure below:



LECTURE 22: Concept of Shear Force and Bending moment in beams



- ❖ Now let us consider the beam as shown in fig 1(a) which is supporting the loads P_1 , P_2 , P_3 and is simply supported at two points creating the reactions R_1 and R_2 respectively.
- ❖ Now let us assume that the beam is to be divided into or imagined to be cut into two portions at a section AA. Now let us assume that the resultant of loads and reactions to the left of AA is 'F' vertically upwards, and since the entire beam is to remain in equilibrium, thus the resultant of forces to the right of AA must also be F, acting downwards. This force 'F' is as a shear force.
- ❖ The shearing force at any x-section of a beam represents the tendency for the portion of the beam to one side of the section to slide or shear laterally relative to the other portion.

- At any x-section of a beam, the shear force 'F' is the algebraic sum of all the lateral components of the forces acting on either side of the x-section.
- **Sign Convention for Shear Force:** The usual sign conventions to be followed for the shear forces have been illustrated in figures 2 and 3.

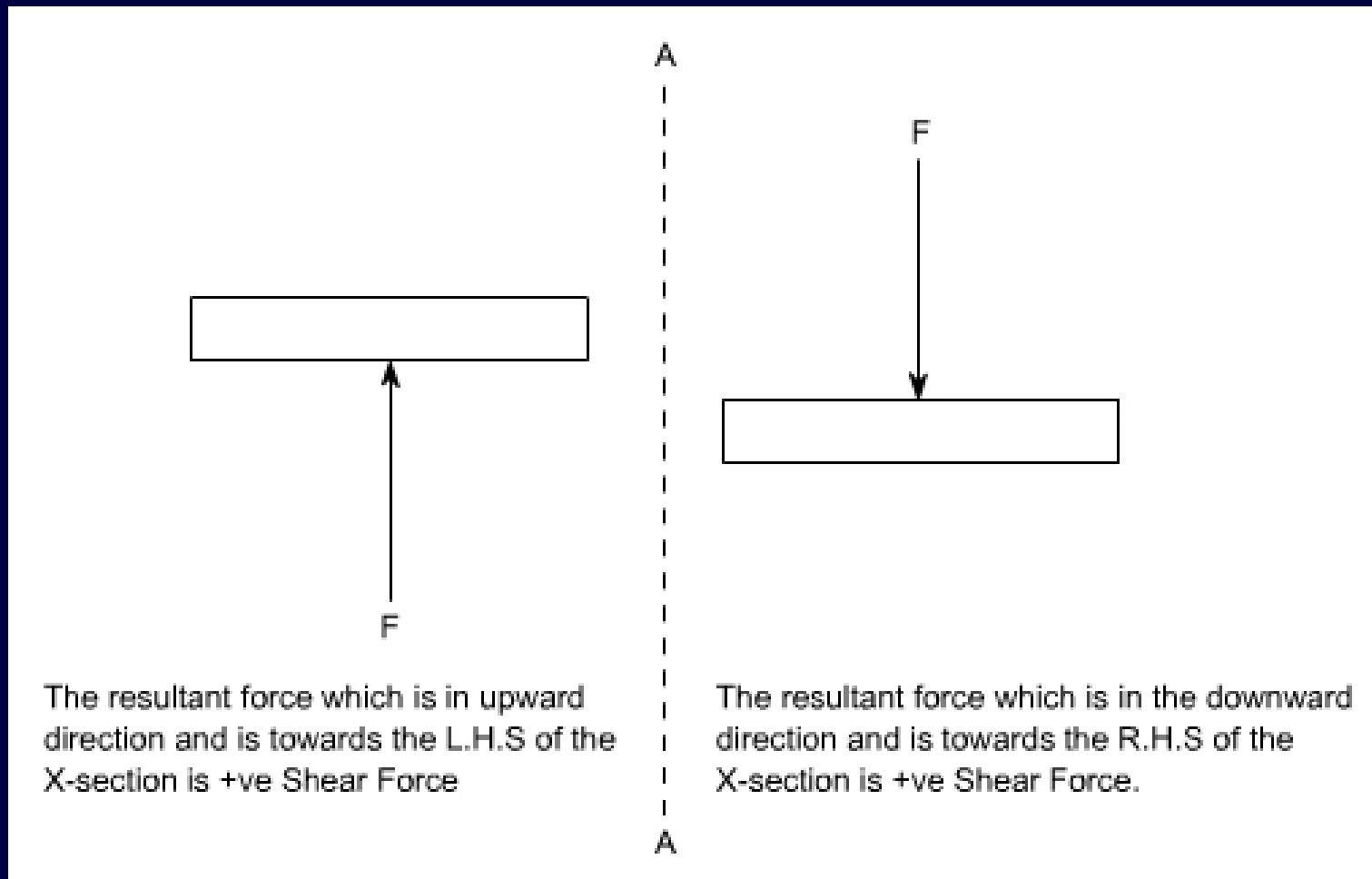


Fig 2: Positive Shear Force

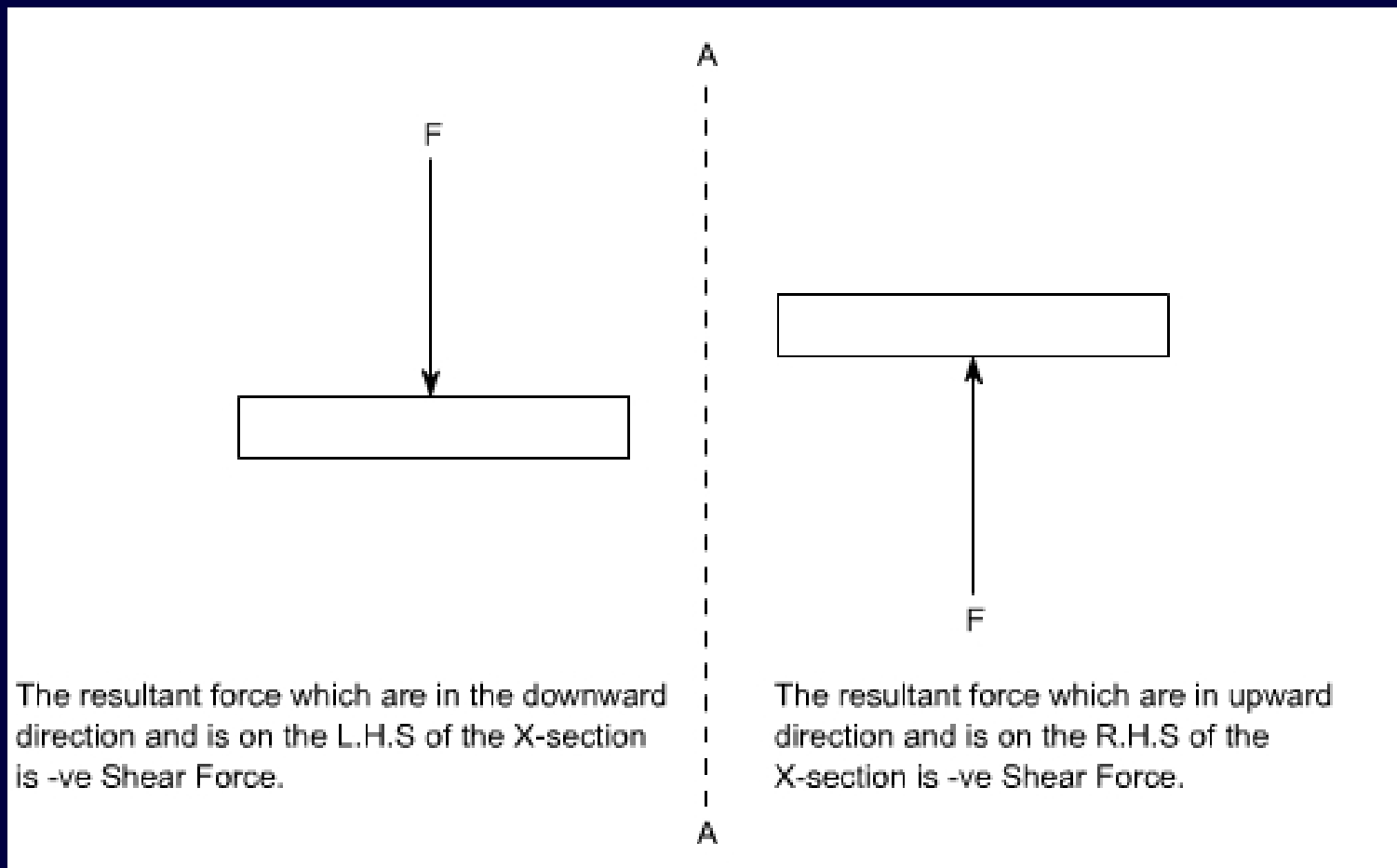
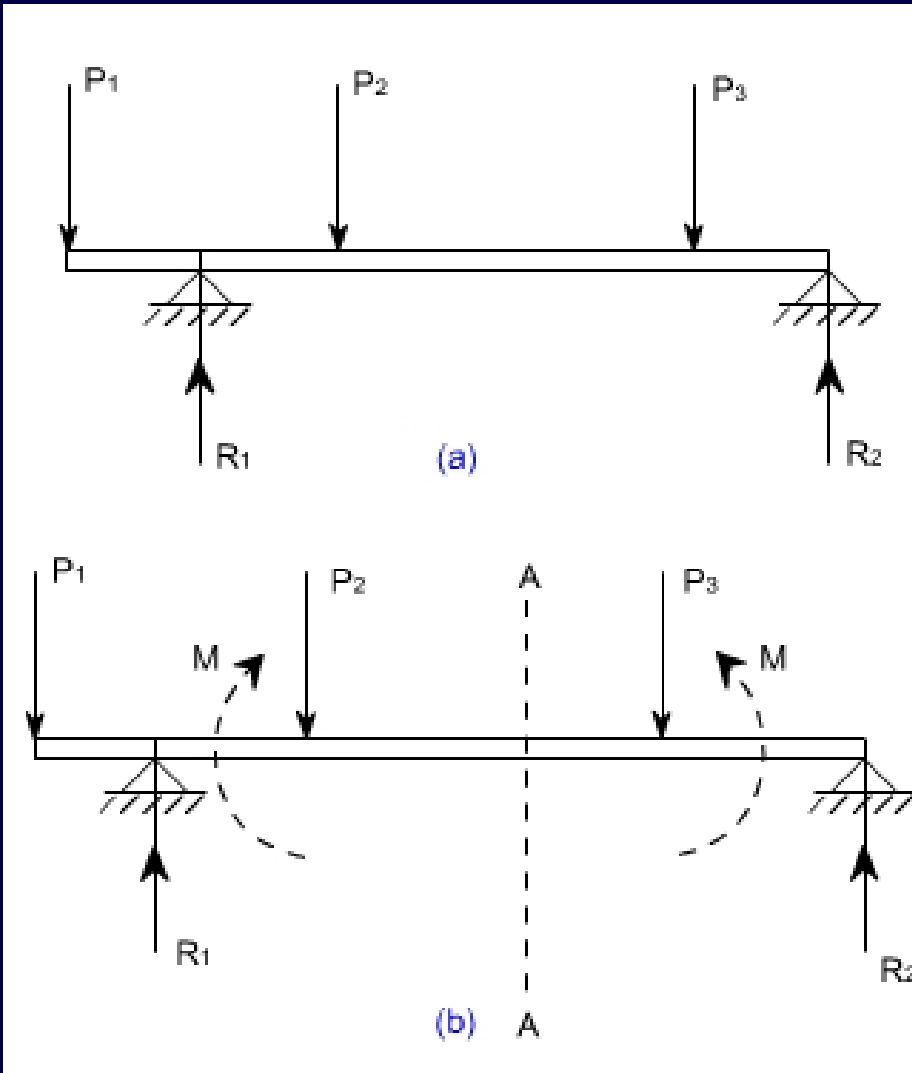


Fig 3: Negative Shear Force

Bending Moment



- ✓ Let us again consider the beam which is simply supported at the two prints, carrying loads P_1 , P_2 and P_3 and having the reactions R_1 and R_2 at the supports Fig 4.
- ✓ Now, let us imagine that the beam is cut into two potions at the x-section AA.
- ✓ Say that the resultant moment about the section AA of all the loads and reactions to the left of the x-section at AA is M in C.W direction, then moment of forces to the right of x-section AA must be ' M ' in C.C.W. Then ' M ' is called as the Bending moment and is abbreviated as B.M.
- ✓ The bending moment to be simply as the algebraic sum of the moments about an x-section of all the forces acting on either side of the section

Sign Conventions for the Bending Moment: For the bending moment, following sign conventions may be adopted as indicated in Fig 5 and Fig 6.

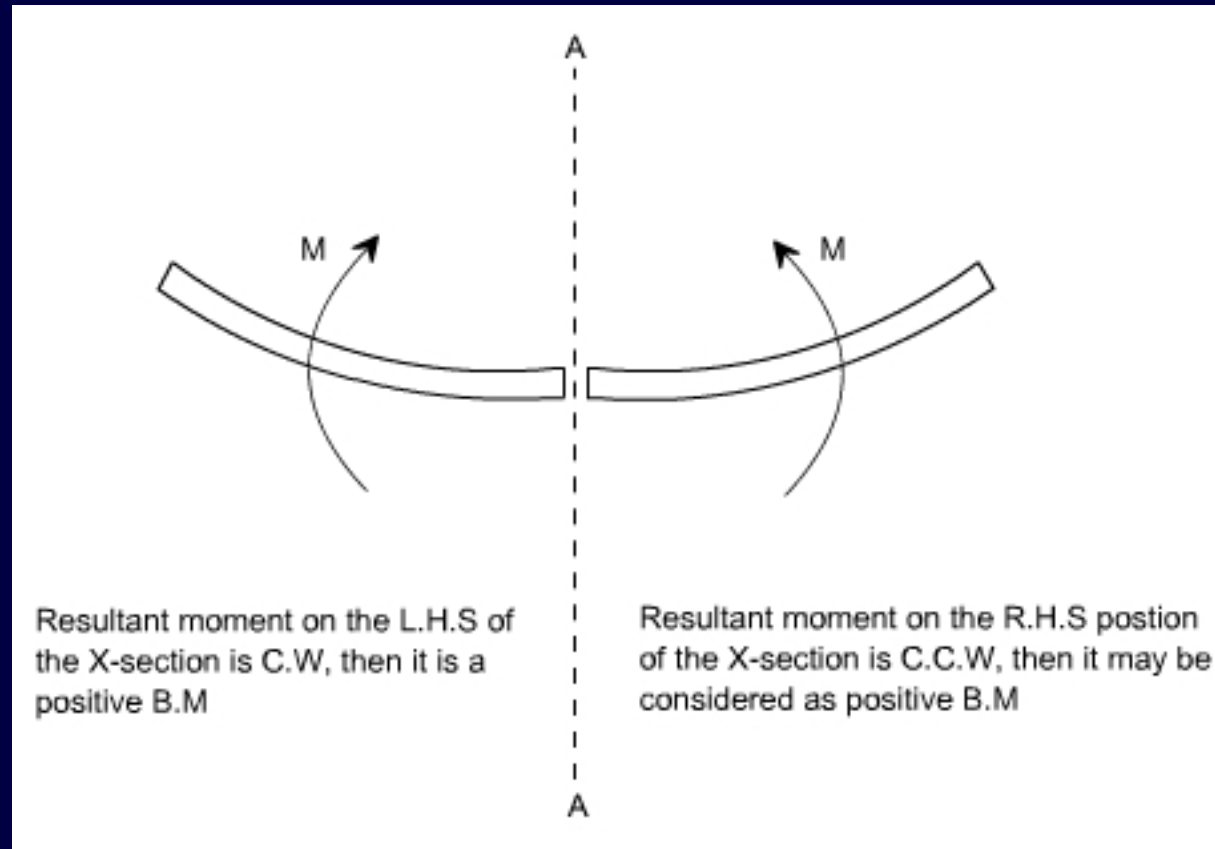


Fig 5: Positive Bending Moment

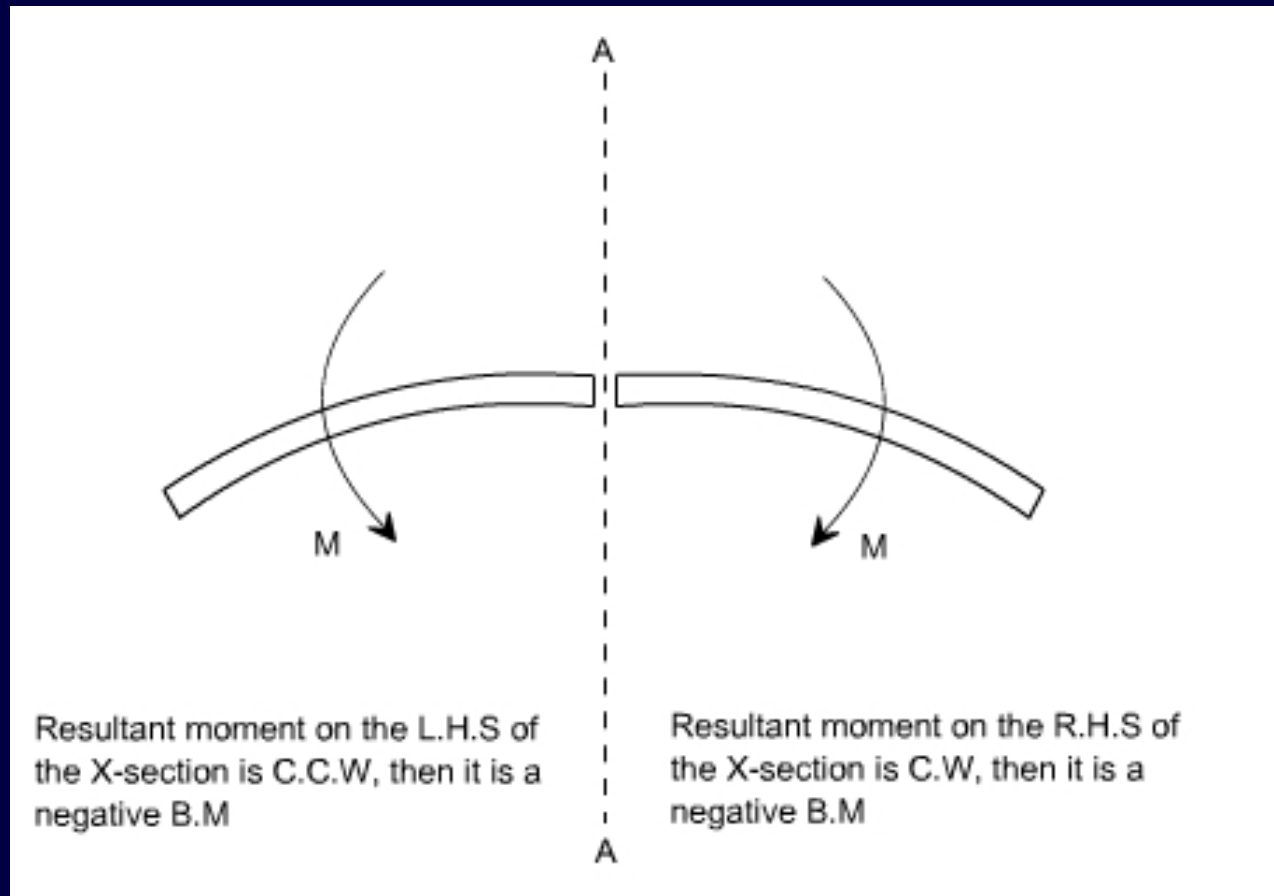


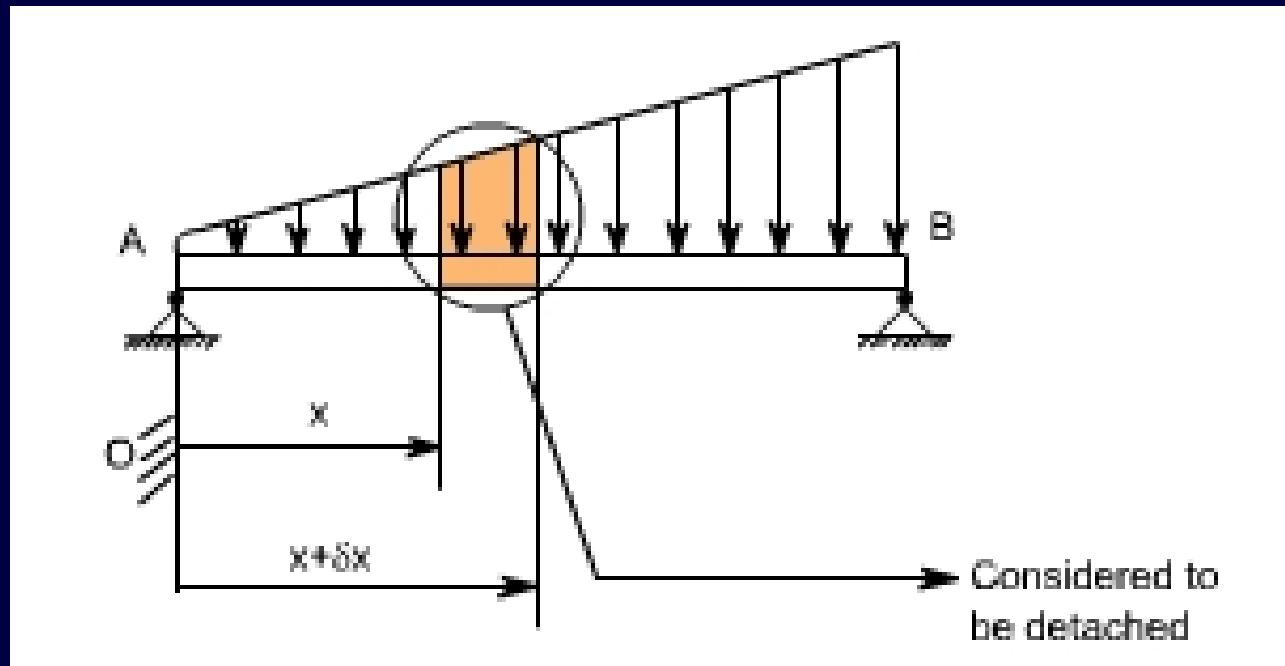
Fig 6: Negative Bending Moment

Some times, the terms 'Sagging' and Hogging are generally used for the positive and negative bending moments respectively.

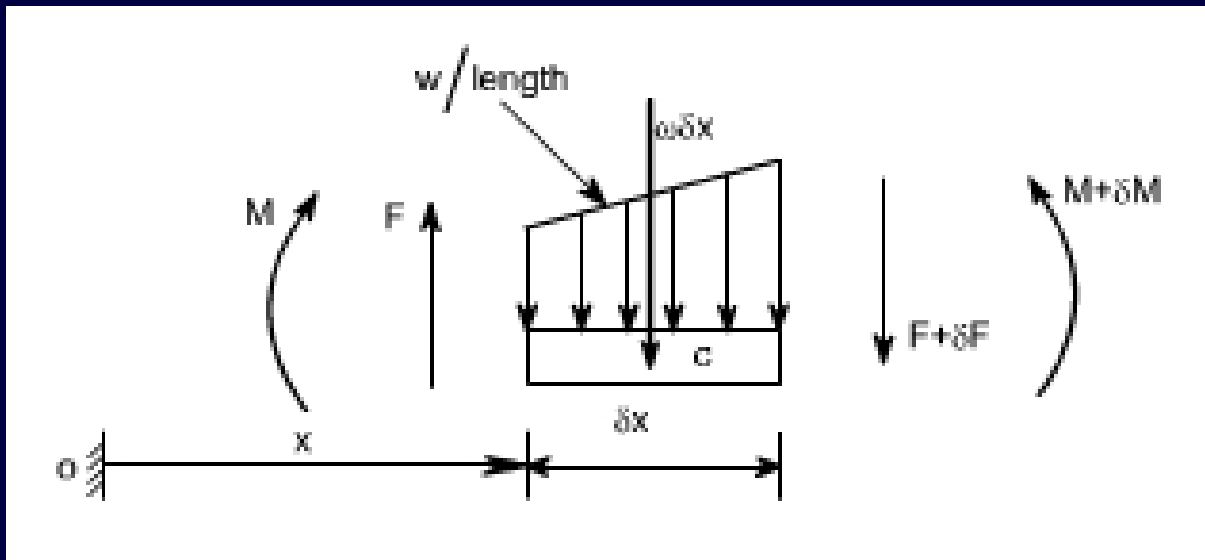
Bending Moment and Shear Force Diagrams: The diagrams which illustrate the variations in B.M and S.F values along the length of the beam for any fixed loading conditions would be helpful to analyze the beam further.

Basic Relationship Between The Rate of Loading, Shear Force and Bending Moment: The construction of the shear force diagram and bending moment diagrams is greatly simplified if the relationship among load, shear force and bending moment is established.

Let us consider a simply supported beam AB carrying a uniformly distributed load w/length . Let us imagine to cut a short slice of length dx cut out from this loaded beam at distance ' x ' from the origin ' O '.



Let us detach this portion of the beam and draw its free body diagram



The forces acting on the free body diagram of the detached portion of this loaded beam are the following

- ❖ The shearing force F and $F + dF$ at the section x and $x + dx$ respectively.
- ❖ The bending moment at the sections x and $x + dx$ be M and $M + dM$ respectively.
- ❖ Force due to external loading, if ' w ' is the mean rate of loading per unit length then the total loading on this slice of length dx is $w \cdot dx$, which is approximately acting through the center ' c '. If the loading is assumed to be uniformly distributed then it would pass exactly through the center ' c '.

This small element must be in equilibrium under the action of these forces and couples. Now let us take the moments at the point 'c'. Such that

$$M + F \cdot \frac{\delta x}{2} + (F + \delta F) \cdot \frac{\delta x}{2} = M + \delta M$$

$$\Rightarrow F \cdot \frac{\delta x}{2} + (F + \delta F) \cdot \frac{\delta x}{2} = \delta M$$

$$\Rightarrow F \cdot \frac{\delta x}{2} + F \cdot \frac{\delta x}{2} + \delta F \cdot \frac{\delta x}{2} = \delta M \quad [\text{Neglecting the product of } \delta F \text{ and } \delta x \text{ being small quantities}]$$

$$\Rightarrow F \cdot \delta x = \delta M$$

$$\Rightarrow F = \frac{\delta M}{\delta x}$$

Under the limits $\delta x \rightarrow 0$

$$\boxed{F = \frac{dM}{dx}} \quad \dots\dots\dots (1)$$

Re solving the forces vertically we get

$$w \cdot \delta x + (F + \delta F) = F$$

$$\Rightarrow w = - \frac{\delta F}{\delta x}$$

Under the limits $\delta x \rightarrow 0$

$$\Rightarrow w = - \frac{dF}{dx} \text{ or } - \frac{d}{dx} \left(\frac{dM}{dx} \right)$$

$$\boxed{w = - \frac{dF}{dx} = - \frac{d^2M}{dx^2}} \quad \dots\dots\dots (2)$$

Conclusions:

- From Equation (1), the area of the shear force diagram between any two points, from the basic calculus is the bending moment diagram
- The slope of bending moment diagram is the shear force
- Thus, if $F=0$; the slope of the bending moment diagram is zero and the bending moment is therefore constant.
- The maximum or minimum Bending moment occurs where shear forces are zero
- The slope of the shear force diagram is equal to the magnitude of the intensity of the distributed loading at any position along the beam. The +ve sign is as a consequence of our particular choice of sign conventions

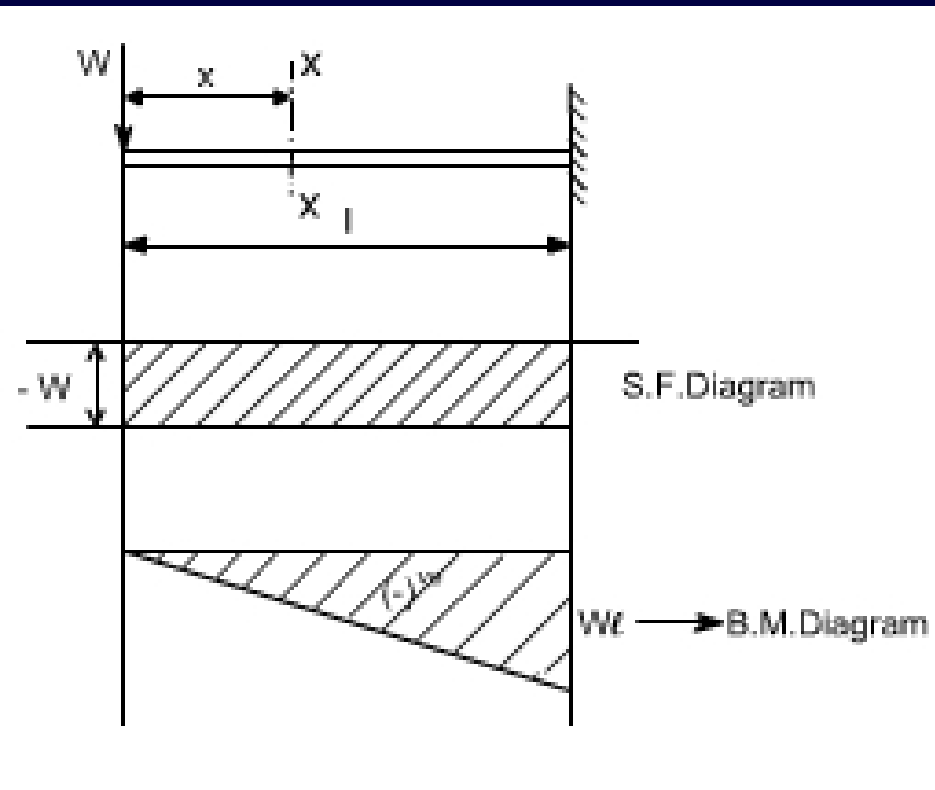
Procedure for drawing shear force and bending moment diagram

- ❑ A shear force diagram (SFD) can be constructed from the loading diagram of the beam. First the reactions must be determined always. Then the vertical components of forces and reactions are successively summed from the left end of the beam to preserve the mathematical sign conventions adopted. The shear at a section is simply equal to the sum of all the vertical forces to the left of the section.
- ❑ When the successive summation process is used, the SFD should end up with the previously calculated shear (reaction at right end of the beam). No shear force acts through the beam just beyond the last vertical force or reaction. If the SFD closes in this fashion, then it gives an important check on mathematical calculations.
- ❑ The bending moment diagram (BMD) is obtained by proceeding continuously along the length of beam from the left hand end and summing up the areas of shear force diagrams giving due regard to sign. The process of obtaining the BMD from the SFD by summation is exactly the same as that for drawing SFD from load diagram.
- ❑ It may also be observed that a constant shear force produces a uniform change in the bending moment, resulting in st. line in the moment diagram. If no shear force exists along a certain portion of a beam, then it indicates that there is no change in moment takes place. It may also further observe that $dM/dx = F$ therefore, from the fundamental theorem of calculus the maximum or minimum moment occurs where the shear is zero. In order to check the validity of the BMD, the terminal conditions for the moment must be satisfied. If the end is free or pinned, the computed sum must be equal to zero. If the end is built in, the moment computed by the summation must be equal to the one calculated initially for the reaction. These conditions must always be satisfied.

Illustrative problems

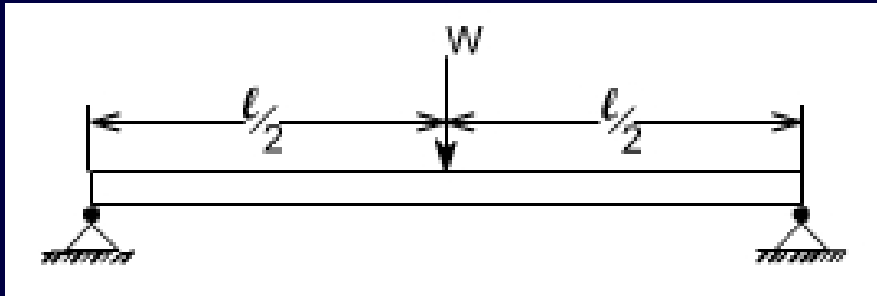
1. A cantilever of length carries a concentrated load 'W' at its free end: Draw SFD & BMD

Solution: At a section a distance x from free end consider the forces to the left, then $F = -W$ (for all values of x) -ve sign means the shear force to the left of the x -section are in downward direction and therefore negative



- ❖ Taking moments about the section gives (obviously to the left of the section)
- ❖ $M = -Wx$ (-ve sign means that the moment on the left hand side of the portion is in the anticlockwise direction and is therefore taken as ' -ve ' according to the sign convention)
- ❖ Maximum bending moment occurs at the fixed end i.e. $M = -Wl$
- ❖ From equilibrium consideration, the fixing moment applied at the fixed end is Wl and the reaction is W . the shear force and bending moment are shown in the figure,

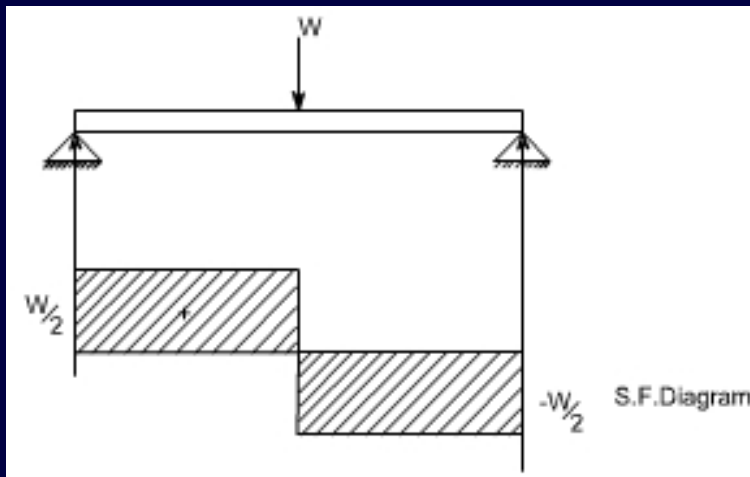
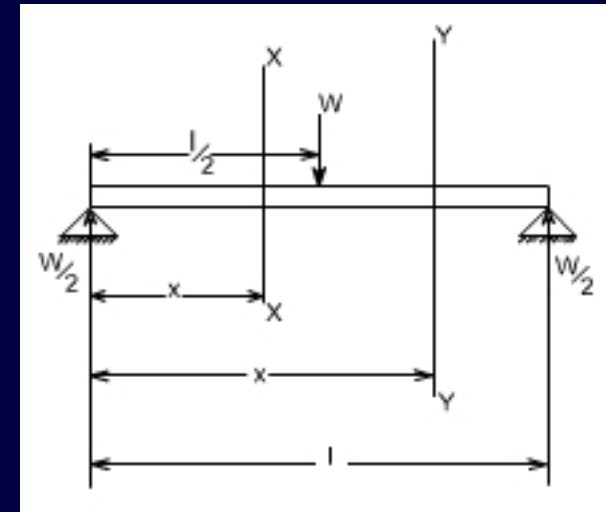
2. Simply supported beam subjected to a central load (i.e. load acting at the mid-way)



By symmetry the reactions at the two supports would be $W/2$ and $W/2$. now consider any section X-X from the left end then, the beam is under the action of following forces.

- ❖ So the shear force at any X-section would be $= W/2$ [Which is constant up to $x < l/2$]
- ❖ If we consider another section Y-Y which is beyond $l/2$ then

$$S.F._{Y-Y} = \frac{W}{2} - W = -\frac{W}{2} \text{ for all values greater } = l/2$$



For B.M diagram: If we just take the moments to the left of the cross-section

$$B.M_{x-x} = \frac{W}{2} x \text{ for } x \text{ lies between } 0 \text{ and } l/2$$

$$B.M_{\text{at } x = \frac{l}{2}} = \frac{W}{2} \cdot \frac{l}{2} \text{ i.e. } B.M_{\text{at } x = 0}$$

$$= \frac{Wl}{4}$$

$$B.M_{y-y} = \frac{W}{2} x - W \left(x - \frac{l}{2} \right)$$

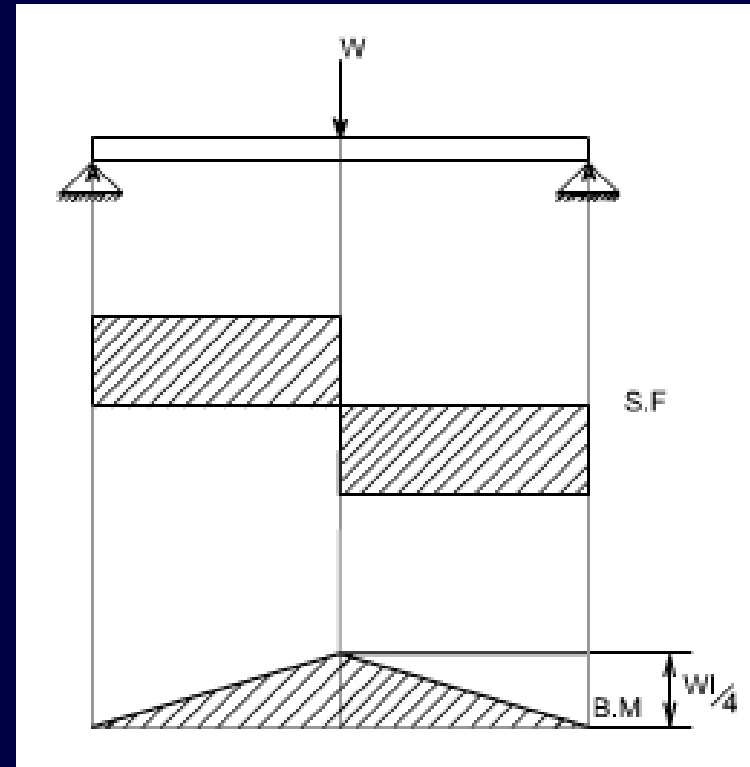
Again

$$= \frac{W}{2} x - Wx + \frac{Wl}{2}$$

$$= -\frac{W}{2} x + \frac{Wl}{2}$$

$$B.M_{\text{at } x = l} = -\frac{Wl}{2} + \frac{Wl}{2}$$

$$= 0$$



It may be observed that at the point of application of load there is an abrupt change in the shear force, at this point the B.M is maximum.

3. A cantilever beam subjected to U.d.L, draw S.F and B.M diagram.

Consider any cross-section XX which is at a distance of x from the free end. If we just take the resultant of all the forces on the left of the X-section, then

$$S.F_{xx} = -Wx \text{ for all values of } x. \text{ ----- (1)}$$

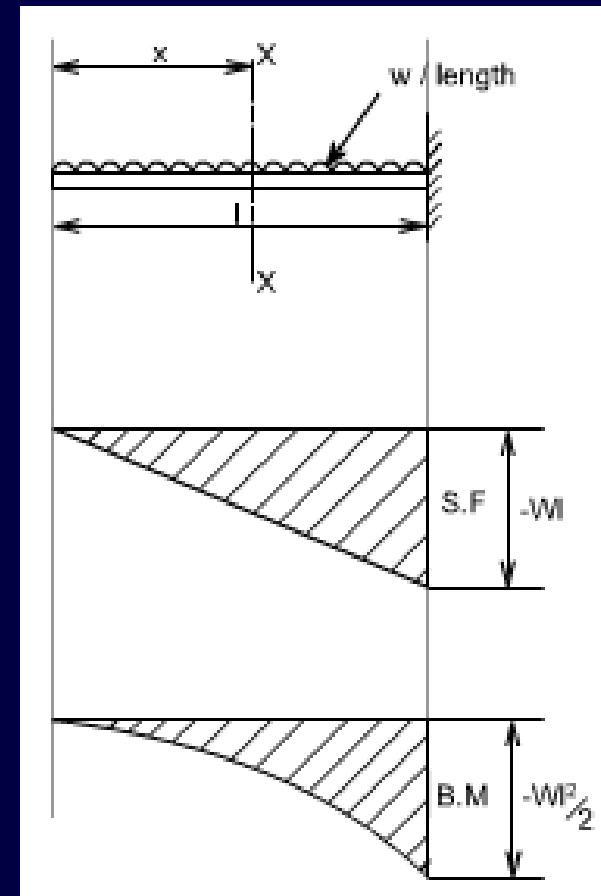
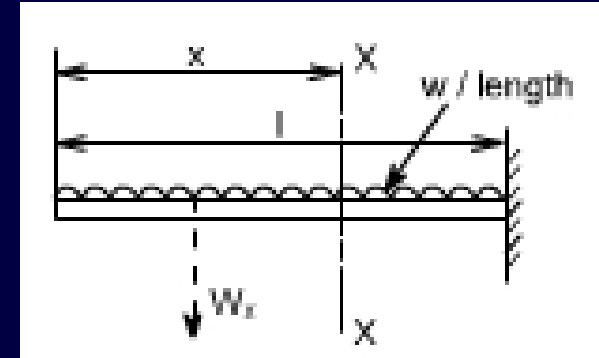
$$S.F_{xx} \Big|_{\text{at } x=0} = 0$$

$$S.F_{xx} \Big|_{\text{at } x=l} = -Wl$$

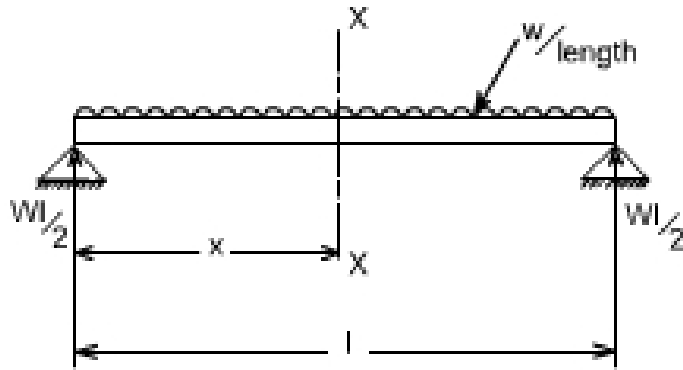
Therefore, the bending moment at any cross-section X-X is

$$B.M_{x-x} = -Wx \frac{x}{2} = -W \frac{x^2}{2}$$

$$B.M_{\text{at } x=l} = -W \frac{l^2}{2}$$



4. Simply supported beam subjected to a uniformly distributed load [U.D.L].



By symmetry the reactions at the two supports would be $w l / 2$ and $w l / 2$. now consider any section $X-X$ from the left end then, the beam is under the action of following forces.

$$S.F_{xx} = \frac{w l}{2} - w x = w \left(\frac{l}{2} - x \right) \text{ ----- (1)}$$

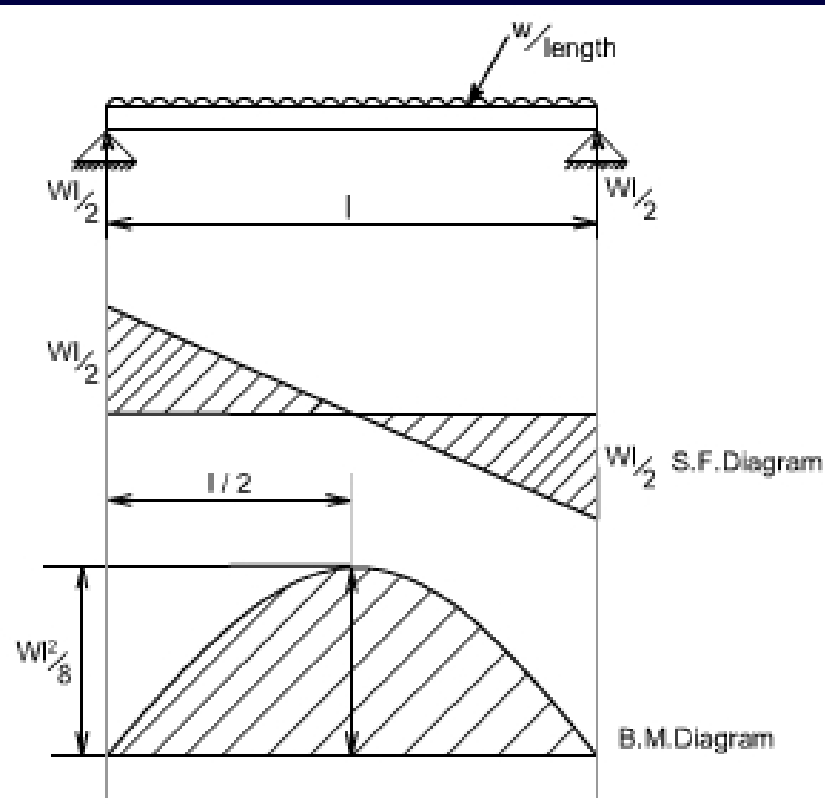
$$S.F_{xx} \big|_{\text{at } x=0} = \frac{w l}{2}; S.F_{xx} \big|_{\text{at } x=l/2} = 0$$

$$\text{and } S.F_{xx} \big|_{\text{at } x=l} = -\frac{w l}{2}$$

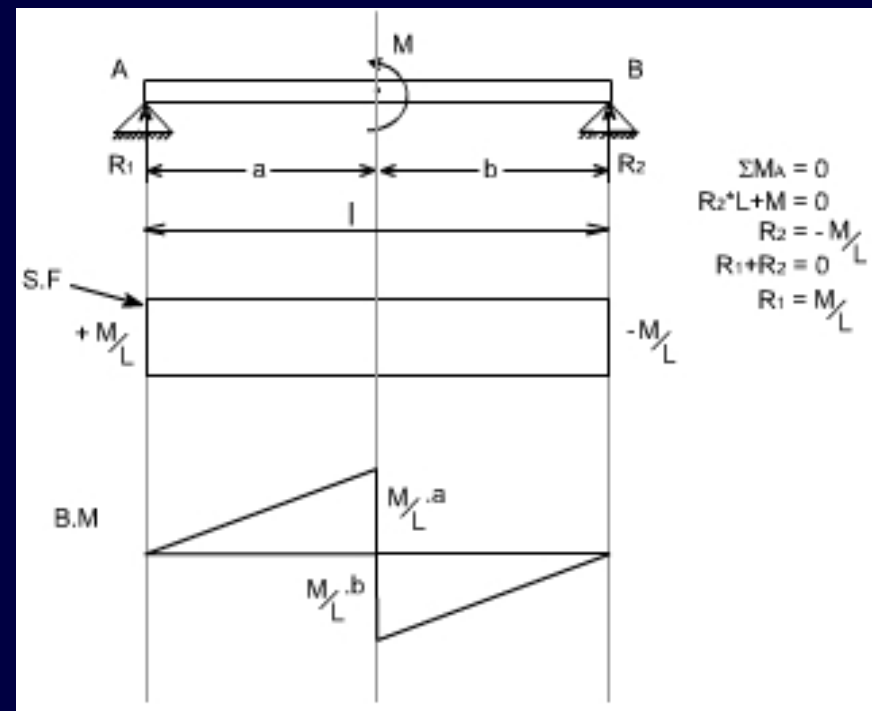
$$B.M_{xx} = \frac{w l}{2} x - w x \frac{x}{2} = w \frac{x}{2} (l - x) \text{ ----- (2)}$$

$$B.M_{\text{at } x=0} = 0; B.M_{\text{at } x=l} = 0$$

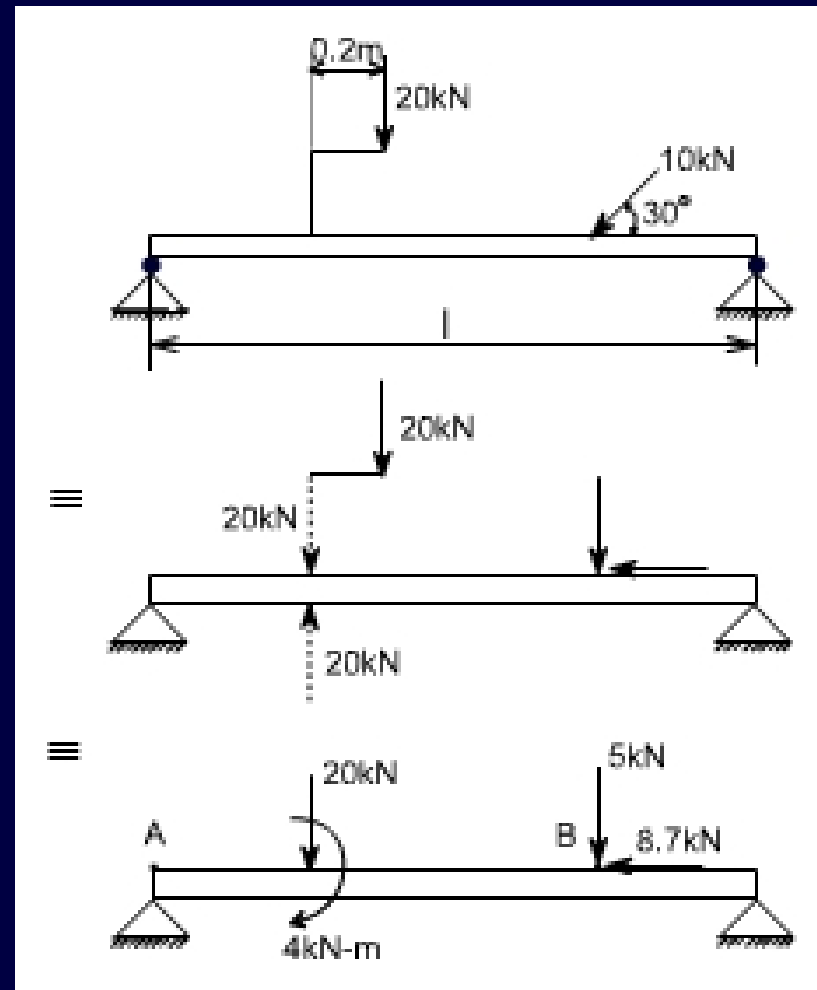
$$\text{and } B.M_{\text{at } x=l/2} = -\frac{w l^2}{8}$$



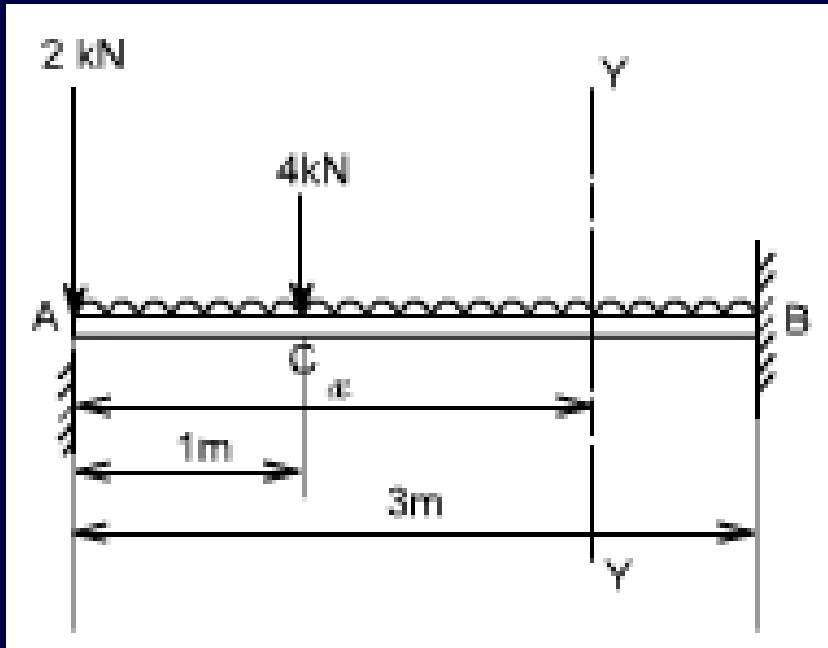
5(a). Couple.



5(b). Eccentric loads.



6. Loading changes or there is an abrupt change of loading:



Problem: A cantilever beam of 3 meters length. It carries a uniformly distributed load of 2 kN/m and a concentrated loads of 2kN at the free end and 4kN at 2 meters from fixed end. Draw S. F. and B.M. diagrams and state the maximum values of the shearing force and bending moment.

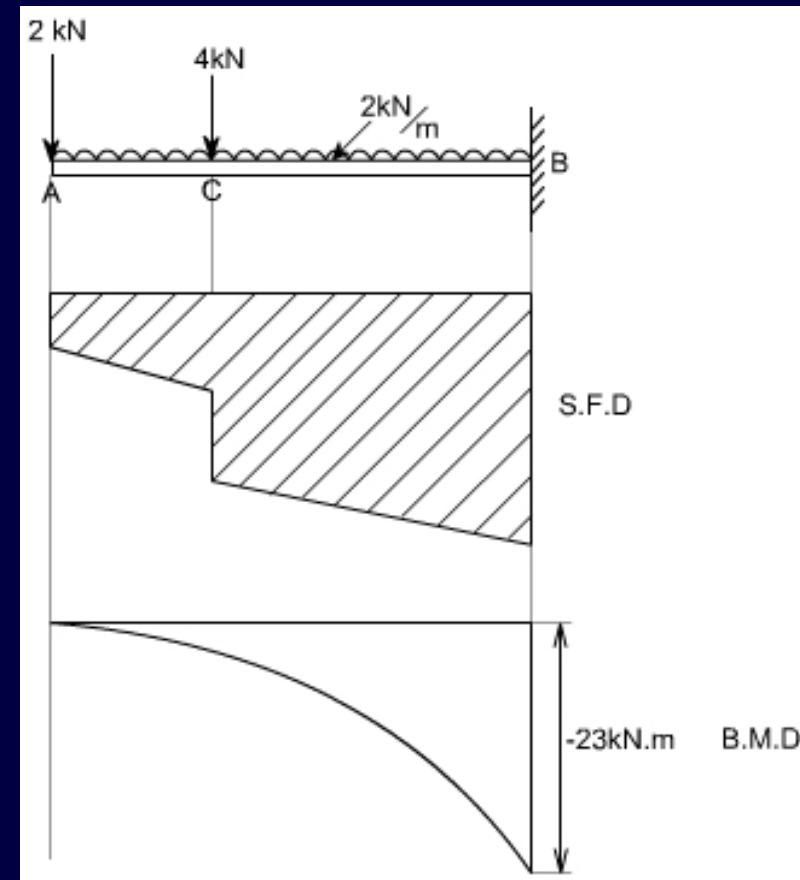
Consider any cross section x-x, at a distance x from the free end

$$\text{At } \rightarrow 0 < x < 1$$

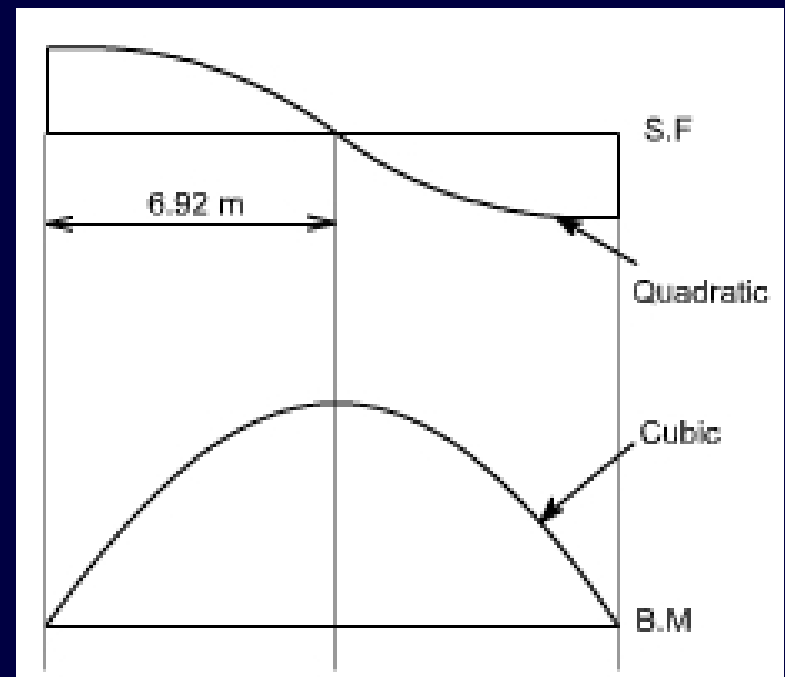
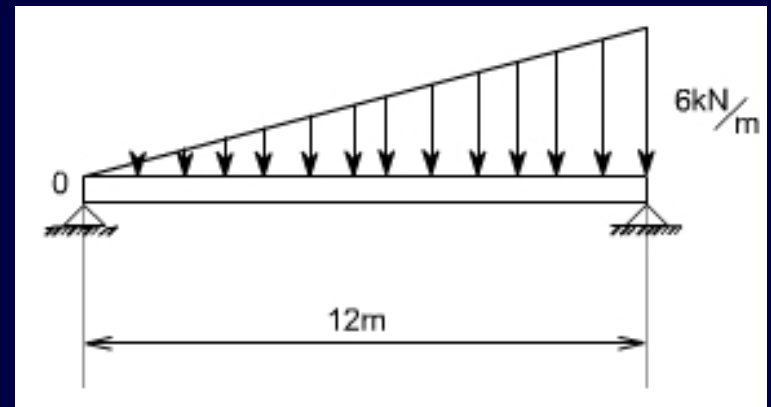
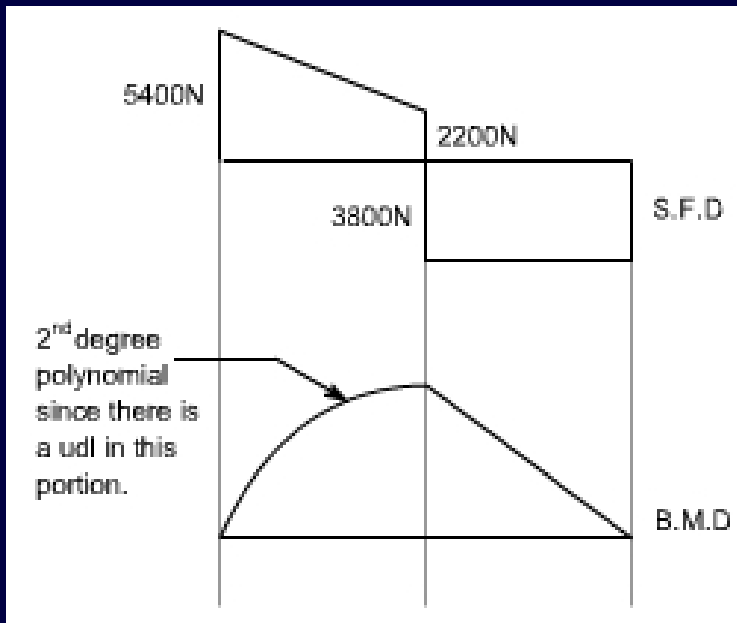
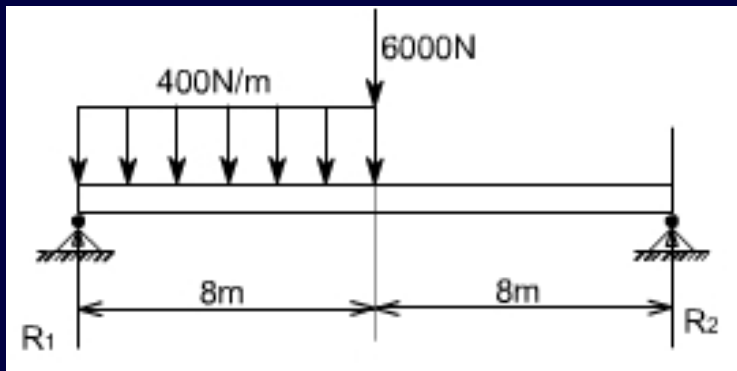
$$S.F._{xx} = -2 - 2x \text{ and } B.M._{xx} = -2x - x^2$$

$$\text{At } \rightarrow 1 < x < 3$$

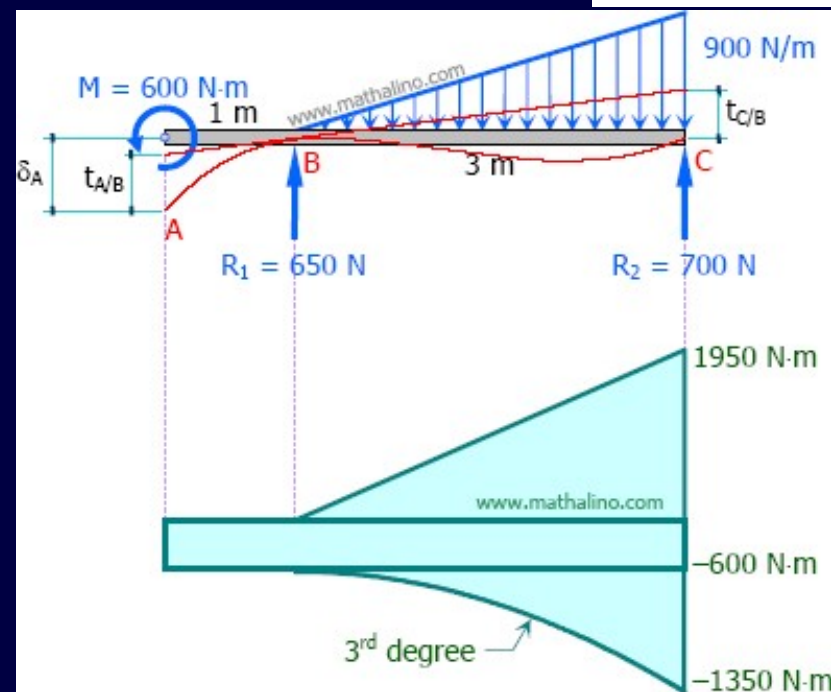
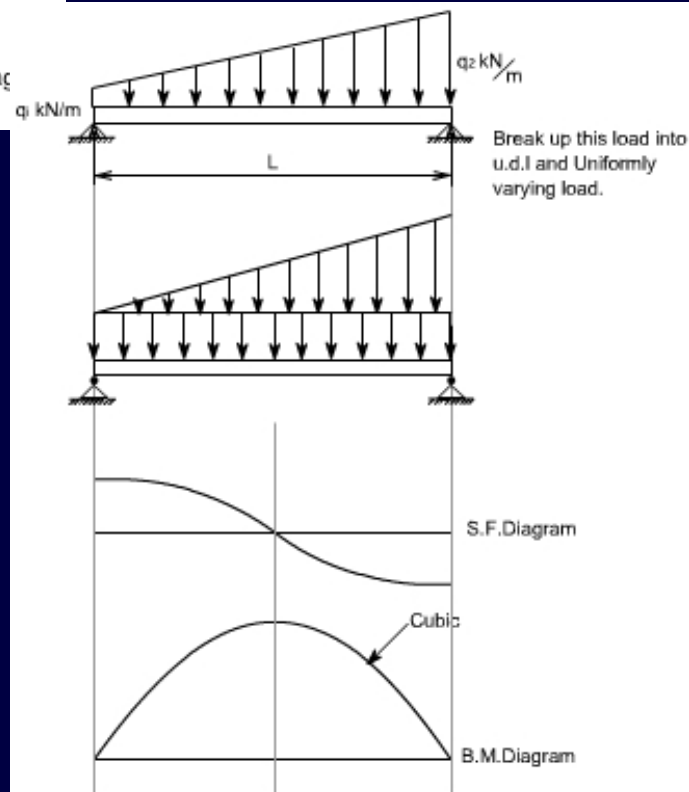
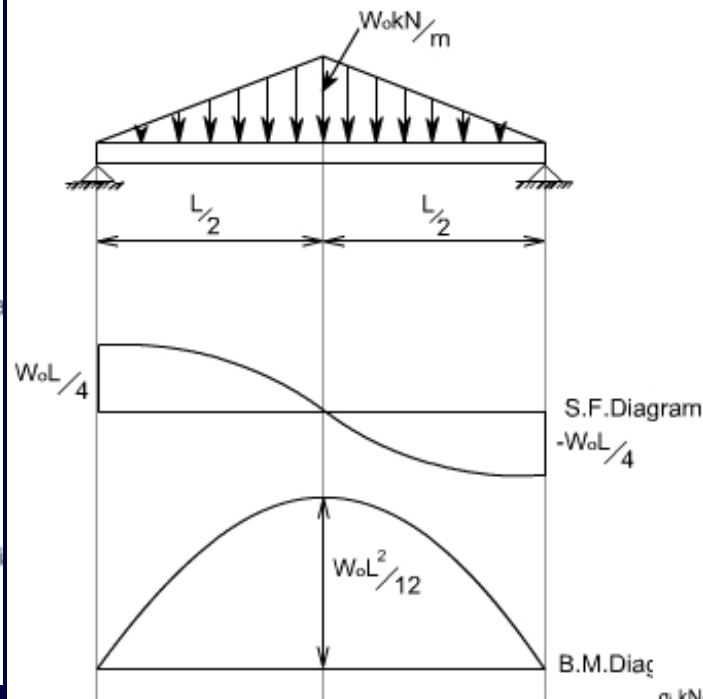
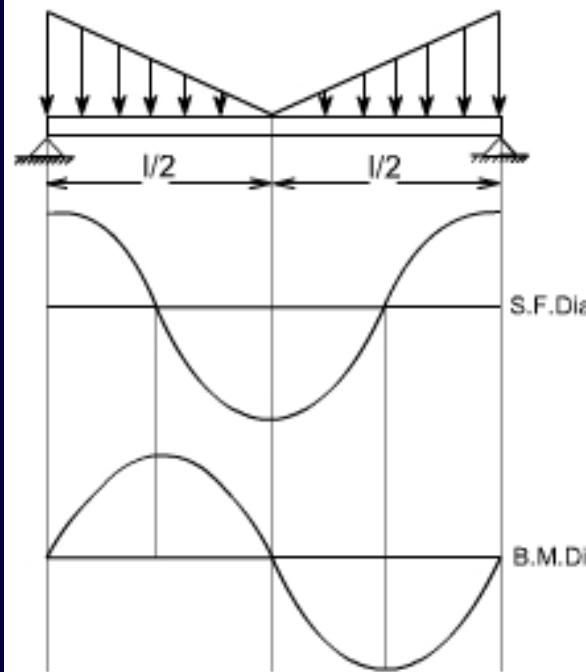
$$S.F._{xx} = -2 - 2x - 4 \text{ and } B.M._{xx} = -2x - x^2 - 4(x-1)$$



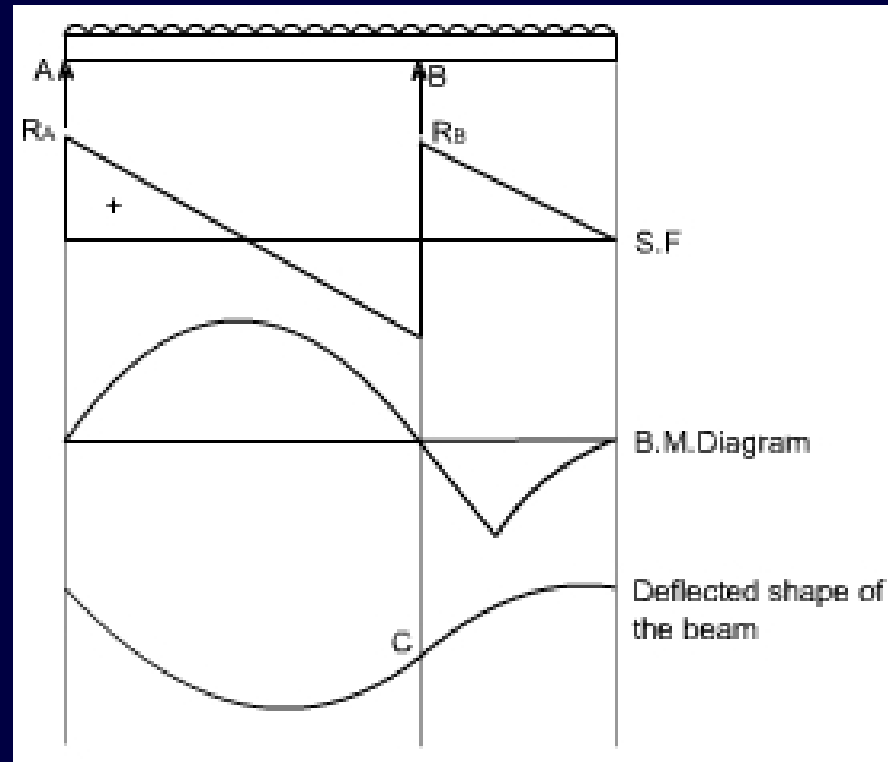
7. Illustrative Examples



Illustrative Examples



Point of Contraflexure: It corresponds to a point where the bending moment changes the sign, hence in order to find the point of contraflexures obviously the B.M would change its sign when it cuts the X-axis therefore to get the points of contraflexure equate the bending moment equation equal to zero. The fibre stress is zero at such sections.

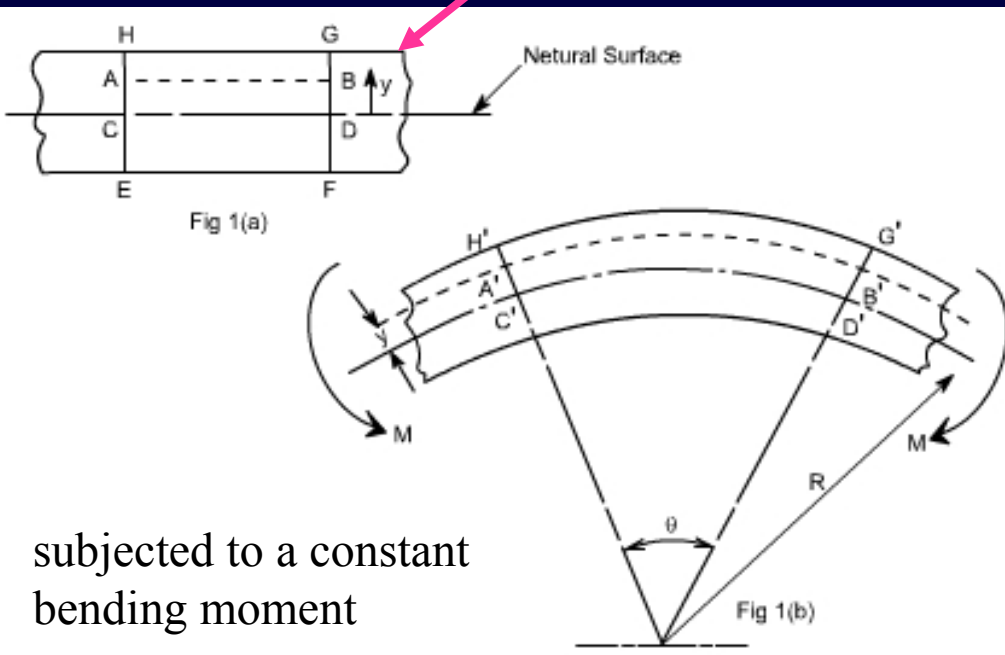


The point C on the beam where the curvature changes from sagging to hogging is a point of contraflexure.

Simple Bending Theory / Theory of Flexure for Initially Straight Beams

The normal stress due to bending are called **flexure stresses**

Unstressed beam



Assumptions:

- ❖ Beam is initially **straight**, and has a **constant cross-section**.
- ❖ Beam is made of **homogeneous material** and the beam has a **longitudinal plane of symmetry**.
- ❖ Resultant of the applied loads lies in the plane of symmetry.
- ❖ The geometry of the overall member is such that bending not buckling is the primary cause of failure.
- ❖ Elastic limit is no-where exceeded and 'E' is same in tension and compression.
- ❖ Plane cross - sections remains plane before and after bending.

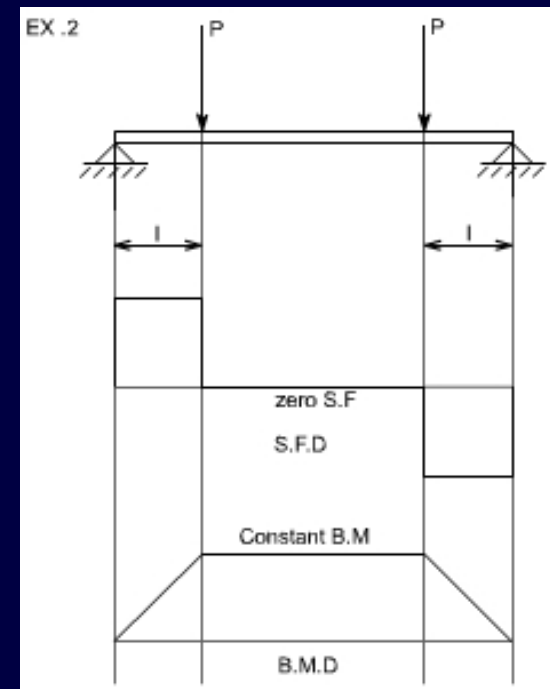
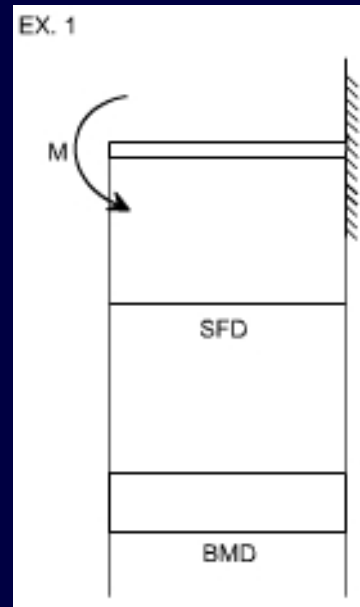
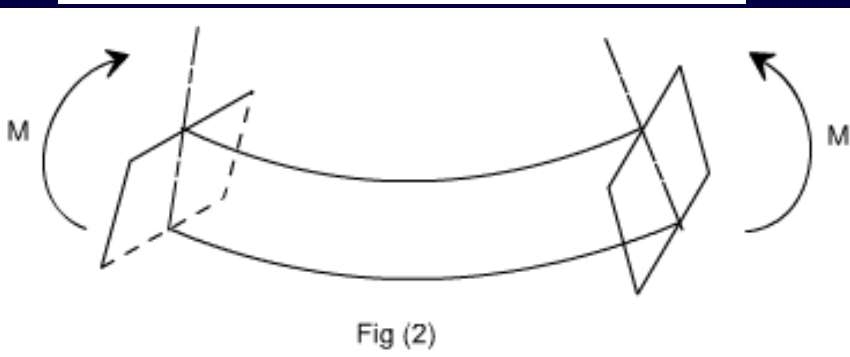
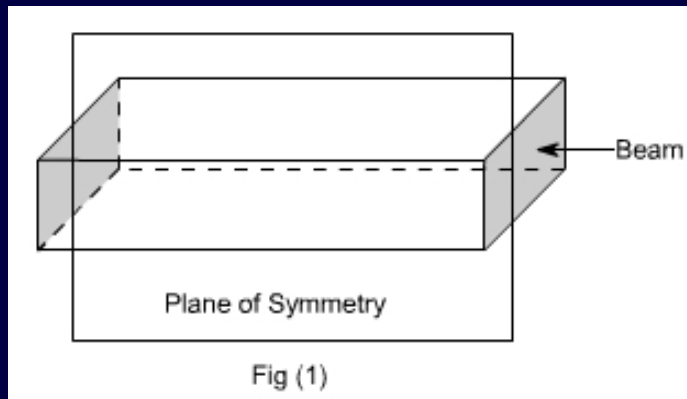
Top fibers of the beam will be subjected to tension and the bottom to compression, therefore, that some where between the two there are points at which the stress is zero. The locus of all such points is known as neutral axis. The radius of curvature R is then measured to this axis. For symmetrical sections the N. A. is the axis of symmetry but what ever the section N. A. will always pass through the centre of the area or centroid.

Concept of pure bending

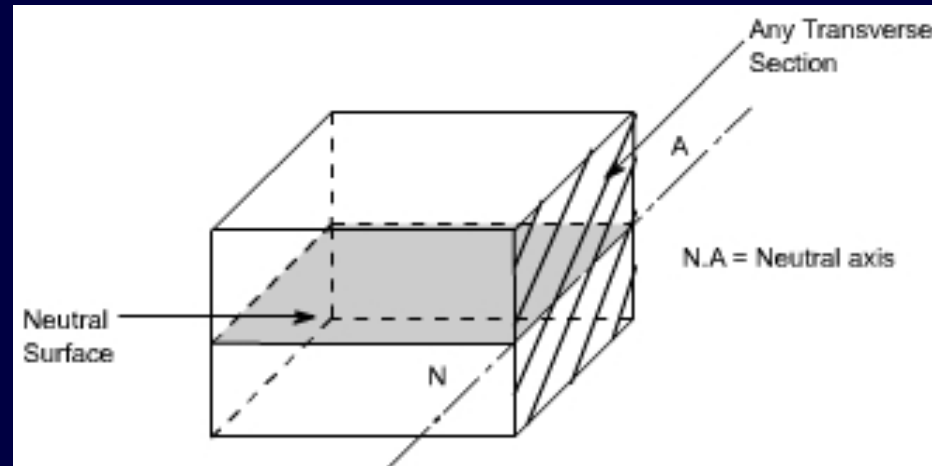
Loading restrictions: Internal reactions developed on any cross-section of a beam may consist of a resultant normal force, a resultant shear force and a resultant couple. In order to ensure that the bending effects alone are investigated, we shall put a constraint on the loading such that the resultant normal and the resultant shear forces are zero on any cross-section perpendicular to the longitudinal axis of the member,

That means $F = 0$ or $dM/dx = F = 0$, i.e., $M = \text{Constant}$

Such a situation may be visualized or envisaged when the beam or some portion of the beam, as been loaded only by pure couples at its ends. It is also to note that the couples are assumed to be loaded in the plane of symmetry.



1. Plane sections originally perpendicular to longitudinal axis of the beam remain plane and perpendicular to the longitudinal axis even after bending , i.e. the cross-section A'E', B'F' (refer Fig 1(a)) do not get warped or curved.
2. In the deformed section, the planes of this cross-section have a common intersection i.e. any time originally parallel to the longitudinal axis of the beam becomes an arc of circle.



We know that when a beam is under bending the fibres at the top will be lengthened while at the bottom will be shortened provided the bending moment M acts at the ends. In between these there are some fibres which remain unchanged in length that is they are not strained, that is they do not carry any stress. The plane containing such fibres is called neutral surface.

The line of intersection between the neutral surface and the transverse exploratory section is called the neutral axis (**N A**) .

Bending Stresses in Beams / Derivation of Elastic Flexural formula

In order to compute the value of bending stresses developed in a loaded beam, let us consider the two cross-sections of a beam **HE** and **GF**, originally parallel as shown in fig 1(a). when the beam is to bend it is assumed that these sections remain parallel i.e. **H'E'** and **G'F'**, the final position of the sections, are still straight lines, they then subtend some angle q .

Consider now fiber **AB** in the material, at a distance y from the N.A, when the beam bends this will stretch to **A'B'**

Therefore,

$$\text{strain in fibre AB} = \frac{\text{change in length}}{\text{original length}}$$

$$= \frac{A'B' - AB}{AB}$$

$$\text{But } AB = CD \text{ and } CD = C'D'$$

refer to fig1(a) and fig1(b)

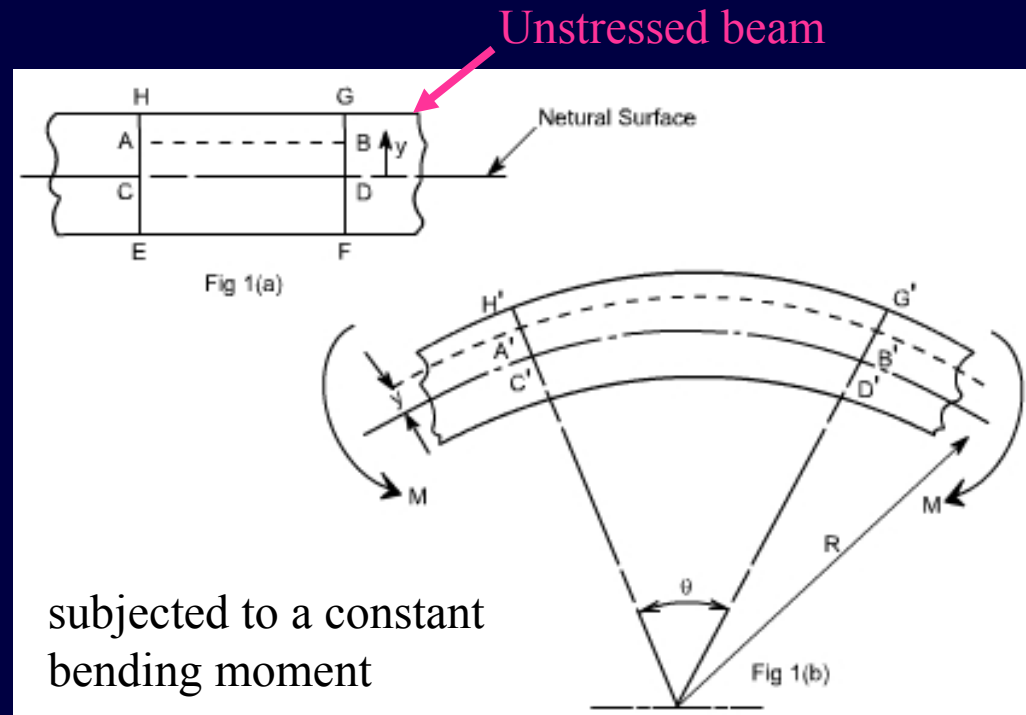
$$\therefore \text{strain} = \frac{A'B' - C'D'}{C'D'}$$

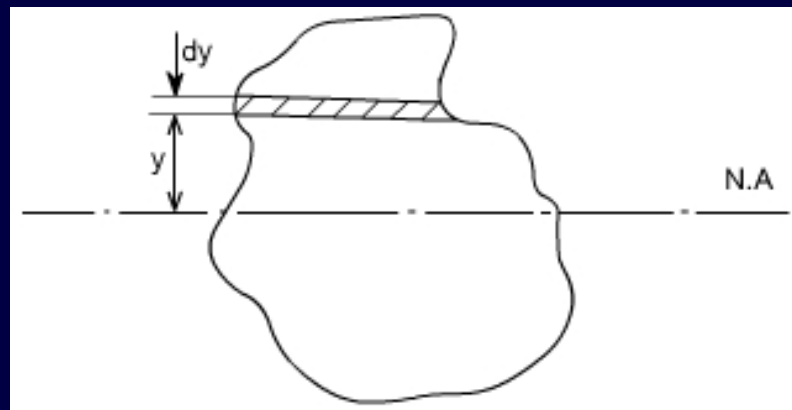
$$= \frac{(R + y)\theta - R\theta}{R\theta} = \frac{R\theta + y\theta - R\theta}{R\theta} = \frac{y}{R}$$

However $\frac{\text{stress}}{\text{strain}} = E$ where E = Young's Modulus of elasticity

Therefore, equating the two strains as obtained from the two relations i.e.,

$$\frac{\sigma}{E} = \frac{y}{R} \text{ or } \frac{\sigma}{y} = \frac{E}{R} \quad \dots\dots\dots(1)$$





$$\sigma = \frac{E}{R} y$$

if the shaded strip is of area ' δA '
then the force on the strip is

$$F = \sigma \delta A = \frac{E}{R} y \delta A$$

Moment about the neutral axis would be $= F \cdot y = \frac{E}{R} y^2 \delta A$

The total moment for the whole
cross-section is therefore equal to

$$M = \sum \frac{E}{R} y^2 \delta A = \frac{E}{R} \sum y^2 \delta A$$

$$M = \frac{E}{R} I \quad \dots\dots\dots (2)$$

combining equation 1 and 2 we get

$$\boxed{\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}}$$

- **This equation is known as the Bending Theory Equation.**
- The above proof has involved the assumption of pure bending without any shear force being present. Therefore this termed as the pure bending equation.
- This equation gives distribution of stresses which are normal to cross-section i.e. in x-direction.

Section Modulus: From simple bending theory equation, the maximum stress obtained in any cross-section is given as

$$\sigma_{\max^m} = \frac{M}{I} y_{\max^m}$$

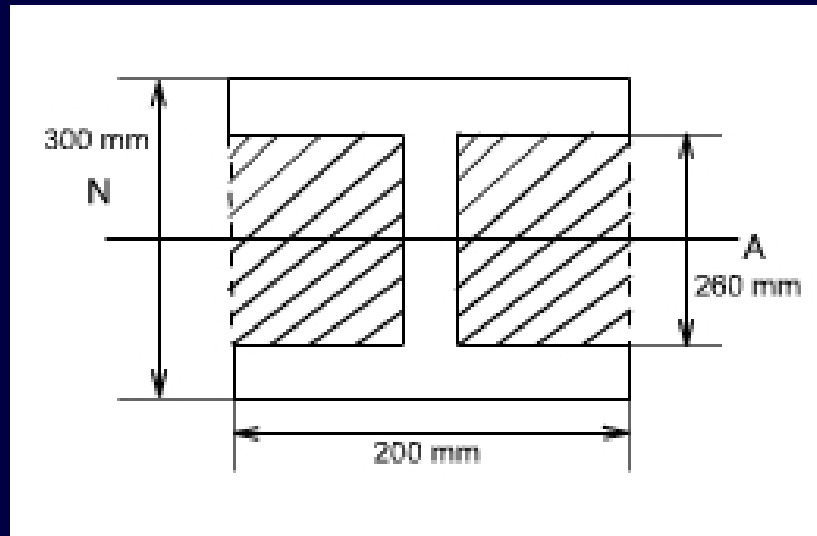
For any given allowable stress the maximum moment which can be accepted by a particular shape of cross-section is therefore

$$M = \frac{I}{y_{\max^m}} \sigma_{\max^m} = Z \sigma_{\max^m}$$

where $Z = \frac{I}{y_{\max^m}}$ is termed as section modulus

Illustrative Problems: An I - section girder, 200mm wide by 300 mm depth flange and web of thickness is 20 mm is used as simply supported beam for a span of 7 m. The girder carries a distributed load of 5 KN /m and a concentrated load of 20 KN at mid-span. Determine the

- The second moment of area of the cross-section of the girder
- The maximum stress set up.



$$I_{\text{girder}} = I_{\text{rectangle}} - I_{\text{shaded portion}}$$

$$= \left[\frac{200 \times 300^3}{12} \right] 10^{-12} - 2 \left[\frac{90 \times 260^3}{12} \right] 10^{-12}$$

$$= (4.5 - 2.64) 10^{-4}$$

$$= 1.86 \times 10^{-4} \text{ m}^4$$

The maximum stress may be found from the simple bending theory by equation

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

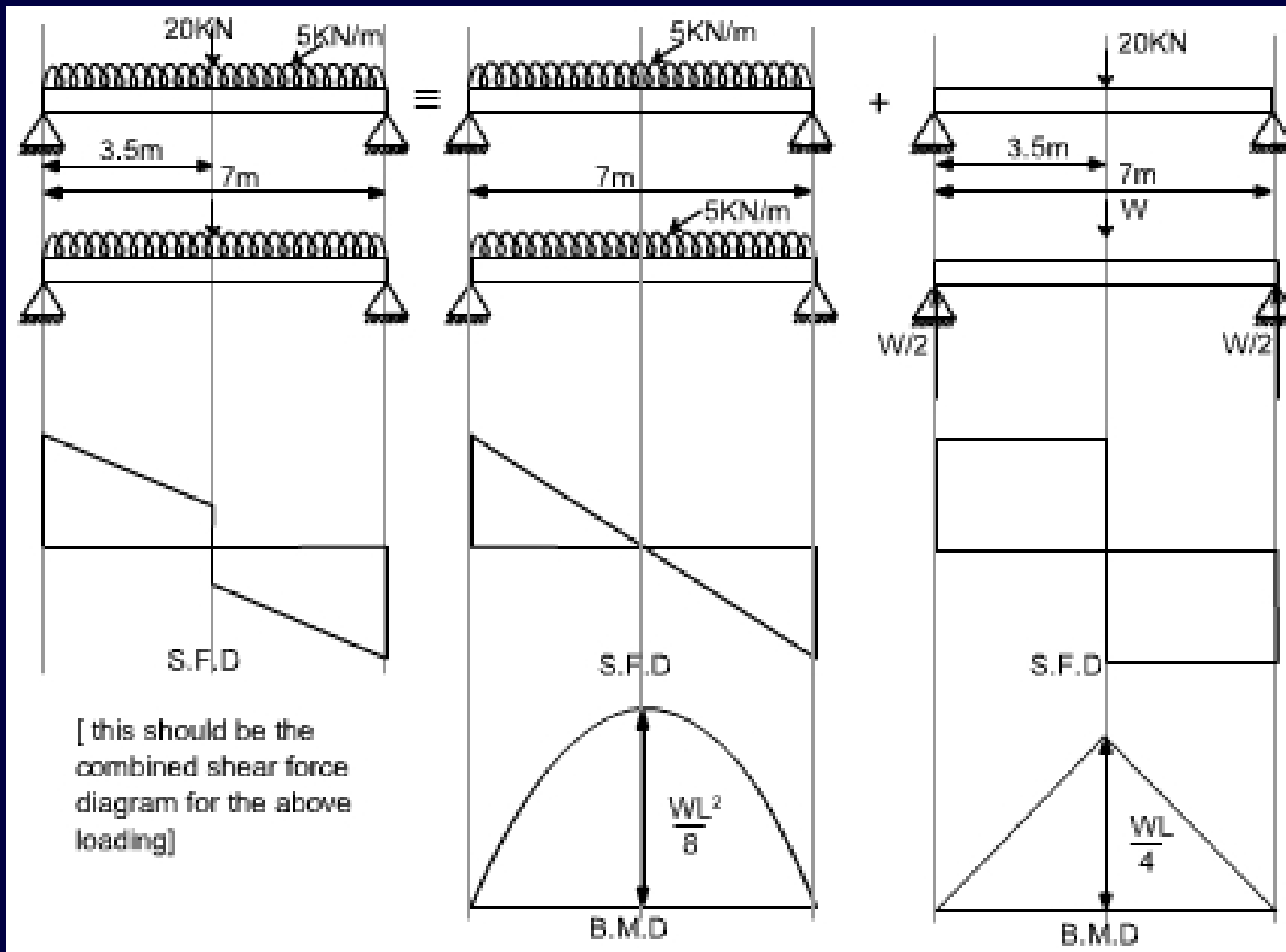
i.e.

$$\sigma_{\text{max}^m} = \frac{M_{\text{max}^m}}{I} y_{\text{max}^m}$$

Computation of Bending Moment: In this case the loading of the beam is of two types

- Uniformly distributed load
- Concentrated Load

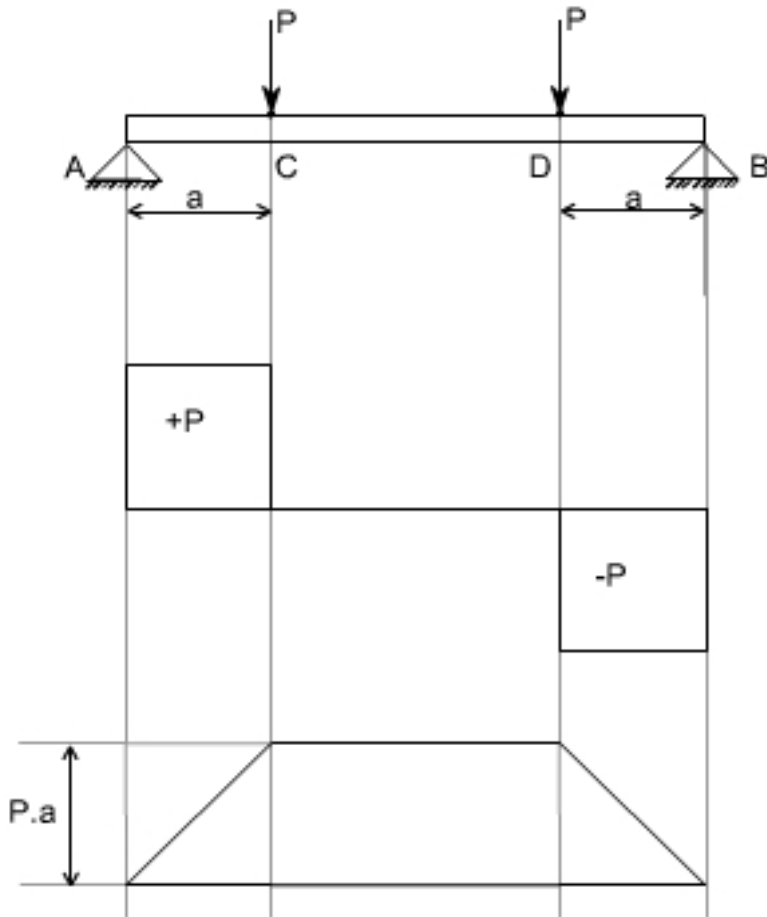
In order to obtain the maximum bending moment the technique will be to consider each loading on the beam separately and get the bending moment due to it as if no other forces acting on the structure and then superimpose the two results.



$$M_{\max^m} = \frac{wL}{4} + \frac{wL^2}{8} = \frac{20 \times 10^2 \times 7}{4} + \frac{5 \times 10^3 \times 7^2}{8} = 65.63 \text{ kN-m}$$

$$\sigma_{\max^m} = \frac{M_{\max^m}}{I} y_{\max^m} = \frac{65.63 \times 10^3 \times 150 \times 10^3}{1.06 \times 10^4} = 51.8 \text{ MN/m}^2$$

Shearing Stresses in Beams

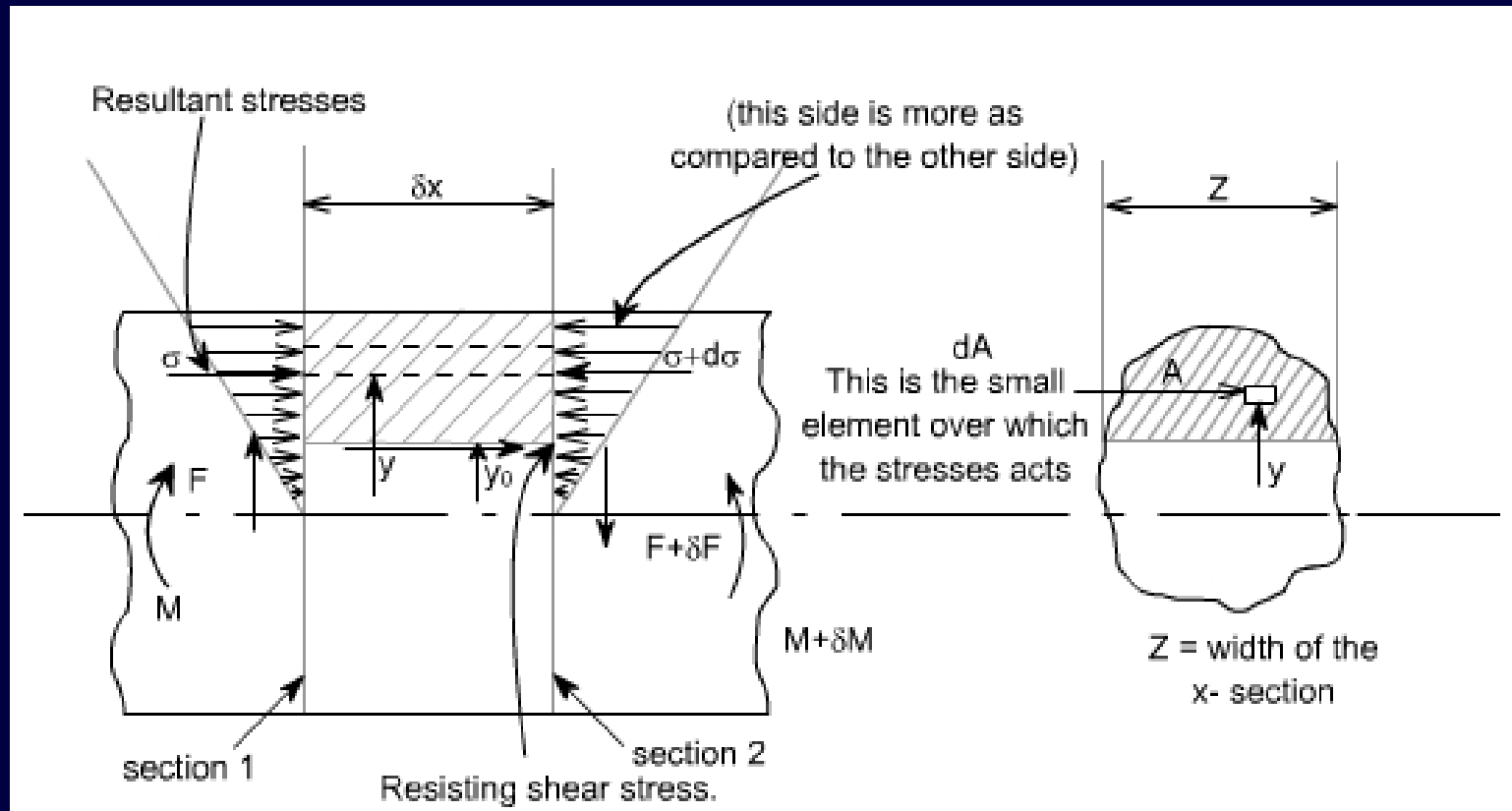


In the case of non-uniform bending of a beam where the bending moment varies from one X-section to another, there is a shearing force on each X-section and shearing stresses are also induced in the material. The deformation associated with those shearing stresses causes warping of the x-section so that the assumption which we assumed while deriving the flexure relation that the plane cross-section after bending remains plane is violated. Now due to warping the plane cross section before bending do not remain plane after bending. This complicates the problem but more elaborate analysis shows that the normal stresses due to bending, as calculated from the equation.

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

The above equation gives the distribution of stresses which are normal to the cross-section that is in x-direction or along the span of the beam are not greatly altered by the presence of these shearing stresses. Thus, it is justifiable to use the theory of pure bending in the case of non uniform bending and it is accepted practice to do so.

Concept of Shear Stresses in Beams: Bending moment represents the resultant of certain linear distribution of normal stresses τ_x over the cross-section. Similarly, the shear force F_x over any cross-section must be the resultant of a certain distribution of shear stresses.



Assumptions :

- ❖ Stress is uniform across the width (i.e. parallel to the neutral axis)
- ❖ The presence of the shear stress does not affect the distribution of normal bending stresses.

$$\text{i.e. } \tau.z\delta x = \int d\sigma.dA$$

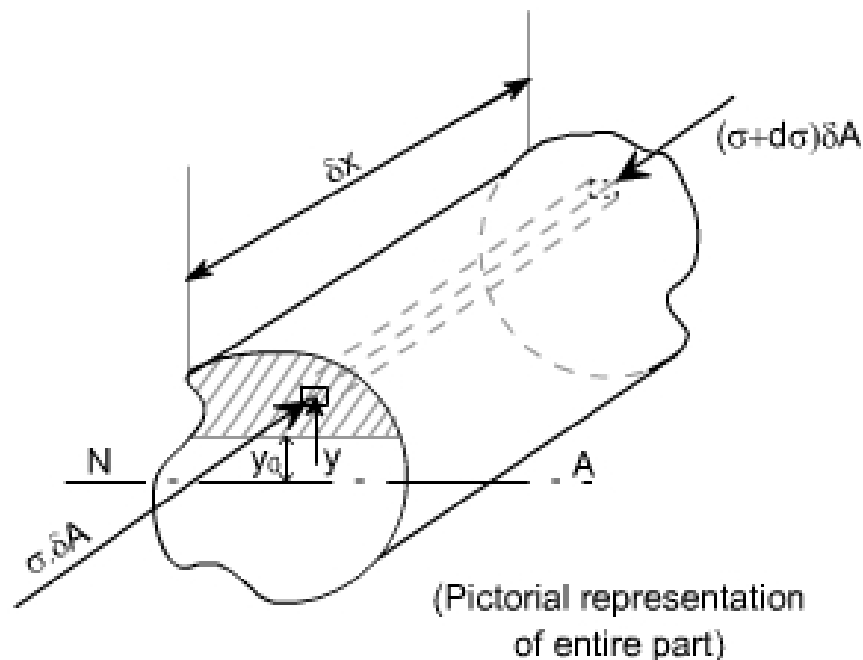
from the bending theory equation

$$\frac{\sigma}{y} = \frac{M}{I}$$

$$\sigma = \frac{M.y}{I}$$

$$\sigma + d\sigma = \frac{(M + \delta M).y}{I}$$

Thus
$$d\sigma = \frac{\delta M.y}{I}$$



$$d\sigma = \frac{\delta M.y}{I}$$

$$\begin{aligned} \tau.z\delta x &= \int d\sigma.dA \\ &= \int \frac{\delta M.y.\delta A}{I} \end{aligned}$$

$$\tau.z\delta x = \frac{\delta M}{I} \int y.\delta A$$

But
$$F = \frac{\delta M}{\delta x}$$

i.e.
$$\tau = \frac{F}{I.z} \int y.\delta A$$

But from definition, $\int y.dA = A\bar{y}$

$\int y.dA$ is the first moment of area of the shaded portion and $\bar{y}$ = centroid of the area 'A'

Hence

$$\tau = \frac{F.A.\bar{y}}{I.z}$$

Shearing stress distribution in typical cross-sections

Rectangular x-section: Consider a rectangular x-section of dimension b and d

$$\tau = \frac{F.A.\bar{y}}{I.z}$$

for this case, $A = b\left(\frac{d}{2} - y\right)$

While $\bar{y} = \left[\frac{1}{2}\left(\frac{d}{2} - y\right) + y \right]$

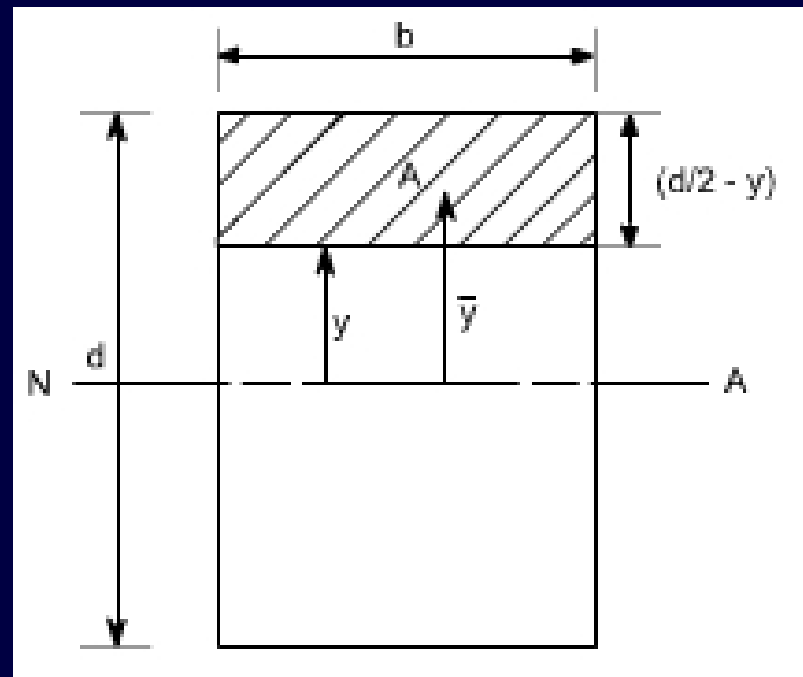
i.e $\bar{y} = \frac{1}{2}\left(\frac{d}{2} + y\right)$ and $z = b; I = \frac{b.d^3}{12}$

substituting all these values, in the formula

$$\begin{aligned} \tau &= \frac{F.A.\bar{y}}{I.z} \\ &= \frac{F.b.\left(\frac{d}{2} - y\right) \cdot \frac{1}{2} \cdot \left(\frac{d}{2} + y\right)}{b \cdot \frac{b.d^3}{12}} \end{aligned}$$

$$= \frac{\frac{F}{2} \cdot \left\{ \left(\frac{d}{2}\right)^2 - y^2 \right\}}{\frac{b.d^3}{12}}$$

$$= \frac{6.F \cdot \left\{ \left(\frac{d}{2}\right)^2 - y^2 \right\}}{b.d^3}$$



Such that
$$\tau_{\max} = \frac{6.F}{b.d^3} \cdot \frac{d^2}{4}$$

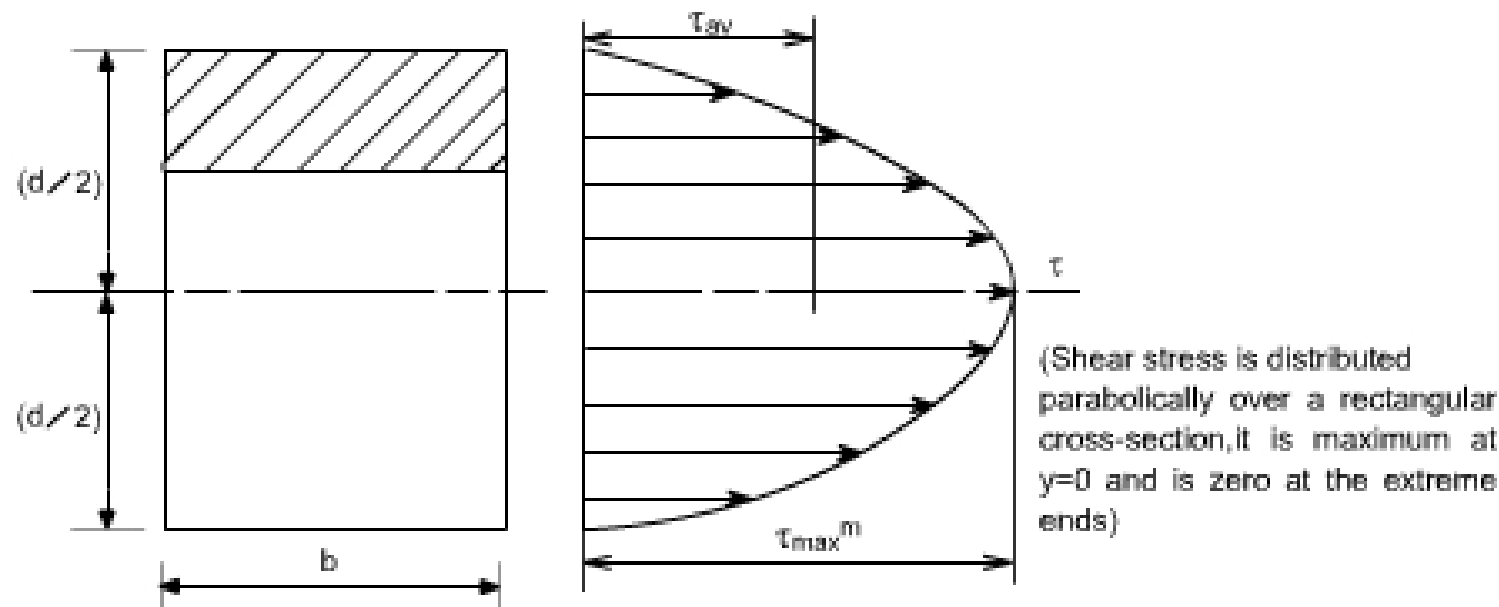
$$= \frac{3.F}{2.b.d}$$

So $\boxed{\tau_{\max} = \frac{3.F}{2.b.d}}$ The value of $\tau_{\max}$ occurs at the neutral axis

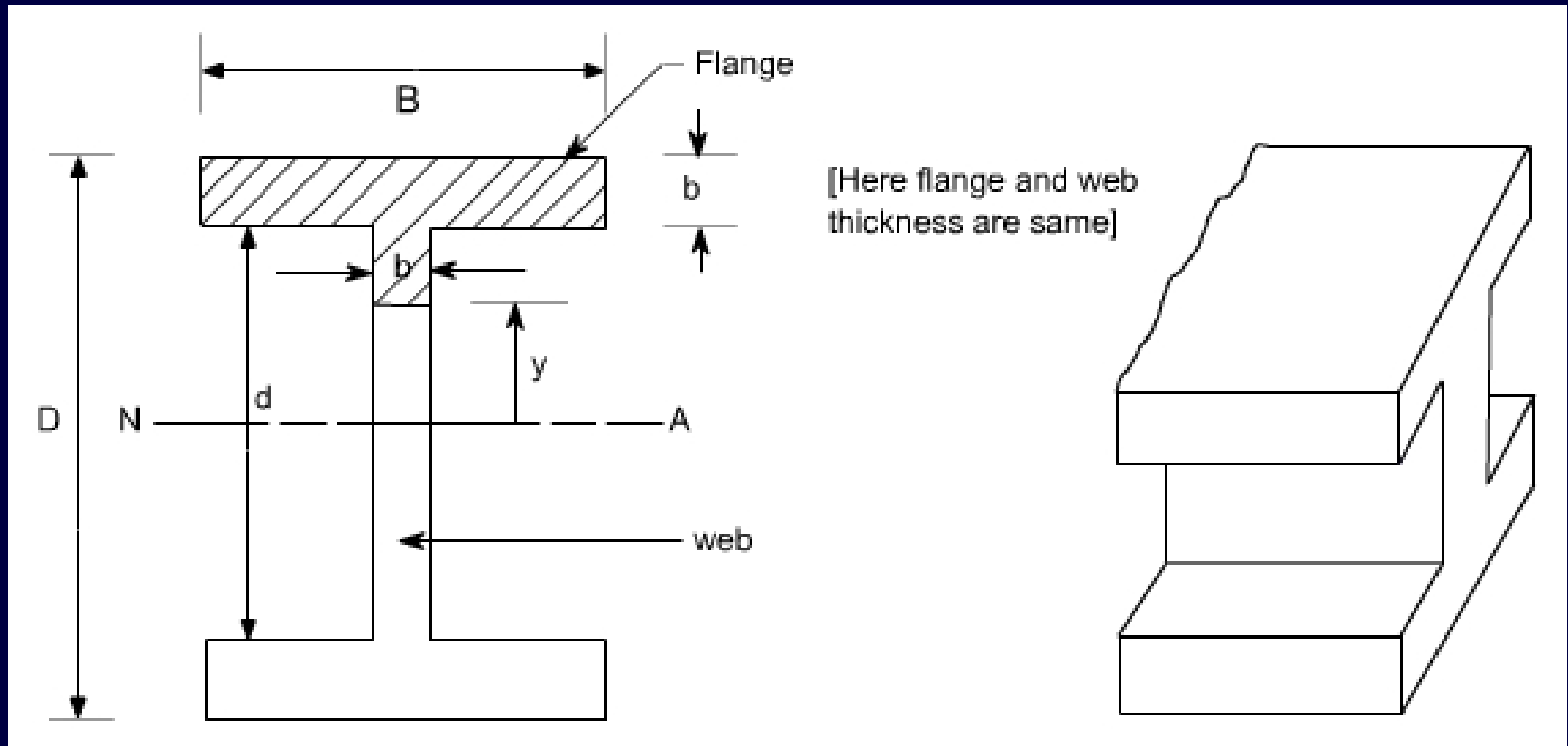
The mean shear stress in the beam is defined as

$$\tau_{\text{mean}} \text{ or } \tau_{\text{avg}} = \frac{F}{A} = \frac{F}{b.d}$$

So $\tau_{\max} = 1.5 \tau_{\text{mean}} = 1.5 \tau_{\text{avg}}$



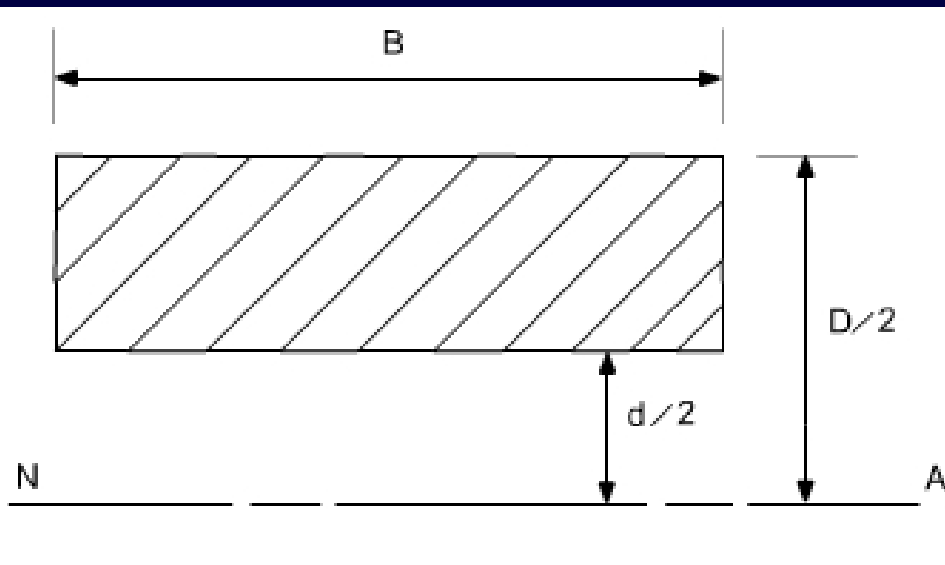
I - section:



$$\tau = \frac{FA\bar{y}}{ZI}$$

Flange area

Web area →



$$\text{Area of the flange} = B \left(\frac{D - d}{2} \right)$$

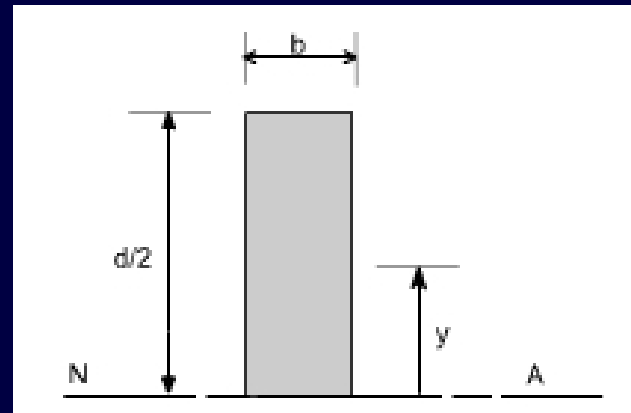
Distance of the centroid of the flange from the N.A

$$\bar{y} = \frac{1}{2} \left(\frac{D - d}{2} \right) + \frac{d}{2}$$

$$\bar{y} = \left(\frac{D + d}{4} \right)$$

Hence,

$$A\bar{y}|_{\text{Flange}} = B \left(\frac{D - d}{2} \right) \left(\frac{D + d}{4} \right)$$



Area of the web

$$A = b \left(\frac{d}{2} - y \right)$$

Distance of the centroid from N.A

$$\bar{y} = \frac{1}{2} \left(\frac{d}{2} - y \right) + y$$

$$\bar{y} = \frac{1}{2} \left(\frac{d}{2} + y \right)$$

Therefore,

$$A\bar{y}|_{\text{web}} = b \left(\frac{d}{2} - y \right) \frac{1}{2} \left(\frac{d}{2} + y \right)$$

Hence,

$$A\bar{y}|_{\text{Total}} = B \left(\frac{D - d}{2} \right) \left(\frac{D + d}{4} \right) + b \left(\frac{d}{2} - y \right) \left(\frac{d}{2} + y \right) \frac{1}{2}$$

Thus,

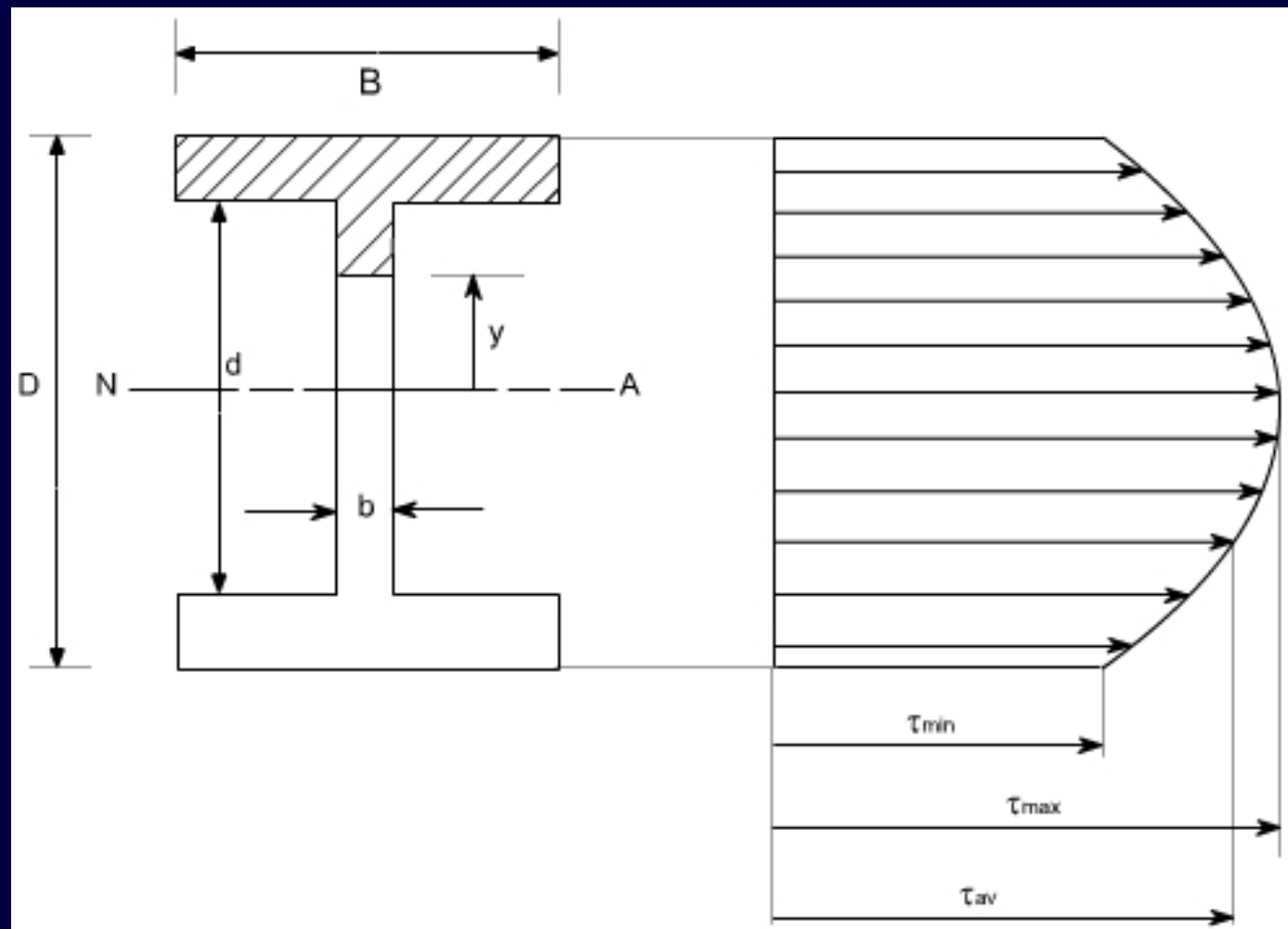
$$A\bar{y}|_{\text{Total}} = B \left(\frac{D^2 - d^2}{8} \right) + \frac{b}{2} \left(\frac{d^2}{4} - y^2 \right)$$

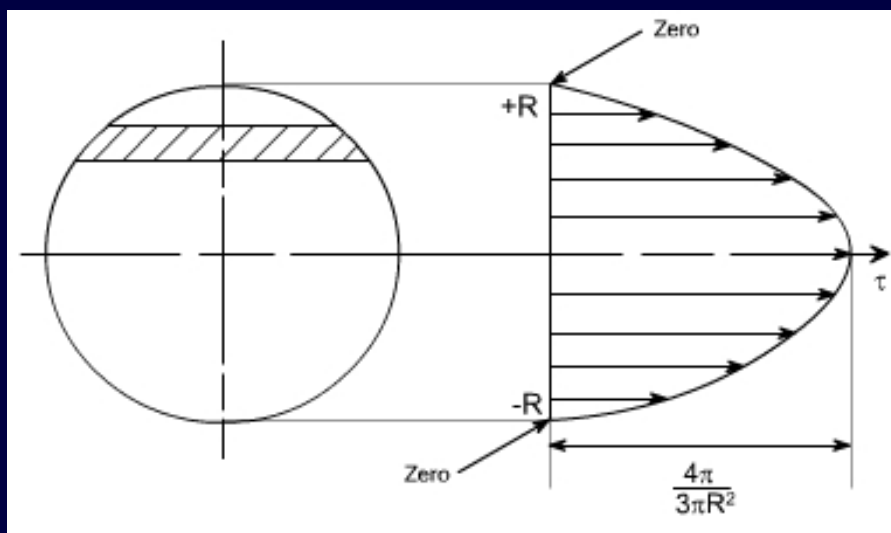
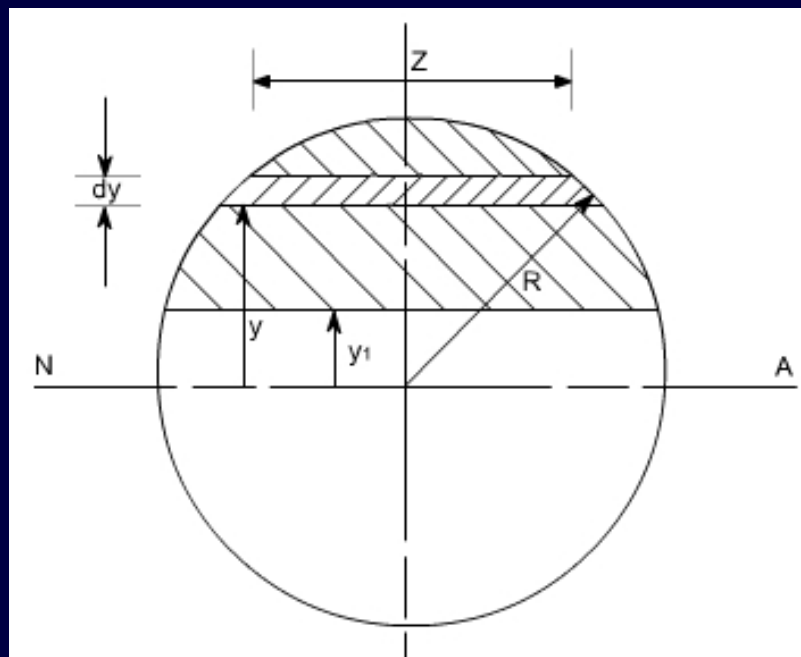
Therefore shear stress,

$$\tau = \frac{F}{bl} \left[\frac{B(D^2 - d^2)}{8} + \frac{b}{2} \left(\frac{d^2}{4} - y^2 \right) \right]$$

$$\tau_{\max} \text{ at } y=0 = \frac{F}{8 b I} \left[B (D^2 - d^2) + b d^2 \right]$$

$$\tau_{\min} \text{ at } y = d/2 = \frac{F}{8 b I} \left[B (D^2 - d^2) \right]$$





$$\left(\frac{Z}{2}\right)^2 + y^2 = R^2$$

$$\left(\frac{Z}{2}\right)^2 = R^2 - y^2 \text{ or } \frac{Z}{2} = \sqrt{R^2 - y^2}$$

$$Z = 2\sqrt{R^2 - y^2}$$

$$dA = Z dy = 2\sqrt{R^2 - y^2} \cdot dy$$

$$I_{N.A. \text{ for a circular cross-section}} = \frac{\pi R^4}{4}$$

Hence,

$$\tau = \frac{F A \bar{y}}{Z I} = \frac{F}{\frac{\pi R^4}{4} \cdot 2\sqrt{R^2 - y^2}} \int_{y_1}^R 2 y \sqrt{R^2 - y^2} dy$$

Where R = radius of the circle.

[The limits have been taken from y_1 to R because we have to find moment of area of the shaded portion]

$$= \frac{4 F}{\pi R^4 \sqrt{R^2 - y_1^2}} \int_{y_1}^R y \sqrt{R^2 - y^2} dy$$

The integration yields the final result to be

$$\tau = \frac{4 F (R^2 - y_1^2)}{3 \pi R^4}$$

Again this is a parabolic distribution of shear stress, having a maximum value when $y_1 = 0$

$$\tau_{\max} |_{y_1 = 0} = \frac{4 F}{3 \pi R^2}$$

Obviously at the ends of the diameter the value of $y_1 = \pm R$ thus $\tau = 0$ so this is again a parabolic distribution; maximum at the neutral axis

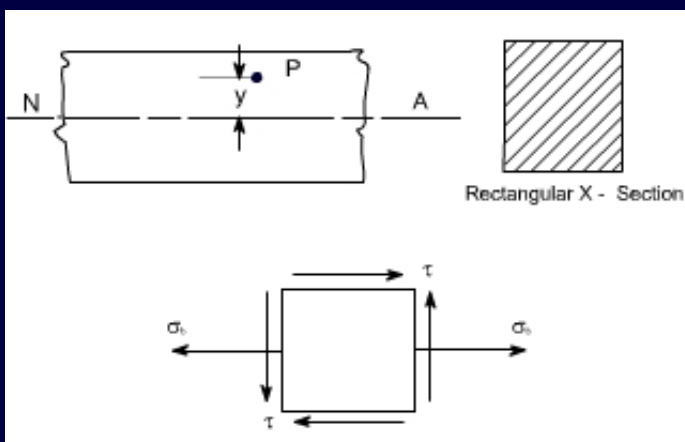
Also

$$\tau_{\text{avg}} \text{ or } \tau_{\text{mean}} = \frac{F}{A} = \frac{F}{\pi R^2}$$

Hence,

$$\tau_{\max} = \frac{4}{3} \tau_{\text{avg}}$$

Principal Stresses in Beams



$\sigma_b = \frac{My}{I}$ for a beam of rectangular cross - section of dimensions b and d; $I = \frac{bd^3}{12}$

$$\sigma_b = \frac{12 My}{bd^3}$$

whereas the value shear stress in the rectangular cross - section is given as

$$\tau = \frac{6 F}{bd^3} \left[\frac{d^2}{4} - y^2 \right]$$

Hence the values of principle stress can be determined from the relations,

$$\sigma_1, \sigma_2 = \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4 \tau_{xy}^2}$$

Letting $\sigma_y = 0$; $\sigma_x = \sigma_b$, the values of σ_1 and σ_2 can be computed as

$$\text{Hence } \sigma_1 / \sigma_2 = \frac{1}{2} \left(\frac{12 My}{bd^3} \right) \pm \frac{1}{2} \sqrt{\left(\frac{12 My}{bd^3} \right)^2 + 4 \left(\frac{6 F}{bd^3} \left(\frac{d^2}{4} - y^2 \right) \right)^2}$$

$$\sigma_1, \sigma_2 = \frac{6}{bd^3} \left[My \pm \sqrt{M^2 y^2 + F^2 \left(\frac{d^2}{4} - y^2 \right)^2} \right]$$

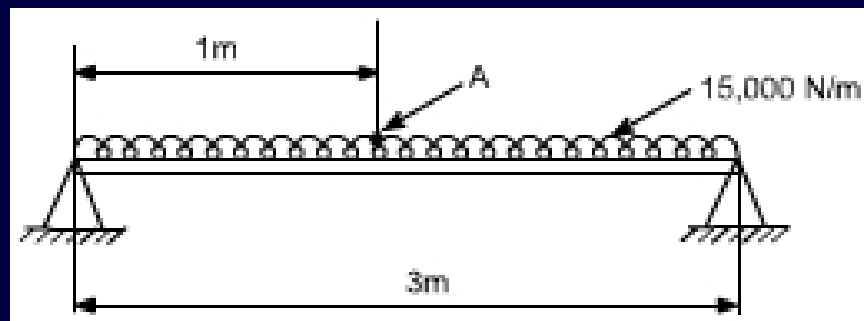
Also,

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \quad \text{putting } \sigma_y = 0$$

we get,

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x}$$

Find the principal stress at a point A in a uniform rectangular beam 200 mm deep and 100 mm wide, simply supported at each end over a span of 3 m and carrying a uniformly distributed load of 15,000 N/m.



$$S.F|_{x=1m} = 7,500 \text{ N}$$

$$B.M|_{x=1m} = 15,000 \text{ N.m}$$

$$\sigma_x = \frac{My}{I}$$

$$= \frac{15,000 \times 5 \times 10^{-2} \times 12}{10 \times 10^{-12} \times (20 \times 10^{-2})^3}$$

$$\sigma_x = 11.25 \text{ MN/m}^2$$

For the computation of shear stresses

$$\tau = \frac{6F}{bd^3} \left[\frac{d^2}{4} - y^2 \right] \quad \text{putting } y = 50 \text{ mm, } d = 200 \text{ mm}$$

$$F = 7500 \text{ N}$$

$$\tau = 0.422 \text{ MN/m}^2$$

$$\sigma_1 = 11.27 \text{ MN/m}^2 \quad \text{and}$$

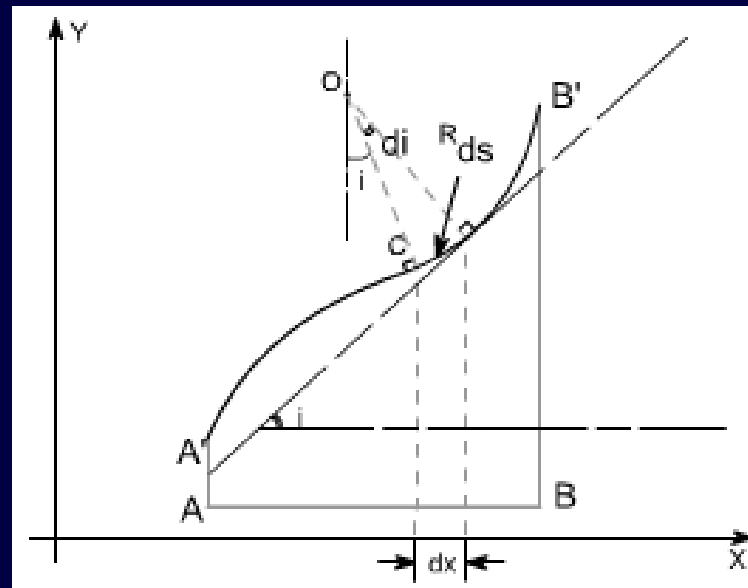
$$\sigma_2 = -0.025 \text{ MN/m}^2$$

Deflection of Beams

Assumptions:

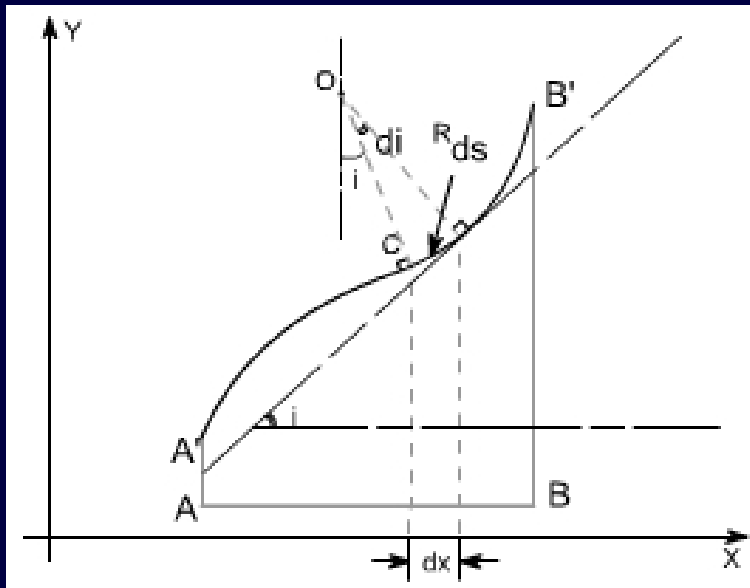
- Stress is proportional to strain i.e. Hooke's law applies. Thus, the equation is valid only for beams that are not stressed beyond the elastic limit.
- The curvature is always small.
- Any deflection resulting from the shear deformation of the material or shear stresses is neglected.

Consider a beam AB which is initially straight and horizontal when unloaded. If under the action of loads the beam deflect to a position A'B' under load or in-fact we say that the axis of the beam bends to a shape A'B'. It is customary to call A'B' the curved axis of the beam as the elastic line or deflection curve.



To express the deflected shape of the beam in rectangular co-ordinates let us take two axes x and y , x -axis coincide with the original straight axis of the beam and the y axis shows the deflection.

Futher, let us consider an element ds of the deflected beam. At the ends of this element let us construct the normal which intersect at point O denoting the angle between these two normal be di



$$\tan i = \frac{dy}{dx} \quad \dots\dots(1) \quad \text{or} \quad i = \frac{dy}{dx} \quad \text{Assuming } \tan i = i$$

Futher

$$ds = R di$$

however,

$$ds = dx \quad [\text{usually for small curvature}]$$

Hence

$$ds = dx = R di$$

$$\text{or} \quad \boxed{\frac{di}{dx} = \frac{1}{R}}$$

substituting the value of i , one get

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{1}{R} \quad \text{or} \quad \frac{d^2 y}{dx^2} = \frac{1}{R}$$

From the simple bending theory

$$\frac{M}{I} = \frac{E}{R} \quad \text{or} \quad M = \frac{EI}{R}$$

so the basic differential equation governing the deflection of beam is

$$\boxed{M = EI \frac{d^2 y}{dx^2}}$$

Relationship between shear force, bending moment and deflection

$$\frac{dM}{dx} = EI \frac{d^3 y}{dx^3} \quad \text{Recalling } \frac{dM}{dx} = F$$

Thus,

$$F = EI \frac{d^3 y}{dx^3}$$

$$\text{i.e } w = - \frac{dF}{dx}$$

$$w = -EI \frac{d^4 y}{dx^4}$$

Therefore if 'y' is the deflection of the loaded beam, then the following important relations can be arrived at

$$\text{slope} = \frac{dy}{dx}$$

$$\text{B.M} = EI \frac{d^2 y}{dx^2}$$

$$\text{Shear force} = EI \frac{d^3 y}{dx^3}$$

$$\text{load distribution} = EI \frac{d^4 y}{dx^4}$$

Methods for finding the deflection

Direct integration method: The governing differential equation is defined as

$$M = EI \frac{d^2 y}{dx^2} \quad \text{or} \quad \frac{M}{EI} = \frac{d^2 y}{dx^2}$$

on integrating one get,

$$\frac{dy}{dx} = \int \frac{M}{EI} dx + A \quad \text{--- this equation gives the slope}$$

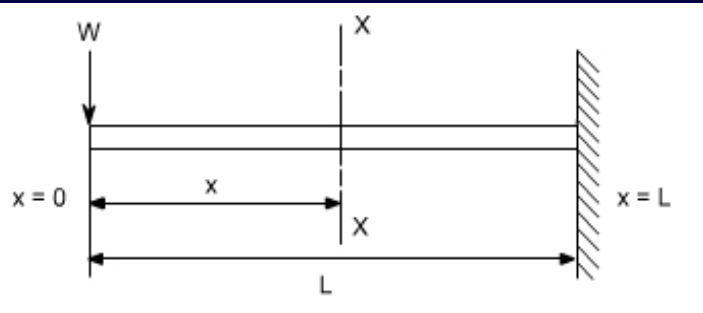
of the loaded beam.

Integrate once again to get the deflection.

$$y = \iint \frac{M}{EI} dx + Ax + B$$

Where A and B are constants of integration to be evaluated from the known conditions of slope and deflections for the particular value of x.

Case 1: Cantilever Beam with Concentrated Load at the end



End Criteria / Boundary Conditions:

i.e at $x = L$; $y = 0$ ----- (1)

at $x = L$; $dy/dx = 0$ ----- (2)

$$\text{S.F}|_{x-x} = -W$$

$$\text{B.M}|_{x-x} = -W \cdot x$$

$$\text{Therefore } M|_{x-x} = -W \cdot x$$

$$\text{the governing equation } \frac{M}{EI} = \frac{d^2 y}{dx^2}$$

substituting the value of M in terms of x then integrating the equation one get

$$\frac{M}{EI} = \frac{d^2 y}{dx^2}$$

$$\frac{d^2 y}{dx^2} = -\frac{Wx}{EI}$$

$$\int \frac{d^2 y}{dx^2} = \int -\frac{Wx}{EI} dx$$

$$\frac{dy}{dx} = -\frac{Wx^2}{2EI} + A$$

Integrating once more,

$$\int \frac{dy}{dx} = \int -\frac{Wx^2}{2EI} dx + \int A dx$$

$$y = -\frac{Wx^3}{6EI} + Ax + B$$

$$A = \frac{WL^2}{2EI}$$

While employing the first condition yields

$$y = -\frac{WL^3}{6EI} + AL + B$$

$$B = \frac{WL^3}{6EI} - AL$$

$$= \frac{WL^3}{6EI} - \frac{WL^3}{2EI}$$

$$= \frac{WL^3 - 3WL^3}{6EI} = -\frac{2WL^3}{6EI}$$

$$B = -\frac{WL^3}{3EI}$$

Substituting the values of A and B we get

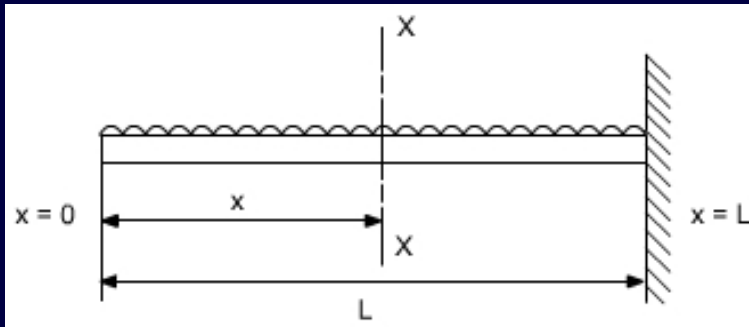
$$y = \frac{1}{EI} \left[-\frac{Wx^3}{6EI} + \frac{WL^2x}{2EI} - \frac{WL^3}{3EI} \right]$$

The slope as well as the deflection would be maximum at the free end hence putting $x=0$ we get,

$$y_{\max} = -\frac{WL^3}{3EI}$$

$$(\text{Slope})_{\max} = +\frac{WL^2}{2EI}$$

Case 2: A Cantilever with Uniformly distributed Loads



End Criteria / Boundary Conditions:

i.e at $x=L$; $y=0$ ----- (1)

at $x=L$; $dy/dx = 0$ ----- (2)

$$S.F|_{x-x} = -w$$

$$B.M|_{x-x} = -w \cdot x \cdot \frac{x}{2} = w \left(\frac{x^2}{2} \right)$$

$$\frac{M}{EI} = \frac{d^2 y}{dx^2}$$

$$\frac{d^2 y}{dx^2} = -\frac{wx^2}{2EI}$$

$$\int \frac{d^2 y}{dx^2} = \int -\frac{wx^2}{2EI} dx$$

$$\frac{dy}{dx} = -\frac{wx^3}{6EI} + A$$

$$\int \frac{dy}{dx} = \int -\frac{wx^3}{6EI} dx + \int A dx$$

$$y = -\frac{wx^4}{24EI} + Ax + B$$

$$A = +\frac{wx^3}{6EI}$$

whereas the first boundary conditions yields

$$B = \frac{wL^4}{24EI} - \frac{wL^4}{6EI}$$

$$B = -\frac{wL^4}{8EI}$$

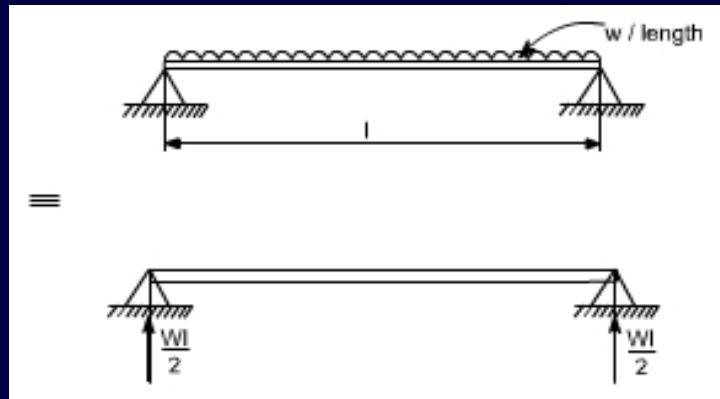
$$\text{Thus, } y = \frac{1}{EI} \left[-\frac{wx^4}{24} + \frac{wL^3 x}{6} - \frac{wL^4}{8} \right]$$

So $y_{\max}$ will be at $x=0$

$$y_{\max} = -\frac{wL^4}{8EI}$$

$$\left(\frac{dy}{dx} \right)_{\max} = \frac{wL^3}{6EI}$$

Case 3: Simply Supported beam with uniformly distributed Loads



End Criteria / Boundary Conditions:

i.e at $x = 0$; $y = 0$ ----- (1)

at $x = L$; $y = 0$ ----- (2)

$$\text{S.F}|_{x-x} = w \left(\frac{l}{2} \right) - w \cdot x$$

$$\text{B.M}|_{x-x} = w \cdot \left(\frac{l}{2} \right) \cdot x - w \cdot x \cdot \left(\frac{x}{2} \right)$$

$$= \frac{wl \cdot x}{2} - \frac{wx^2}{2}$$

The differential equation which gives the elastic curve for the deflected beam is

$$\frac{d^2 y}{dx^2} = \frac{M}{EI} = \frac{1}{EI} \left[\frac{wl \cdot x}{2} - \frac{wx^2}{2} \right]$$

$$\frac{dy}{dx} = \int \frac{wl \cdot x}{2EI} dx - \int \frac{wx^2}{2EI} dx + A$$

$$= \frac{wlx^2}{4EI} - \frac{wx^3}{6EI} + A$$

Integrating, once more one gets

$$y = \frac{wlx^3}{12EI} - \frac{wx^4}{24EI} + A \cdot x + B \quad \text{----- (1)}$$

$$0 = \frac{wl^4}{12EI} - \frac{wl^4}{24EI} + A \cdot l$$

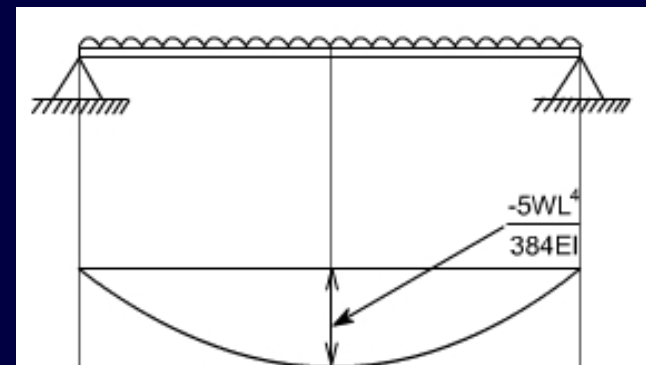
$$A = -\frac{wl^3}{24EI}$$

So the equation which gives the deflection curve is

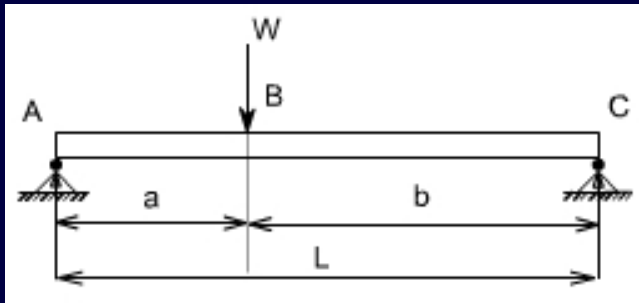
$$y = \frac{1}{EI} \left[\frac{wLx^3}{12} - \frac{wx^4}{24} - \frac{wL^3x}{24} \right]$$

Then $y_{\max} = \frac{1}{EI} \left[\frac{wL}{12} \left(\frac{L^3}{8} \right) - \frac{w}{24} \left(\frac{L^4}{16} \right) - \frac{wL^3}{24} \left(\frac{L}{2} \right) \right]$

$$y_{\max} = -\frac{5wL^4}{384EI}$$



Case 4: Simply Supported beam with Concentrated Load



$$R_1 = \frac{Wb}{a+b}$$

Hence,

$$EI \frac{d^2 y}{dx^2} = \frac{Wb}{(a+b)} x \quad 0 \leq x \leq a \text{ -----(1)}$$

$$EI \frac{d^2 y}{dx^2} = \frac{Wb}{(a+b)} x - W(x - a) \quad a \leq x \leq l \text{ -----(2)}$$

integrating (1) and (2) we get,

$$EI \frac{dy}{dx} = \frac{Wb}{2(a+b)} x^2 + k_1 \quad 0 \leq x \leq a \text{ -----(3)}$$

$$EI \frac{dy}{dx} = \frac{Wb}{2(a+b)} x^2 - \frac{W(x-a)^2}{2} + k_2 \quad a \leq x \leq l \text{ -----(4)}$$

B.M for the portion AB

$$M|_{AB} = R_1 \cdot x \quad 0 \leq x \leq a$$

B.M for the portion BC

$$M|_{BC} = R_1 \cdot x - W(x - a) \quad a \leq x \leq l$$

so the differential equation for the two cases would be,

$$EI \frac{d^2 y}{dx^2} = R_1 x$$

$$EI \frac{d^2 y}{dx^2} = R_1 x - W(x - a)$$

Boundary Conditions:

at $x = 0$; $y = 0$ in the portion AB i.e. $0 \leq x \leq a$

at $x = l$; $y = 0$ in the portion BC i.e. $a \leq x \leq l$

at $x = a$; dy/dx , the slope is same for both portion

at $x = a$; y , the deflection is same for both portion

Using Condition (c)

$$EI \frac{dy}{dx} = \frac{Wb}{2(a+b)} x^2 + k \quad 0 \leq x \leq a \text{-----(3)}$$

$$EI \frac{dy}{dx} = \frac{Wb}{2(a+b)} x^2 - \frac{W(x-a)^2}{2} + k \quad a \leq x \leq l \text{-----(4)}$$

Integrating again equation (3) and (4) we get

$$EIy = \frac{Wb}{6(a+b)} x^3 + kx + k_3 \quad 0 \leq x \leq a \text{-----(5)}$$

$$EIy = \frac{Wb}{6(a+b)} x^3 - \frac{W(x-a)^3}{6} + kx + k_4 \quad a \leq x \leq l \text{-----(6)}$$

Utilizing condition (a) in equation (5) yields

$$k_3 = 0$$

Utilizing condition (b) in equation (6) yields

$$0 = \frac{Wb}{6(a+b)} l^3 - \frac{W(l-a)^3}{6} + kl + k_4$$

$$k_4 = -\frac{Wb}{6(a+b)} l^3 + \frac{W(l-a)^3}{6} - kl$$

But $a+b=l$,

Thus,

$$k_4 = -\frac{Wb(a+b)^2}{6} + \frac{Wb^3}{6} - k(a+b)$$

Using Condition (d)

Therefore $y|_{\text{from equation 5}} = y|_{\text{from equation 6}}$

or

$$\frac{Wb}{6(a+b)}x^3 + kx + k_3 = \frac{Wb}{6(a+b)}x^3 - \frac{W(x-a)^3}{6} + kx + k_4$$

$$\frac{Wb}{6(a+b)}a^3 + ka + k_3 = \frac{Wb}{6(a+b)}a^3 - \frac{W(a-a)^3}{6} + ka + k_4$$

Thus, $k_4 = 0$;

OR

$$k_4 = -\frac{Wb(a+b)^2}{6} + \frac{Wb^3}{6} - k(a+b) = 0$$

$$k(a+b) = -\frac{Wb(a+b)^2}{6} + \frac{Wb^3}{6}$$

$$k = -\frac{Wb(a+b)}{6} + \frac{Wb^3}{6(a+b)}$$

so the deflection equations for each portion of the beam are

$$Ely = \frac{Wb}{6(a+b)}x^3 + kx + k_3$$

$$Ely = \frac{Wbx^3}{6(a+b)} - \frac{Wb(a+b)x}{6} + \frac{Wb^3x}{6(a+b)} \quad \text{-----for } 0 \leq x \leq a \text{----- (7)}$$

and for other portion

$$Ely = \frac{Wb}{6(a+b)}x^3 - \frac{W(x-a)^3}{6} + kx + k_4$$

Substituting the value of 'k' in the above equation

$$Ely = \frac{Wbx^3}{6(a+b)} - \frac{W(x-a)^3}{6} - \frac{Wb(a+b)x}{6} + \frac{Wb^3x}{6(a+b)} \quad \text{For for } a \leq x \leq l \text{----- (8)}$$

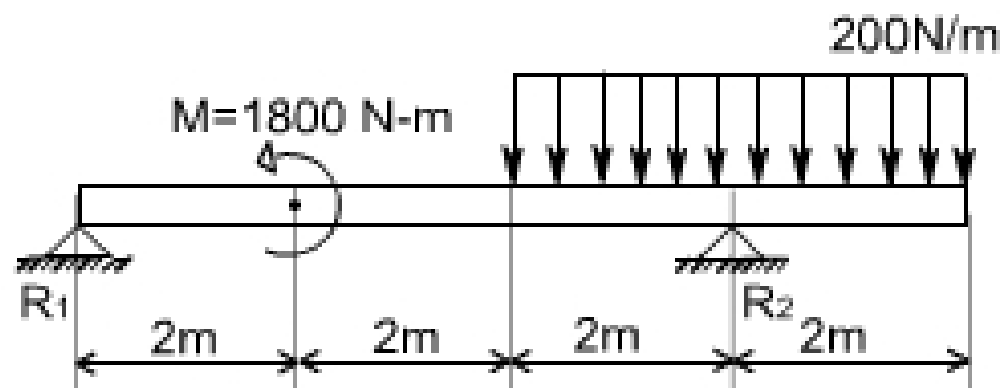
so either of the equation (7) or (8) may be used to find the deflection at $x = a$

hence substituting $x = a$ in either of the equation we get

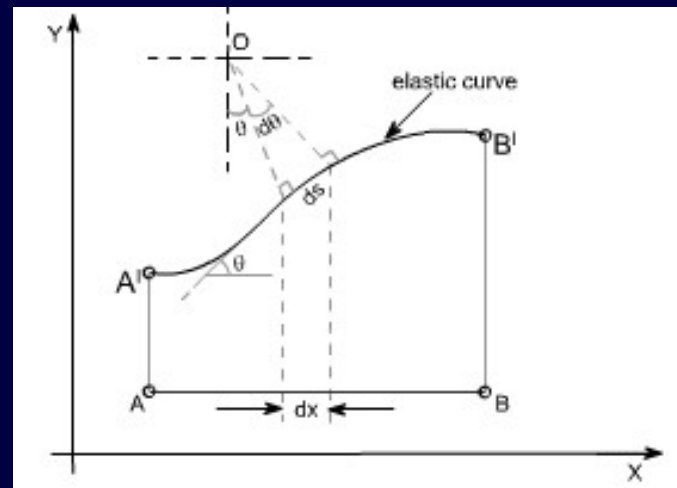
$$Y|_{x=a} = -\frac{Wa^2b^2}{3EI(a+b)}$$

OR if $a = b = l/2$

$$Y_{\text{max}} = -\frac{WL^3}{48EI}$$



THE AREA-MOMENT / MOMENT-AREA METHODS



$$\frac{d\theta}{ds} = \frac{1}{R} = \frac{M}{EI}$$

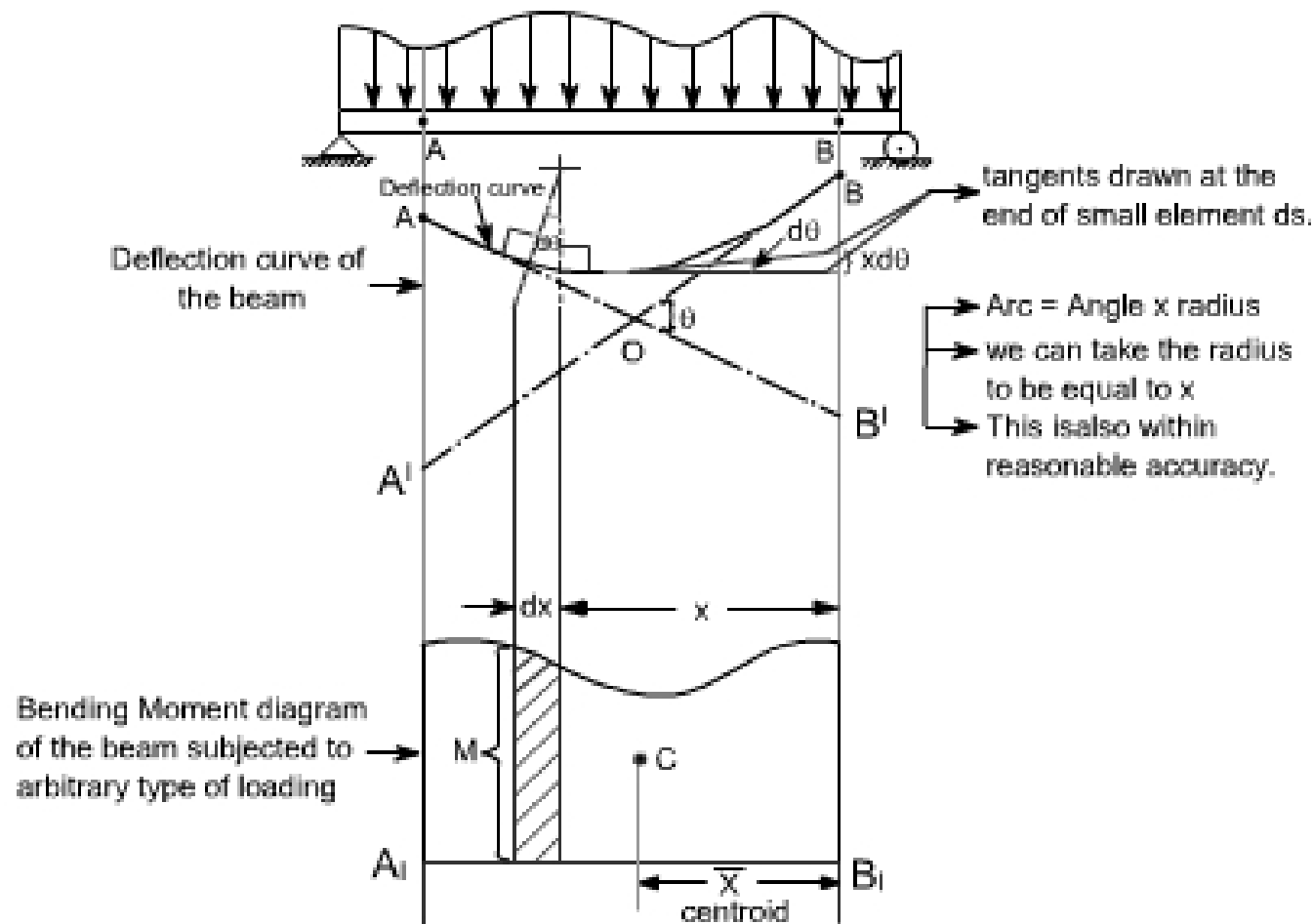
$$\frac{d\theta}{ds} = \frac{M}{EI}$$

But for small curvature [but θ is the angle, slope is $\tan\theta = \frac{dy}{dx}$ for small

angles $\tan\theta \approx \theta$, hence $\theta \approx \frac{dy}{dx}$ so we get $\frac{d^2y}{dx^2} = \frac{M}{EI}$ by putting $ds \approx dx$]

Hence,

$$\frac{d\theta}{dx} = \frac{M}{EI} \text{ or } \boxed{\frac{d\theta}{dx} = \frac{M}{EI}} \text{ ----- (1)}$$

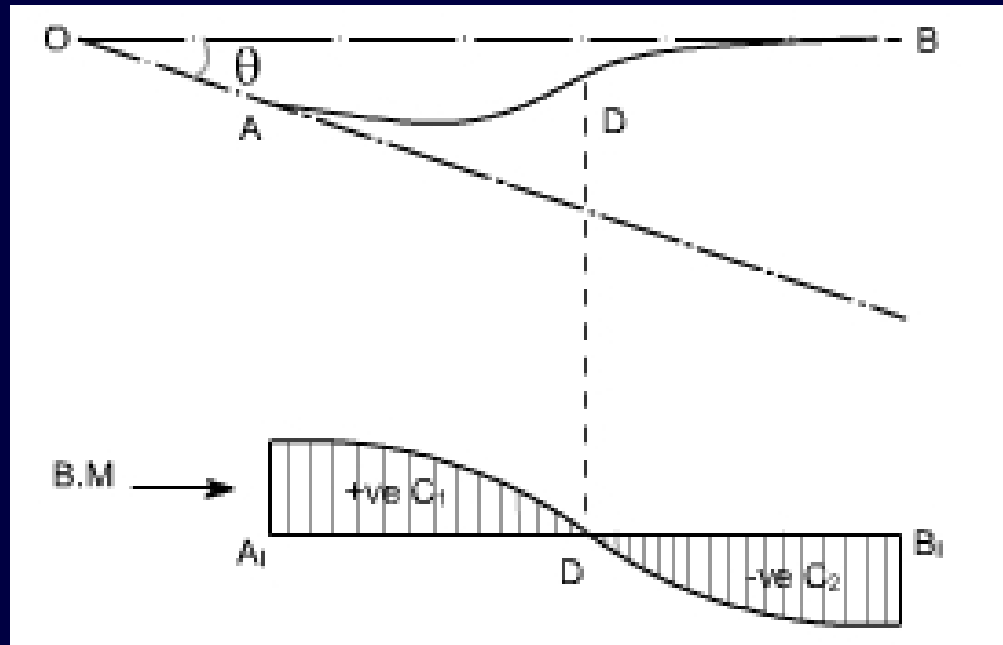


$$\theta = \int_A^B \frac{M dx}{EI} = \frac{1}{EI} \int_A^B M dx$$

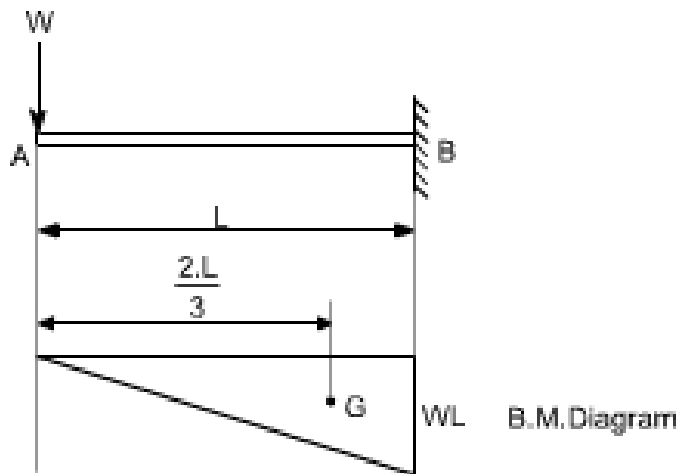
$$\left\{ \begin{array}{l} \text{slope or } \theta \\ \text{between any two points} \end{array} \right\} = \left\{ \frac{1}{EI} \times \text{area of B.M diagram between} \right. \\ \left. \text{corresponding portion of B.M diagram} \right\}$$

Theorem I: The angle θ between tangents at any two points A and B on the elastic line is equal to the total area of corresponding portion of the bending moment diagram, divided by EI.

$$\delta = \int_A^B x d\theta = \frac{1}{EI} \times \left\{ \begin{array}{l} \text{first moment of area with respect} \\ \text{to point B, of the total B.M diagram} \end{array} \right\}$$



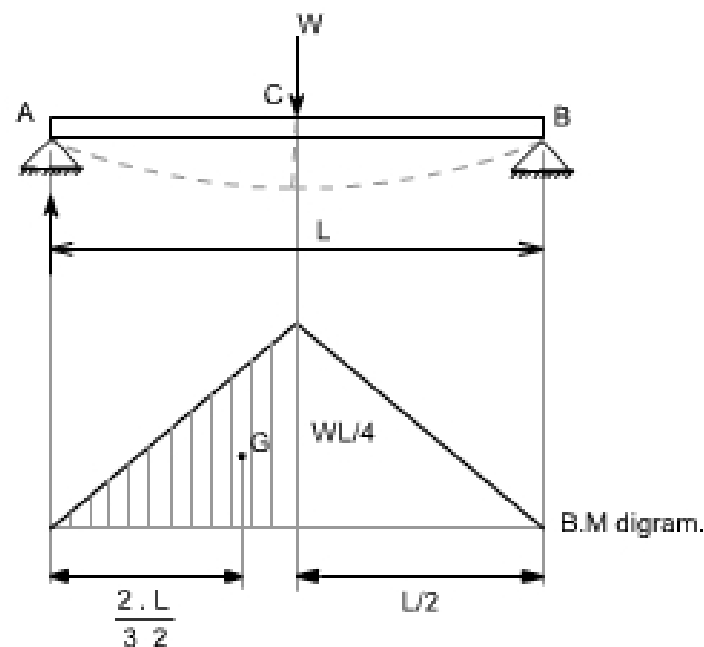
Theorem II: The deflection of B away from the tangent at A is equal to the statical moment, with respect to B, of the bending moment area between A and B, divided by EI.



$$\begin{aligned}
 \text{slope at A} &= \frac{1}{EI} [\text{Area of B.M diagram between the points A and B}] \\
 &= \frac{1}{EI} \left[\frac{1}{2} L \cdot WL \right] \\
 &= \frac{WL^2}{2EI}
 \end{aligned}$$

Thus,

$$\begin{aligned}
 \delta &= \frac{1}{EI} [\text{first moment of area of B.M diagram between A and B about A}] \\
 &= \frac{1}{EI} [A\bar{y}] \\
 &= \frac{1}{EI} \left[\left(\frac{1}{2} L \cdot WL \right) \frac{2}{3} L \right] \\
 &= \frac{WL^3}{3EI}
 \end{aligned}$$



slope at A = $\frac{1}{EI}$ [Area of B. M diagram between A and C]

$$= \frac{1}{EI} \left[\left(\frac{1}{2} \right) \left(\frac{L}{2} \right) \left(\frac{WL}{4} \right) \right] \quad \text{we are taking half area of the B.M because we have to work out this relative to a zero slope}$$

$$= \frac{WL^2}{16EI}$$

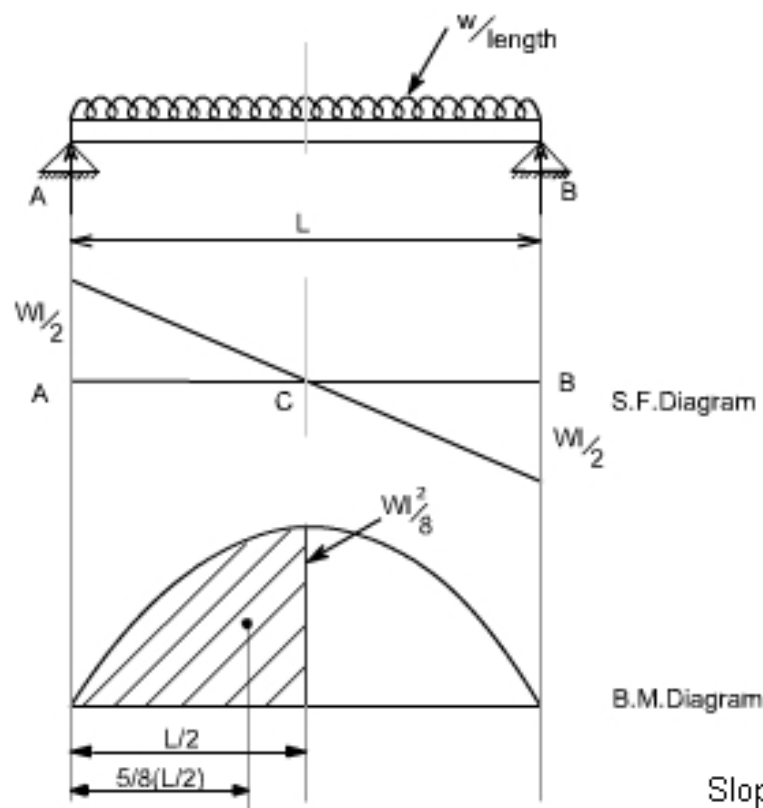
Deflection of A relative to C = central deflection of C

or

$$\delta_C = \frac{1}{EI} [\text{Moment of B.M diagram between points A and C about A}]$$

$$= \frac{1}{EI} \left[\left(\frac{1}{2} \right) \left(\frac{L}{2} \right) \left(\frac{WL}{4} \right) \frac{2}{3} L \right]$$

$$= \frac{WL^3}{48EI}$$



Slope at point C w.r.t point A = $\frac{1}{EI}$ [Area of B.M diagram between point A and C]

$$= \frac{1}{EI} \left[\left(\frac{2}{3} \right) \left(\frac{wL^2}{8} \right) \left(\frac{L}{2} \right) \right]$$

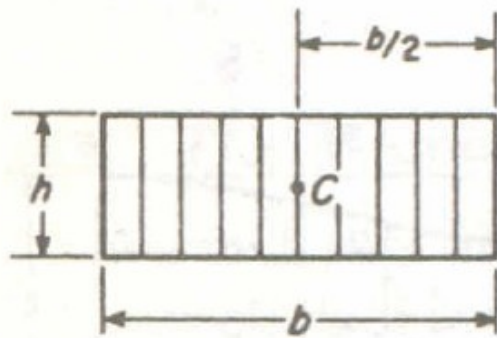
$$= \frac{wL^3}{24EI}$$

Deflection at point C
relative to A

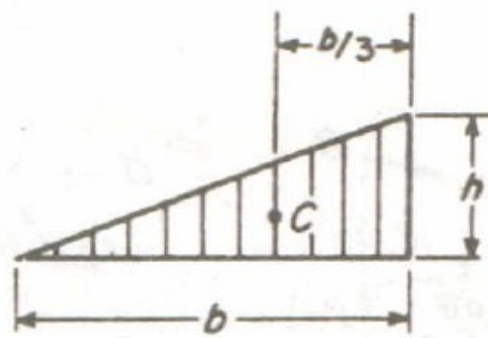
$$= \frac{1}{EI} [A \bar{y}]$$

$$= \frac{1}{EI} \left[\left(\frac{wL^3}{24} \right) \left(\frac{5}{8} \right) \left(\frac{L}{2} \right) \right]$$

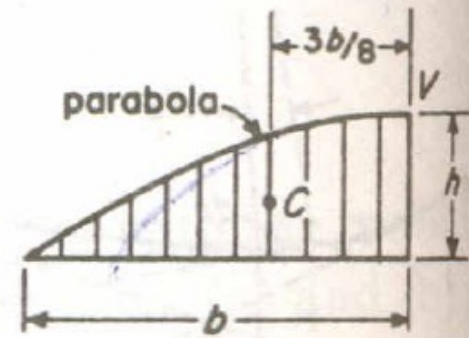
$$= \frac{5}{384EI} . wL^4$$



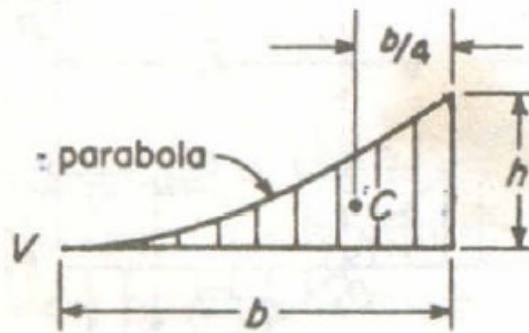
$$\text{Area} = bh$$



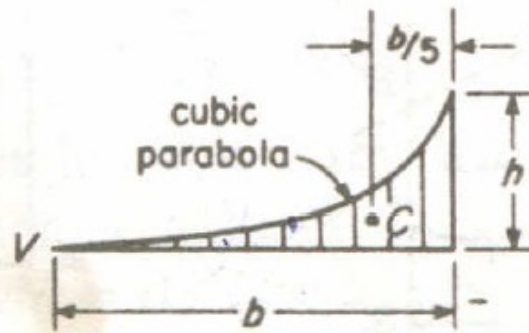
$$\text{Area} = bh/2$$



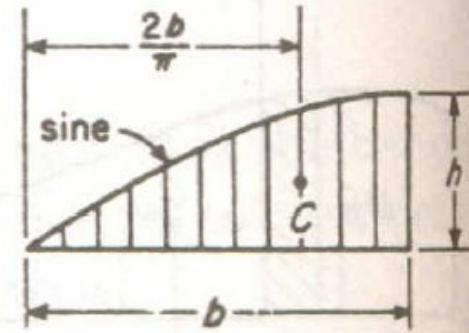
$$\text{Area} = 2bh/3$$



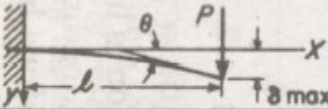
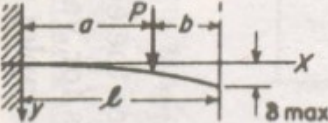
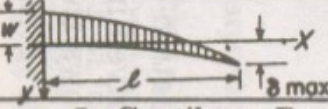
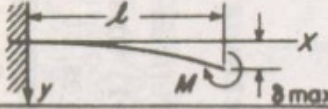
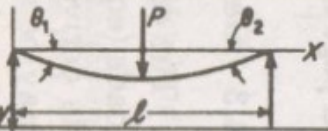
$$\text{Area} = bh/3$$

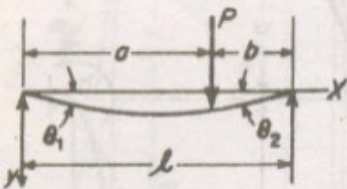
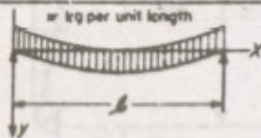
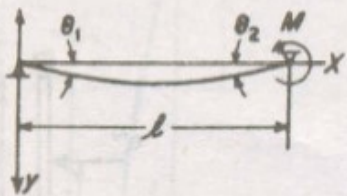


$$\text{Area} = bh/4$$



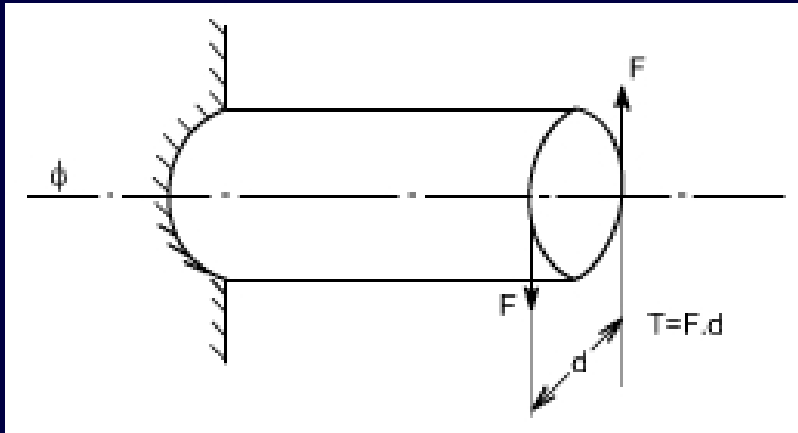
$$\text{Area} = 2bh/\pi$$

BEAM TYPE	SLOPE AT FREE END	DEFLECTION AT ANY SECTION IN TERMS OF x : y IS POSITIVE DOWNWARD	MAXIMUM DEFLECTION
1. Cantilever Beam — Concentrated load P at the free end			
	$\theta = \frac{Pl^2}{2EI}$	$y = \frac{Px^2}{6EI} (3l - x)$	$\delta_{\max} = \frac{Pl^3}{3EI}$
2. Cantilever Beam — Concentrated load P at any point			
	$\theta = \frac{Pa^2}{2EI}$	$y = \frac{Px^2}{6EI} (3a - x) \text{ for } 0 < x < a$ $y = \frac{Pa^2}{6EI} (3x - a) \text{ for } a < x < l$	$\delta_{\max} = \frac{Pa^2}{6EI} (3l - a)$
3. Cantilever Beam — Uniformly distributed load of w kg per unit length			
	$\theta = \frac{wl^3}{6EI}$	$y = \frac{wx^2}{24EI} (x^2 + 6l^2 - 4lx)$	$\delta_{\max} = \frac{wl^4}{8EI}$
4. Cantilever Beam — Uniformly varying load; maximum intensity w kg per unit length			
	$\theta = \frac{wl^3}{24EI}$	$y = \frac{wx^2}{120lEI} (10l^3 - 10l^2x + 5lx^2 - x^3)$	$\delta_{\max} = \frac{wl^4}{30EI}$
5. Cantilever Beam — Couple M applied at the free end			
	$\theta = \frac{Ml}{EI}$	$y = \frac{Mx^2}{2EI}$	$\delta_{\max} = \frac{Ml^2}{2EI}$
6. Beam Freely Supported at Ends — Concentrated load P at the center			
	$\theta_1 = \theta_2 = \frac{Pl^2}{16EI}$	$y = \frac{Px}{12EI} \left(\frac{3l^2}{4} - x^2 \right) \text{ for } 0 < x < \frac{l}{2}$	$\delta_{\max} = \frac{Pl^3}{48EI}$

BEAM TYPE	SLOPE AT ENDS	DEFLECTION AT ANY SECTION IN TERMS OF x : y IS POSITIVE DOWNWARD	MAXIMUM AND CENTER DEFLECTION
7. Beam Freely Supported at Ends — Concentrated load at any point			
	Left End. $\theta_1 = \frac{Pb(l^2 - b^2)}{6EI}$ Right End. $\theta_2 = \frac{Pab(2l - b)}{6EI}$	$y = \frac{Pbx}{6EI} (l^2 - x^2 - b^2) [0 < x < a]$ $y = \frac{Pb}{6EI} \left[\frac{l}{b}(x - a)^3 + (l^2 - b^2)x - x^3 \right] [a < x < l]$	$\delta_{\max} = \frac{Pb(l^2 - b^2)^{3/2}}{9\sqrt{3}lEI}$ at $x = \sqrt{\frac{l^2 - b^2}{3}}$ At center, if $a > b$ $\delta = \frac{Pb}{48EI}(3l^2 - 4b^2)$
8. Beam Freely Supported at Ends — Uniformly distributed load of w kg per unit length			
	$\theta_1 = \theta_2 = \frac{wl^3}{24EI}$	$y = \frac{wx}{24EI} (l^3 - 2lx^2 + x^3)$	$\delta_{\max} = \frac{5wl^4}{384EI}$
9. Beam Freely Supported at Ends — Couple M at the right end			
	$\theta_1 = \frac{Ml}{6EI}$ $\theta_2 = \frac{Ml}{3EI}$	$y = \frac{Mlx}{6EI} \left(1 - \frac{x^2}{l^2} \right)$	$\delta_{\max} = \frac{Ml^2}{9\sqrt{3}EI}$ at $x = l/\sqrt{3}$ At center $\delta = \frac{Ml^2}{16EI}$
10. Beam Freely Supported at Ends — Uniformly varying load: max. intensity w			
	$\theta_1 = \frac{7wl^3}{360EI}$ $\theta_2 = \frac{wl^3}{45EI}$	$y = \frac{wx}{360EI} (7l^4 - 10l^2x^2 + 3x^4)$	$\delta_{\max} = .00652 \frac{wl^4}{EI}$ at $x = 0.519l$ At center $\delta = .00651 \frac{wl^4}{EI}$

Members Subjected to Torsional Loads

Definition of Torsion: Consider a shaft rigidly clamped at one end and twisted at the other end by a torque $T = F.d$ applied in a plane perpendicular to the axis of the bar such a shaft is said to be in torsion.



Effects of Torsion: The effects of a torsional load applied to a bar are

- To impart an angular displacement of one end cross section with respect to the other end.
- To setup shear stresses on any cross section of the bar perpendicular to its axis.

GENERATION OF SHEAR STRESSES: The physical understanding of the phenomena of setting up of shear stresses in a shaft subjected to a torsion may be understood from the figures 1-3.

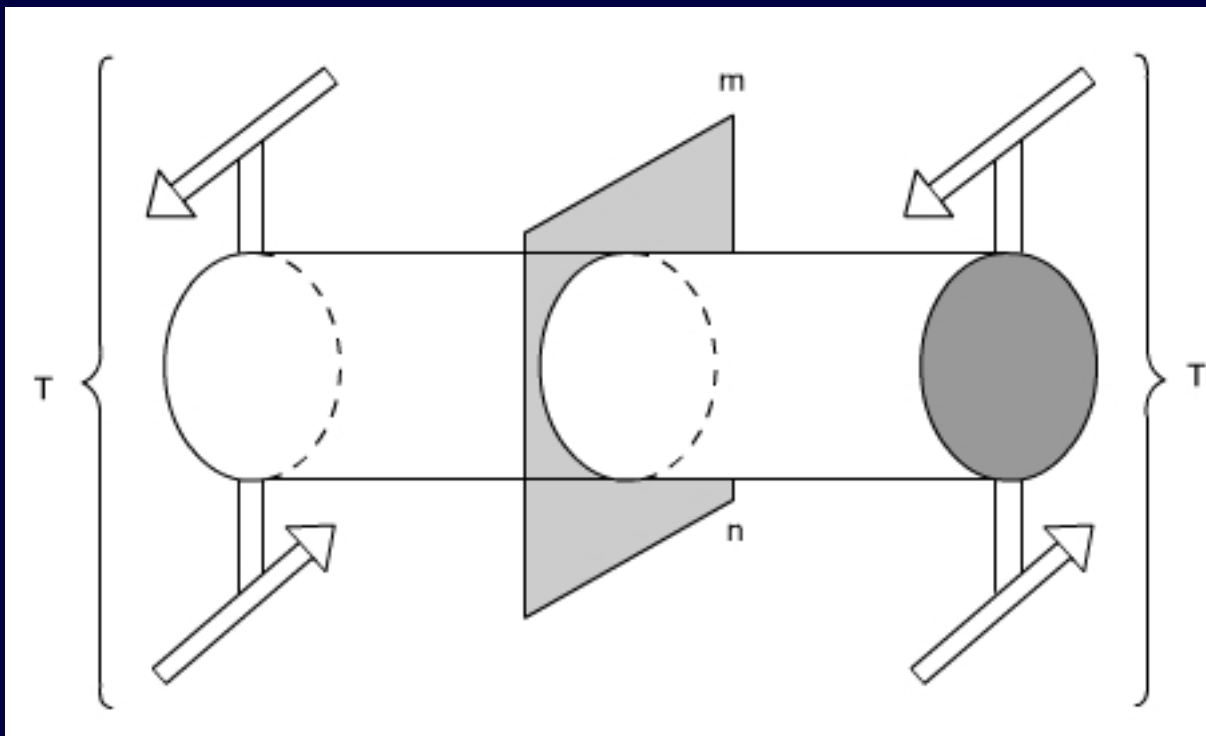
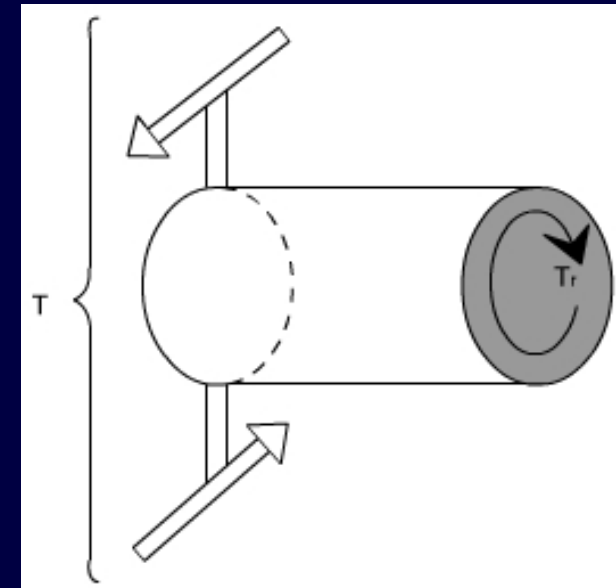


Fig 1: Here the cylindrical member or a shaft is in static equilibrium where T is the resultant external torque acting on the member. Let the member be imagined to be cut by some imaginary plane 'mn'.

Fig 2: When the plane 'mn' cuts remove the portion on R.H.S. and we get a fig 2. Now since the entire member is in equilibrium, therefore, each portion must be in equilibrium. Thus, the member is in equilibrium under the action of resultant external torque T and developed resisting Torque T_r .



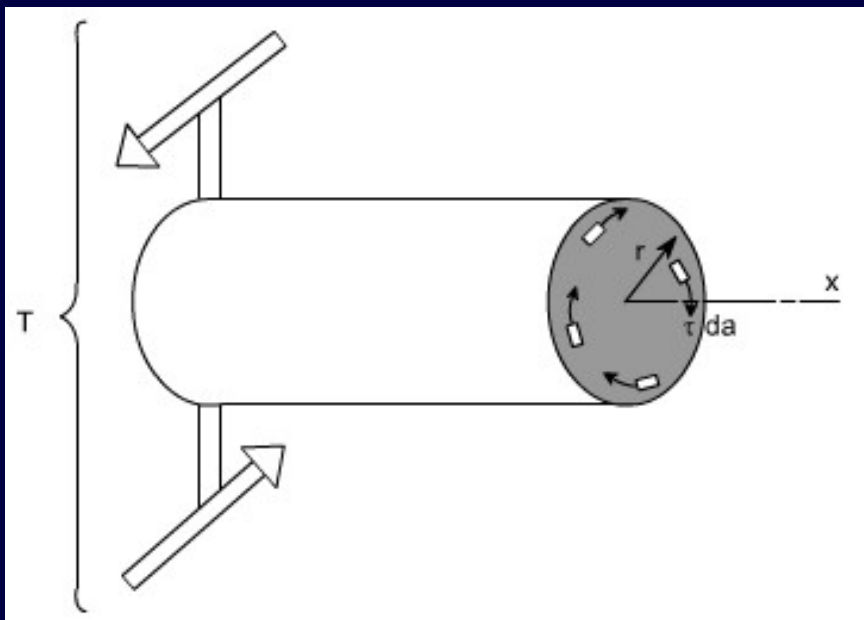
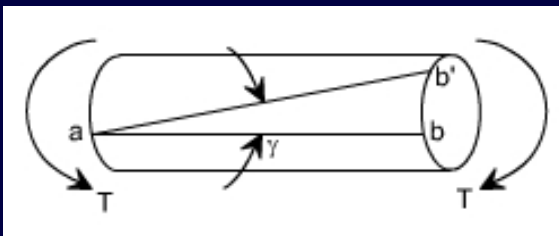


Fig 3: The Figure shows that how the resisting torque T_r is developed. The resisting torque T_r is produced by virtue of an infinitesimal shear forces acting on the plane perpendicular to the axis of the shaft. Obviously such shear forces would be developed by virtue of shear stresses.

Shaft: The shafts are the machine elements which are used to transmit power in machines.

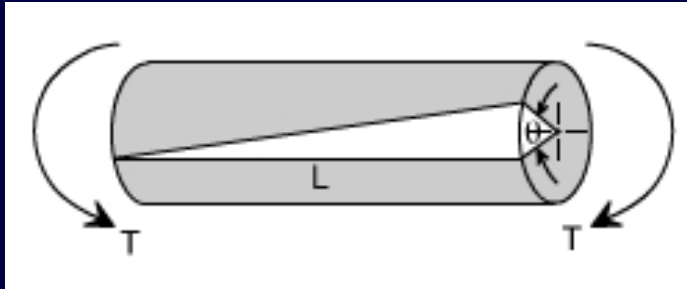
Twisting Moment: The twisting moment for any section along the bar / shaft is defined to be the algebraic sum of the moments of the applied couples that lie to one side of the section under consideration. The choice of the side in any case is of course arbitrary.

Shearing Strain: If a generator ab is marked on the surface of the unloaded bar, then after the twisting moment ' T ' has been applied this line moves to $a'b'$. The angle γ measured in radians, between the final and original positions of the generators is defined as the shearing strain at the surface of the bar or shaft. The same definition will hold at any interior point of the bar.



Modulus of Elasticity in shear: The ratio of the shear stress to the shear strain is called the modulus of elasticity in shear OR Modulus of Rigidity and is represented by the symbol

$$G = \frac{\tau}{\gamma}$$



Angle of Twist: If a shaft of length L is subjected to a constant twisting moment T along its length, then the angle θ through which one end of the bar will twist relative to the other is known as the angle of twist.

- Despite the differences in the forms of loading, we see that there are number of similarities between bending and torsion, including for example, a linear variation of stresses and strain with position.
- In torsion the members are subjected to moments (couples) in planes normal to their axes.
- For the purpose of designing a circular shaft to withstand a given torque, we must develop an equation giving the relation between twisting moment, maximum shear stress produced, and a quantity representing the size and shape of the cross-sectional area of the shaft.
- Many torque carrying engineering members are cylindrical in shape. Examples are drive shafts, bolts and screw drivers.

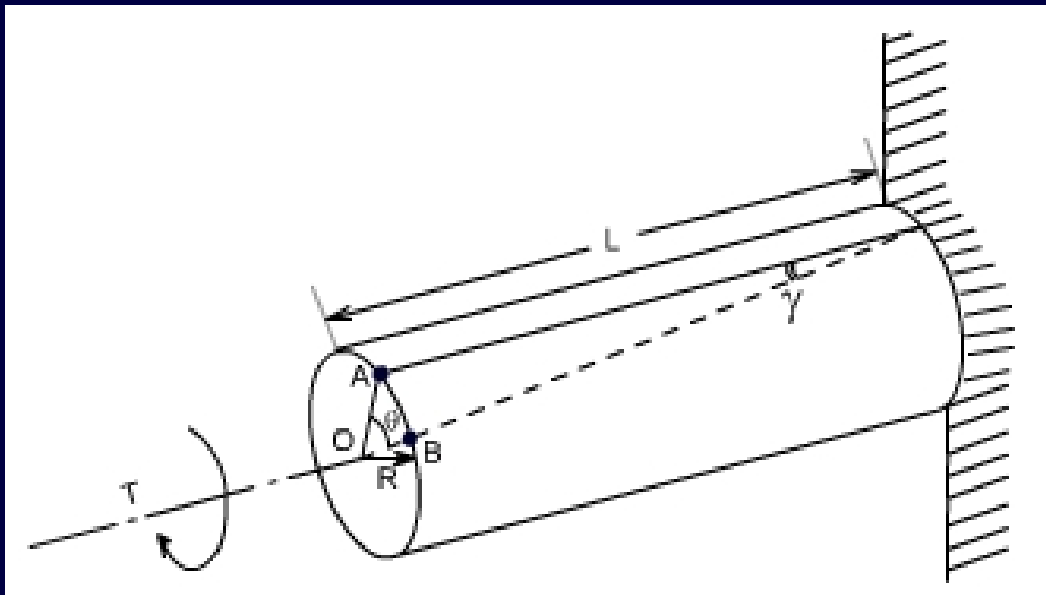
Relationship in Torsion

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G \cdot \theta}{L}$$

Torsion Theory or Development of Torsion Formula

Assumption:

- The material is homogenous i.e. of uniform elastic properties exists throughout the material.
- The material is elastic, follows Hook's law, with shear stress proportional to shear strain.
- The stress does not exceed the elastic limit.
- The circular section remains circular
- Cross section remain plane.
- Cross section rotate as if rigid i.e. every diameter rotates through the same angle.



Consider now the solid circular shaft of radius R subjected to a torque T at one end, the other end being fixed. Under the action of this torque a radial line at the free end of the shaft twists through an angle γ , point A moves to B , and AB subtends an angle ' γ ' at the fixed end. This is then the angle of distortion of the shaft i.e. the shear strain.

Since angle in radians = arc / Radius

arc AB = $R\theta = L \gamma$ [since L and γ also constitute the arc AB]

Thus, $\gamma = R\theta / L$

(1)

From the definition of Modulus of rigidity or Modulus of elasticity in shear

$$G = \frac{\text{shear stress}(\tau)}{\text{shear strain}(\gamma)}$$

where γ is the shear strain set up at radius R.

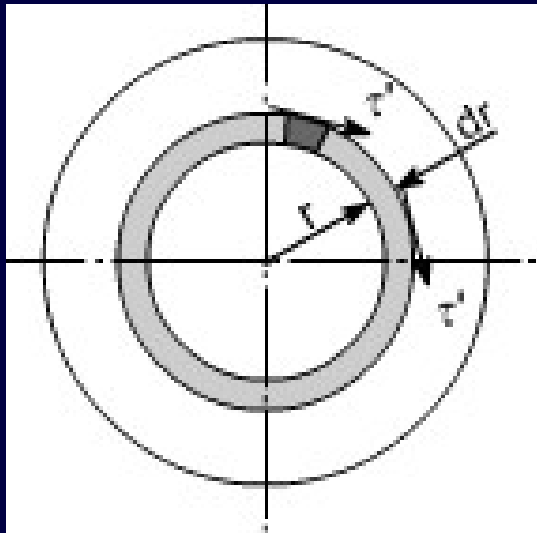
$$\text{Then } \frac{\tau}{G} = \gamma$$

Equating the equations (1) and (2) we get $\frac{R\theta}{L} = \frac{\tau}{G}$

$$\frac{\tau}{R} = \frac{G\theta}{L} \left(= \frac{\tau'}{r} \right) \text{ where } \tau' \text{ is the shear stress at any radius } r.$$

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G.\theta}{L}$$

Stresses: Let us consider a small strip of radius r and thickness dr which is subjected to shear stress τ .



The force set up on each element = stress $\times$ area = $\tau \times 2\pi r dr$ (approximately)

This force will produce a moment or torque about the center axis of the shaft
 $= \tau \times 2\pi r dr \cdot r = \tau \times 2\pi r^2 dr$

The total torque T on the section, will be the sum of all the contributions

$$T = \int_0^R 2\pi \tau r^2 dr$$

$$\text{i.e. } \tau' = \frac{G\theta r}{L}$$

$$\text{we get } T = \int_0^R 2\pi \frac{G\theta}{L} r^3 dr$$

$$T = \frac{2\pi G\theta}{L} \int_0^R r^3 dr$$

$$= \frac{2\pi G\theta}{L} \left[\frac{R^4}{4} \right]_0^R$$

$$= \frac{G\theta}{L} \cdot \frac{2\pi R^4}{4}$$

$$= \frac{G\theta}{L} \cdot \frac{\pi R^4}{2}$$

$$= \frac{G\theta}{L} \left[\frac{\pi d^4}{32} \right] \text{ now substituting } R = d/2$$

$$= \frac{G\theta}{L} J$$

$$\text{since } \frac{\pi d^4}{32} = J \text{ the polar moment of inertia}$$

$$\text{or } \frac{T}{J} = \frac{G\theta}{L} \quad \dots\dots(2)$$

if we combine the equation no.(1) and (2) we get $\boxed{\frac{T}{J} = \frac{\tau'}{r} = \frac{G\theta}{L}}$

Tensional Stiffness: The tensional stiffness k is defined as the torque per radius twist
i.e, $k = T / \theta = GJ / L$

J = Polar moment of Inertia

$$= \frac{\pi d^4}{32} \text{ for solid shaft}$$

$$= \frac{\pi(D^4 - d^4)}{32} \text{ for a hollow shaft.}$$

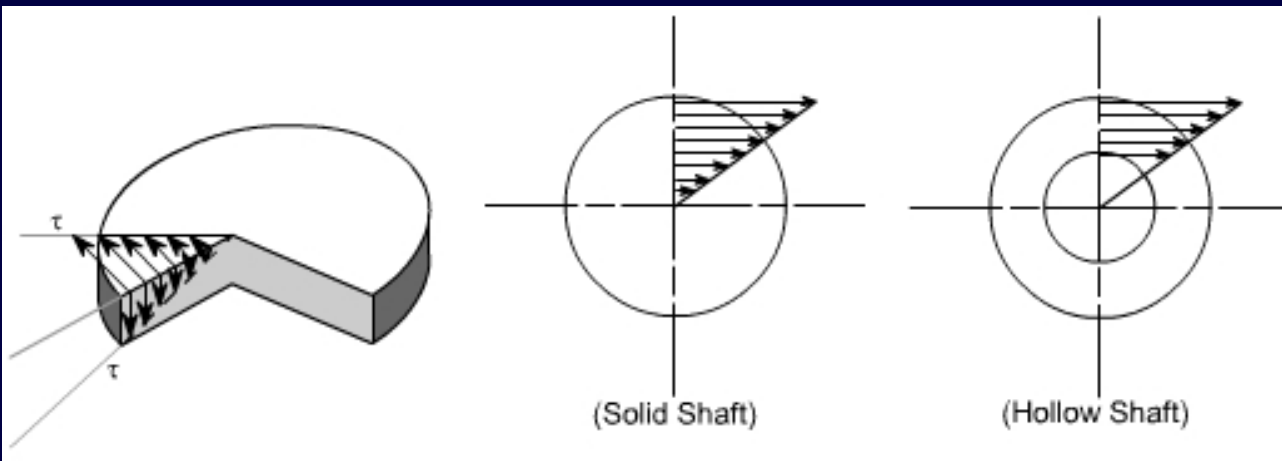
Distribution of
shear stresses in
circular Shafts
subjected to torsion

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

or

$$\tau = \frac{G\theta \cdot r}{L}$$

This states that the shearing stress varies directly as the distance 'r' from the axis of the shaft and the following is the stress distribution in the plane of cross section and also the complementary shearing stresses in an axial plane.



$$\frac{\tau}{r} = \frac{T}{J}$$

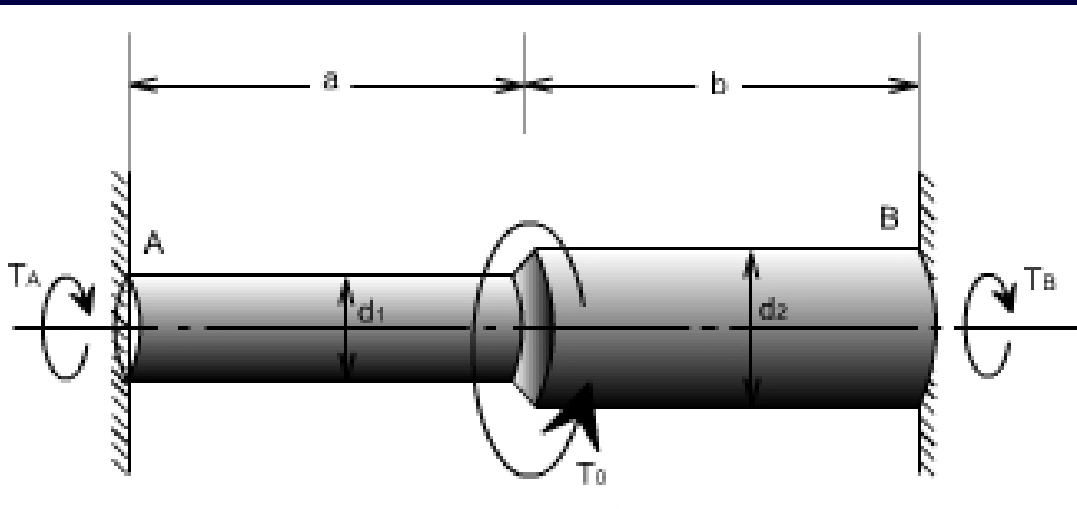
$$\tau_{\max} \Big|_{r=d/2} = \frac{T \cdot R}{J} = \frac{T}{\frac{\pi d^4}{32}} \cdot \frac{d}{2}$$

where d = diameter of solid shaft

$$\text{or } \tau_{\max} = \frac{16T}{\pi d^3}$$

Hence the maximum shear stress occurs on the outer surface of the shaft where $r = R$

Problem 1: A stepped solid circular shaft is built in at its ends and subjected to an externally applied torque. T_0 at the shoulder as shown in the figure. Determine the angle of rotation θ_0 of the shoulder section where T_0 is applied?



$$\text{Thus } T_A + T_B = T_0 \text{ ----- (1)}$$

$$\theta_a = \theta_b = \theta_0$$

$$\frac{T}{J} = \frac{G \cdot \theta}{l}$$

$$\text{or } \theta_A = \frac{T_A a}{J_A G}$$

$$\theta_B = \frac{T_B b}{J_B G}$$

$$\Rightarrow \frac{T_A a}{J_A G} = \frac{T_B b}{J_B G} = \theta_0 \quad \text{or} \quad \frac{T_A}{T_B} = \frac{J_A}{J_B} \cdot \frac{b}{a} \quad (2)$$

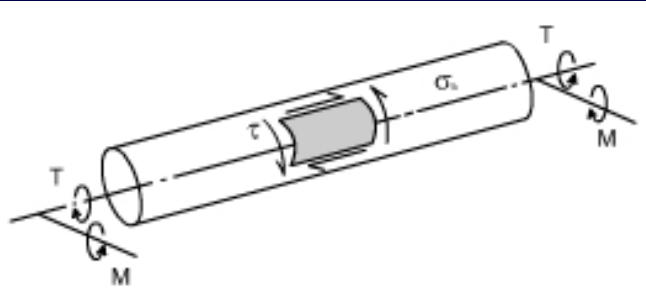
$$T_A = \frac{T_0}{1 + \frac{J_B a}{J_A b}}$$

$$T_B = \frac{T_0}{1 + \frac{J_A b}{J_B a}}$$

Using either of these values in (2) we have the angle of rotation θ_0 at the junction

$$\theta_0 = \frac{T_0 \cdot a \cdot b}{[J_A \cdot b + J_B \cdot a] G}$$

Members Subjected to Combined Loads



$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$$\sigma_b = \frac{M.y}{I}$$

$$\sigma_b|_{\max^m} = \frac{M}{I} \cdot y_{\max^m}$$

$$\begin{aligned} \sigma_b|_{\max^m} &= \frac{M}{\pi d^4 / 64} \cdot \frac{d}{2} \\ \sigma_b|_{\max^m} &= \frac{32M}{\pi d^3} \end{aligned} \quad (1)$$

$$\begin{aligned} \frac{T}{J} &= \frac{\tau'}{r} = \frac{G.\theta}{L} \\ \Rightarrow \frac{\tau'}{r} &= \frac{T}{J} \end{aligned}$$

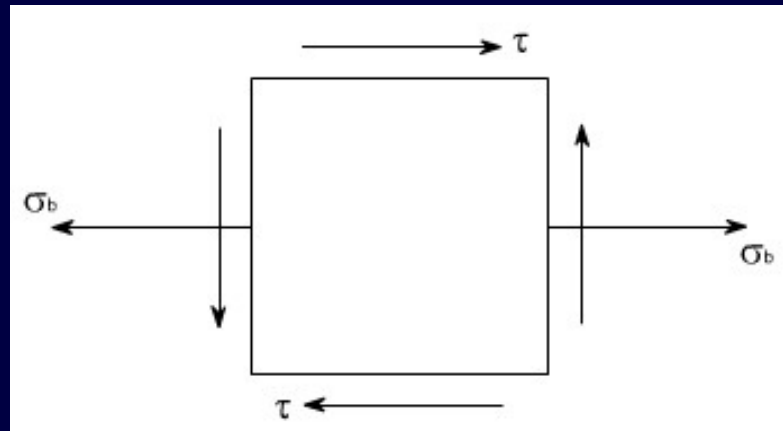
$$\text{Thus } \tau_{\max^m} = \frac{T}{J} \cdot \frac{d}{2}$$

$$J = \frac{\pi d^4}{32}$$

substituting the value of J, we get

$$\tau_{\max^m} = \frac{16T}{\pi d^3} \quad (2)$$

This can now be treated as the two dimensional stress system in which the loading in a vertical plane is zero i.e. $\sigma_y = 0$ and $\sigma_x = \sigma_b$ and is shown below:



$$\sigma_1, \sigma_2 = \left(\frac{\sigma_x + \sigma_y}{2} \right) \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

or

$$\begin{aligned} \sigma_1 &= \frac{\sigma_b}{2} + \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau_{\max}^2} \\ &= \frac{32M}{\pi d^3 \cdot 2} + \frac{1}{2} \sqrt{\left(\frac{32M}{\pi d^3} \right)^2 + 4 \left(\frac{16T}{\pi d^3} \right)^2} \\ &= \frac{16M}{\pi d^3 \cdot 2} + \frac{1}{2} \sqrt{\left(\frac{32M}{\pi d^3} \right)^2 + \left(\frac{2 \cdot 16T}{\pi d^3} \right)^2} \\ &= \frac{16}{\pi d^3} \left[M + \sqrt{M^2 + T^2} \right] \end{aligned}$$

Equivalent Bending Moment

$$\sigma_1 = \frac{32M_e}{\pi d^3}$$

Futher

$$\sigma_1 = \frac{16}{\pi d^3} \left[M + \sqrt{M^2 + T^2} \right]$$

Thus, equating the two we get

$$M_e = \frac{1}{2} \left[M + \sqrt{M^2 + T^2} \right]$$

Equivalent Torque

$$\sigma_1, \sigma_2 = \frac{16}{\pi d^3} \left\{ M \pm \sqrt{M^2 + T^2} \right\}$$

$$\text{or } \sigma_1 = \frac{16}{\pi d^3} \left[\left\{ M + \sqrt{M^2 + T^2} \right\} \right]$$

$$\sigma_2 = \frac{16}{\pi d^3} \left[\left\{ M - \sqrt{M^2 + T^2} \right\} \right]$$

$$\text{As } \tau_{\max} = \frac{\sigma_1 - \sigma_2}{2}$$

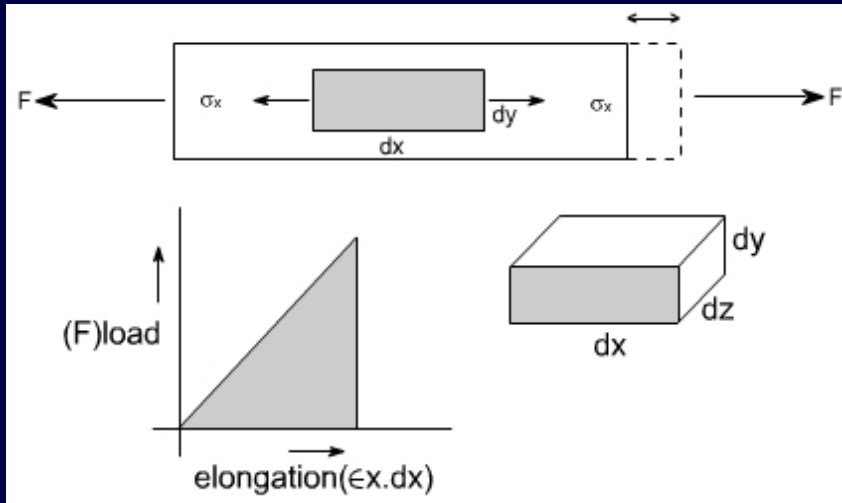
$$\text{so } \tau_{\max} = \frac{16}{\pi d^3} \left[\left\{ M + \sqrt{M^2 + T^2} \right\} \right] - \frac{16}{\pi d^3} \left[\left\{ M - \sqrt{M^2 + T^2} \right\} \right] \bigg/ 2$$

$$\tau_{\max} = \frac{16}{\pi d^3} \sqrt{M^2 + T^2} = \frac{16}{\pi d^3} \cdot T_e$$

$$T_e = \sqrt{M^2 + T^2}$$

THEORIES OF ELASTIC FAILURE

Strain Energy (U), is the energy stored within a material when the material is deformed. It is the area covered under the triangle as shown below.



$$\begin{aligned}
 U &= \frac{1}{2} F \cdot \epsilon_x \cdot dx \\
 &= \frac{1}{2} \sigma_x \cdot dy \cdot dz \cdot dx \cdot \epsilon_x \\
 &= \frac{1}{2} \sigma_x \cdot \epsilon_x \cdot dx \cdot dy \cdot dz \\
 &= \frac{1}{2} \sigma_x \left(\frac{\sigma_x}{E} \right) \cdot dx \cdot dy \cdot dz
 \end{aligned}$$

$$\frac{U}{\text{volume}} = \frac{1}{2} \frac{\sigma_x^2}{E}$$

Different Theories of Failure:

- Maximum Principal stress theory (due to Rankine)
- Maximum shear stress theory (Guest - Tresca)
- Maximum Principal strain (Saint - venant) Theory
- Total strain energy per unit volume (Haigh) Theory
- Shear strain energy per unit volume Theory (Von Mises & Hencky)

σ_{yp} = stress at the yield point in the simple tensile test.

$\sigma_1, \sigma_2, \sigma_3$ - the three principal stresses in the three dimensional complex state of stress systems in order of magnitude.

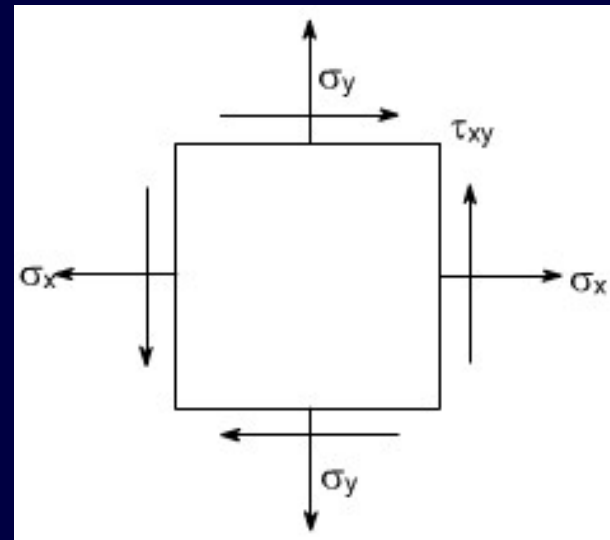
(a) Maximum Principal stress theory

This theory assume that when the maximum principal stress in a complex stress system reaches the elastic limit stress in a simple tension, failure will occur.

Therefore the criterion for failure would be $\sigma_1 = \sigma_{yp}$

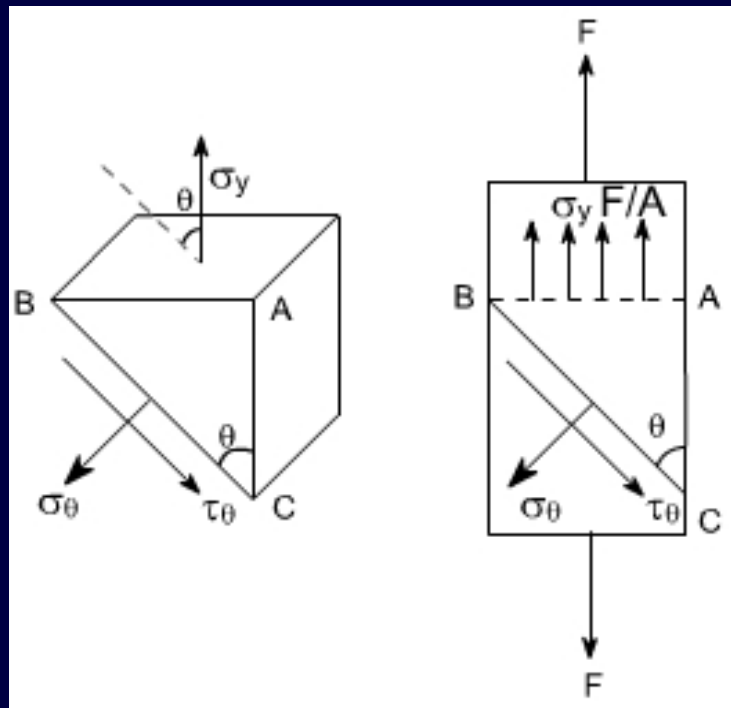
For a two dimensional complex stress system σ_1 is expressed as

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$
$$= \sigma_{yp}$$



(b) Maximum shear stress theory

This theory states that the failure can be assumed to occur when the maximum shear stress in the complex stress system is equal to the value of maximum shear stress in simple tension.



$$\sigma_{\theta} = \sigma_y \sin^2 \theta$$

$$\tau_{\theta} = \frac{1}{2} \sigma_y \sin 2\theta$$

$$\tau_{\theta} |_{\max} = \frac{1}{2} \sigma_y \quad \text{or}$$

$$\tau_{\max} = \frac{1}{2} \sigma_{yp}$$

whereas for the two dimensional complex stress system

$$\tau_{\max} = \left(\frac{\sigma_1 - \sigma_2}{2} \right)$$

where σ_1 = maximum principle stress

σ_2 = minimum principal stress

$$\text{so} \quad \frac{\sigma_1 - \sigma_2}{2} = \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2_{xy}}$$

$$\frac{\sigma_1 - \sigma_2}{2} = \frac{1}{2} \sigma_{yp} \Rightarrow \sigma_1 - \sigma_2 = \sigma_y$$

$$\Rightarrow \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2_{xy}} = \sigma_{yp}$$

becomes the criterion for the failure.

(c) Maximum Principal strain theory

This Theory assumes that failure occurs when the maximum strain for a complex state of stress system becomes equals to the strain at yield point in the tensile test for the three dimensional complex state of stress system.

For a 3 - dimensional state of stress system the total strain energy U_t per unit volume is equal to the total work done by the system and given by the equation

$$U_t = 1/2\sigma_1 \epsilon_1 + 1/2\sigma_2 \epsilon_2 + 1/2\sigma_3 \epsilon_3$$

substituting the values of ϵ_1, ϵ_2 and ϵ_3

$$\epsilon_1 = \frac{1}{E} [\sigma_1 - \gamma(\sigma_2 + \sigma_3)]$$
$$\epsilon_2 = \frac{1}{E} [\sigma_2 - \gamma(\sigma_1 + \sigma_3)]$$
$$\epsilon_3 = \frac{1}{E} [\sigma_3 - \gamma(\sigma_1 + \sigma_2)]$$

Thus, the failure criterion becomes

$$\left(\frac{\sigma_1}{E} - \gamma \frac{\sigma_2}{E} - \gamma \frac{\sigma_3}{E} \right) = \frac{\sigma_{yp}}{E}$$

or

$$\boxed{\sigma_1 - \gamma\sigma_2 - \gamma\sigma_3 = \sigma_{yp}}$$

(d) Total strain energy per unit volume theory

The theory assumes that the failure occurs when the total strain energy for a complex state of stress system is equal to that at the yield point a tensile test.

$$\frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] = \frac{\sigma_{yp}^2}{2E}$$
$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) = \sigma_{yp}^2$$

(e) Maximum shear strain energy per unit volume theory

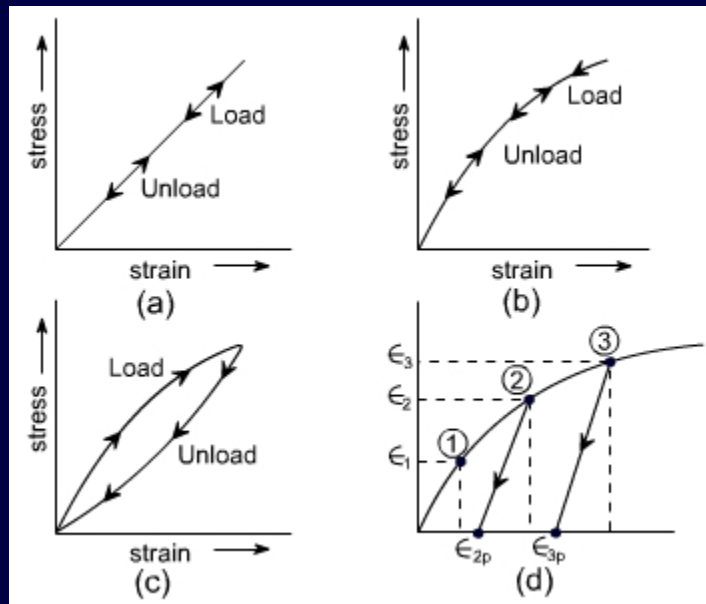
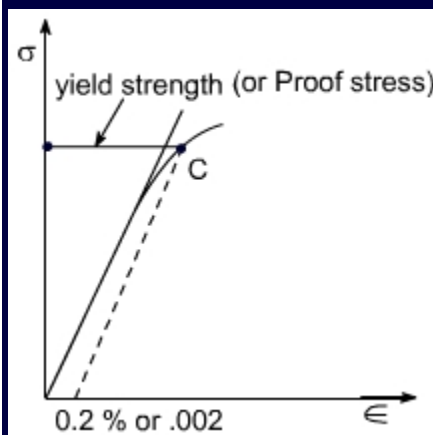
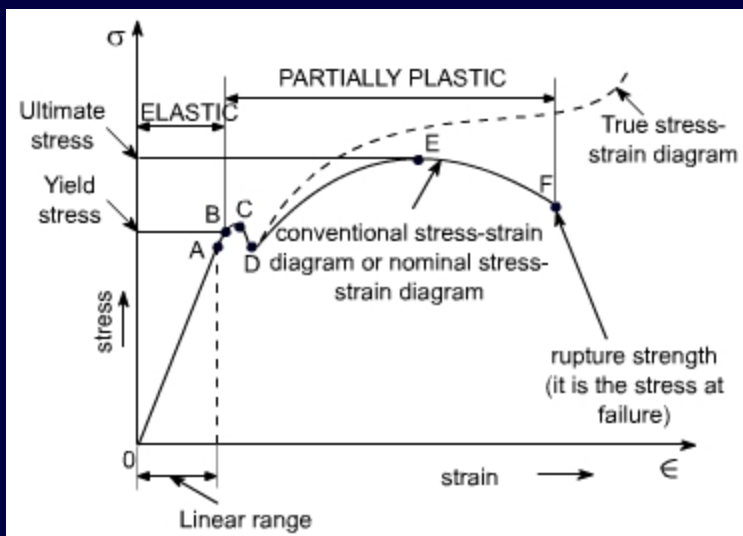
This theory states that the failure occurs when the maximum shear strain energy component for the complex state of stress system is equal to that at the yield point in the tensile test.

$$\frac{1}{12G} \left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right] = \frac{\sigma_{yp}^2}{6G}$$

Where G = shear modulus of rigidity

$$\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right] = 2\sigma_{yp}^2$$

Elastic Action: The elastic is an adjective meaning capable of recovering size and shape after deformation. **Elastic range** is the range of stress below the elastic limit.



- Many engineering materials behave as indicated in Fig(a) however, some behaves as shown in figures in (b) and (c) while in elastic range.
- When a material behaves as in (c), the σ vs ϵ is not single valued since the strain corresponding to any particular ' σ ' will depend upon loading history.
- Fig (d): It illustrates the idea of elastic and plastic strain. If a material is stressed to level (1) and then released the strain will return to zero beyond this plastic deformation remains.
- If a material is stressed to level (2) and then released, the material will recover the amount $(\epsilon_2 - \epsilon_{2p})$, where ϵ_{2p} is the plastic strain remaining after the load is removed. Similarly for level (3) the plastic strain will be ϵ_{3p} .

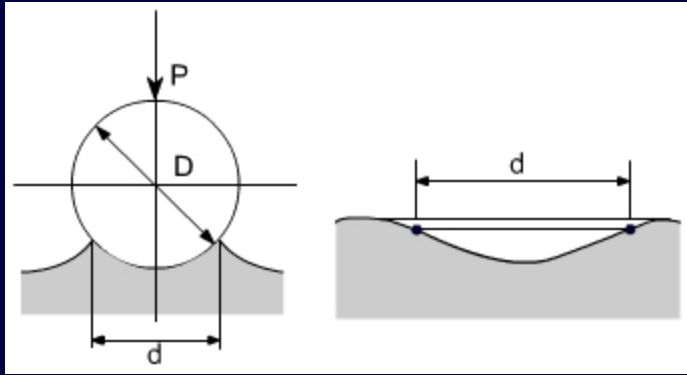
- ❑ **Ductile Materials:** If we just examine the earlier tension curve one can notice that the extension of the materials over the plastic range is considerably in excess of that associated with elastic loading. The Capacity of materials to allow these large deformations or large extensions without failure is termed as **ductility**. The materials with high ductility are termed as ductile materials.
- ❑ **Brittle Materials:** A brittle material is one which exhibits a relatively small extensions or deformations to fracture, so that the partially plastic region of the tensile test graph is much reduced.

Conditions Affecting Mechanical Properties:

- ❖ It has been established that lowering the temperature or increasing the rate of deformation considerably increases the resistance to plastic deformation. Thus, at low temperature (or higher rates of deformation), metals and alloys, which are ductile at normal room temperature may fail with brittle fracture.
- ❖ Notches i.e., sharp changes in cross sections have a great effect on the mechanical properties of the metals. A Notch will cause a non uniform distribution of stresses. They will always contribute lowering the ductility of the materials. A notch reduces the ultimate strength of the high strength materials. Because of the non uniform distribution of the stress or due to stress concentration.
- ❖ **Grain Size:** The grain size also affects the mechanical properties.

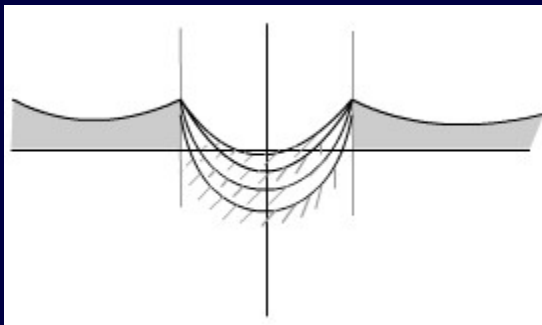
Hardness: Hardness is the resistance of a metal to the penetration of another harder body which does not receive a permanent set.

Hardness Tests consists in measuring the resistance to plastic deformation of layers of metals near the surface of the specimen i.e. there are Ball indentation Tests



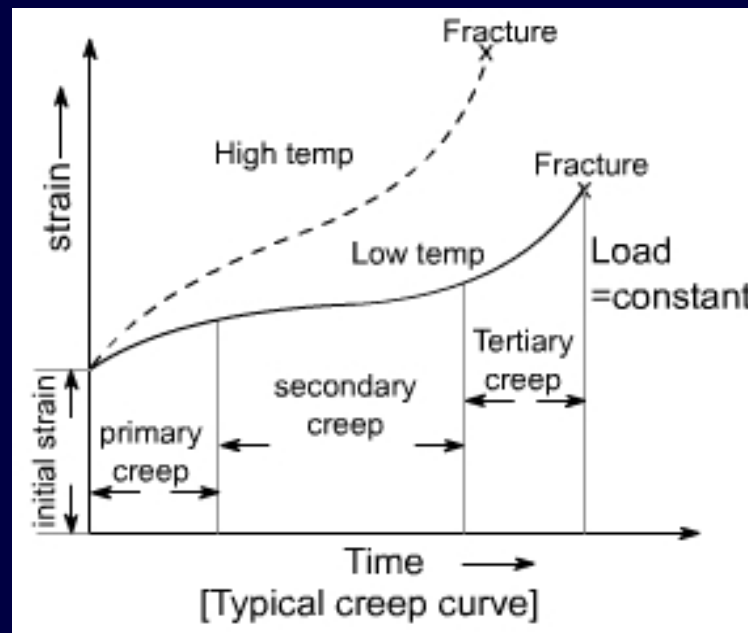
$$Bhn = \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})}$$

Disadvantage of Brinell Hardness Test: The main disadvantage of the Brinell Hardness test is that the Brinell hardness number is not independent of the applied load. This can be realized from. Considering the geometry of indentations for increasing loads. As the ball is pressed into the surface under increasing load the geometry of the indentation changes.



Toughness: It is defined as the ability of the material to withstand crack i.e to prevent the transfer or propagation of cracks across its section hence causing failures. Cracks are propagated due to stress concentration.

Creep: Creep is the gradual increase of plastic strain in a material with time at constant load. Particularly at elevated temperatures some materials are susceptible to this phenomena and even under the constant load, mentioned strains can increase continually until fractures. This form of failure is particularly relevant to the turbines blades, nuclear reactors, furnaces rocket motors etc.



Closed Coiled helical springs subjected to axial loads

Definition: A spring may be defined as an elastic member whose primary function is to deflect or distort under the action of applied load; it recovers its original shape when load is released.

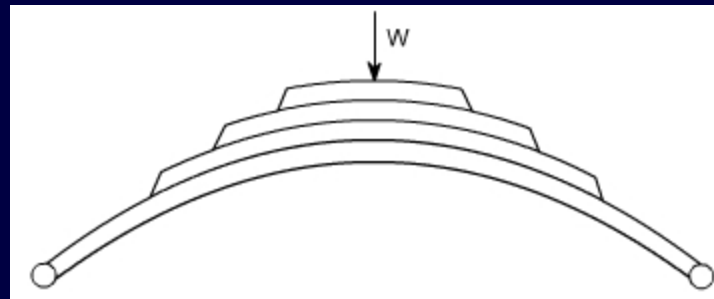
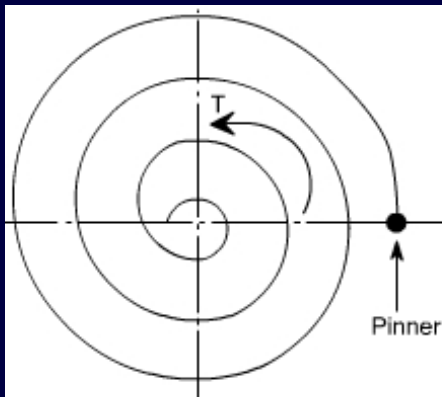
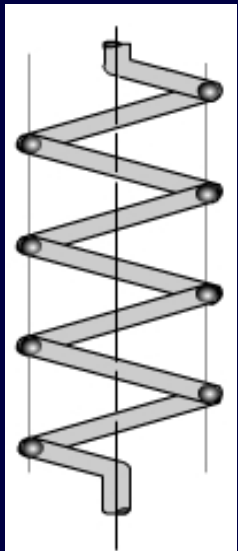
or

Springs are energy absorbing units whose function is to store energy and to restore it slowly or rapidly depending on the particular application.

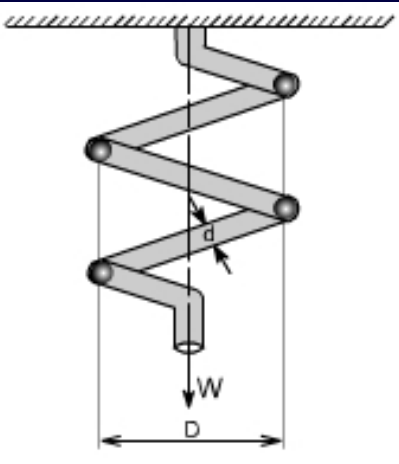
(i) helical spring: They are made of wire coiled into a helical form, the load being applied along the axis of the helix. In these type of springs the major stresses is torsional shear stress due to twisting. They are both used in tension and compression.

(ii) Spiral springs: They are made of flat strip of metal wound in the form of spiral and loaded in torsion. In this the major stresses are tensile and compression due to bending.

(iii) Leaf springs: They are composed of flat bars of varying lengths clamped together so as to obtain greater efficiency. Leaf springs may be full elliptic, semi elliptic or cantilever types, In these type of springs the major stresses which come into picture are tensile & compressive.

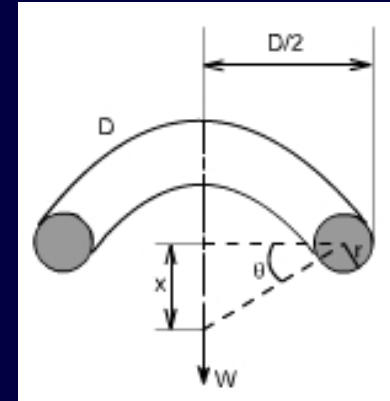


Derivation of the Formula: In order to derive a necessary formula which governs the behaviour of springs, consider a closed coiled spring subjected to an axial load W.



Let,
 W = axial load
 D = mean coil diameter
 d = diameter of spring wire
 n = number of active coils
 C = spring index = D / d For circular wires
 l = length of spring wire

G = modulus of rigidity
 x = deflection of spring
 θ = Angle of twist



$$\frac{T}{J} = \frac{\tau}{r} = \frac{G \cdot \theta}{l}$$

and substituting $J = \frac{\pi d^4}{32}$; $T = w \cdot \frac{d}{2}$

$$\theta = \frac{2 \cdot x}{D}; l = \pi D \cdot n$$

SPRING DEFLECTION

$$\frac{W \cdot D / 2}{\pi d^4 / 32} = \frac{G \cdot 2x / D}{\pi D n}$$

$$\Rightarrow x = \frac{8WD^3 n}{Gd^4}$$

Spring stiffness: The stiffness is defined as the load per unit deflection therefore

$$k = \frac{W}{x} = \frac{W}{\frac{8w \cdot D^3 \cdot n}{G \cdot d^4}}$$

Therefore

$$k = \frac{G \cdot d^4}{8 \cdot D^3 \cdot n}$$

Shear stress

$$\frac{W \cdot D / 2}{\pi d^4 / 32} = \frac{\tau''}{d / 2}$$

$$\Rightarrow \tau'' = \frac{8WD}{\pi d^3}$$

Shearing Stress due to compressive load W

$$\tau' = \frac{4W}{\pi d^2}$$

Therefore, the combined load $\tau = \tau' + \tau'' = \frac{4W}{\pi d^2} + \frac{8WD}{\pi d^3} = \frac{8WD}{\pi d^3} \left(1 + \frac{d}{2D} \right)$

$$\tau_{\max} = \frac{16.T.k}{\pi d^3}$$

WAHL'S FACTOR (K): It is defined as

$$K = \frac{4c - 1}{4c - 4} + \frac{0.615}{c}$$

Where C = spring index
= D/d

Strain Energy :

$$U = \frac{T^2 L}{2EI}$$

$$L = \pi D n$$

$$I = \frac{\pi d^4}{64}$$

so after substitution we get

$$U = \frac{32T^2 D n}{E.d^4}$$

Example: A close coiled helical spring is to carry a load of 5000N with a deflection of 50 mm and a maximum shearing stress of 400 N/mm² .if the number of active turns or active coils is 8.Estimate the following:

- wire diameter
- mean coil diameter
- weight of the spring.

Assume $G = 83,000 \text{ N/mm}^2$; $r = 7700 \text{ kg/m}^3$

for wire diameter if
W is the axial load,
then

$$\frac{w.d/2}{\frac{\pi d^4}{32}} = \frac{\tau_{\max}}{d/2}$$

$$D = \frac{400}{d/2} \cdot \frac{\pi d^4}{32} \cdot \frac{2}{W}$$

$$D = \frac{400 \cdot \pi d^3 \cdot 2}{5000 \cdot 16}$$

$$D = 0.0314 d^3$$

$$x = \frac{8wD^3.n}{G.d^4}$$

on substituting the relevant parameters we get

$$50 = \frac{8 \cdot 5000 \cdot (0.0314 d^3)^3 \cdot 8}{83,000 \cdot d^4}$$

$$d = 13.32 \text{ mm}$$

$$D = 0.0314 \times (13.317)^3 \text{ mm} = 74.15 \text{ mm}$$

Weight

mass or weight = volume . density
 = area . length of the spring . density of spring material

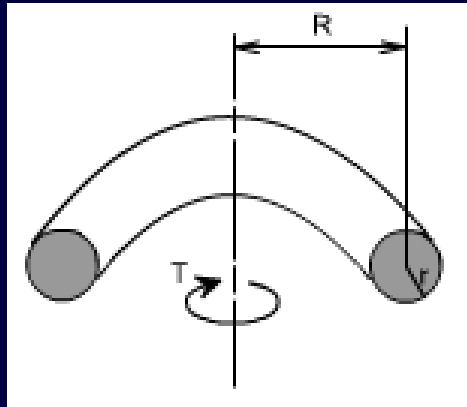
$$= \frac{\pi d^2}{4} \cdot \pi D n \cdot \rho$$

On substituting the relevant parameters we get

$$\text{Weight} = 1.996 \text{ kg}$$

$$= 2.0 \text{ kg}$$

Close coiled helical spring subjected to axial torque T or axial couple.



In this case the material of the spring is subjected to pure bending which tends to reduce Radius R of the coils. In this case the bending moment is constant through out the spring and is equal to the applied axial Torque T. The stresses i.e. maximum bending stress may thus be determined from the bending theory.

Deflection or wind up angle:

$$\theta = \int_0^L \frac{M dL}{EI} \text{ but } M = T$$
$$= \int_0^L \frac{T \cdot dL}{EI} = \frac{T}{EI} \int_0^L dL$$

Thus, as 'T' remains constant

$$\theta = \frac{T \cdot L}{EI}$$

Further

$$L = \pi D \cdot n$$

$$I = \frac{\pi d^4}{64}$$

Therefore, on substitution, the value of θ obtained is

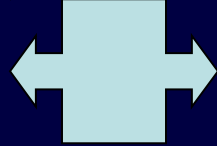
$$\theta = \frac{64 T \cdot D \cdot n}{E \cdot d^4}$$

$$\sigma_{\max} = \frac{M \cdot y}{I}$$
$$= \frac{T \cdot d/2}{\frac{\pi d^4}{64}}$$
$$\sigma_{\max} = \frac{32 T}{\pi d^3}$$

Elastic Stability Of Columns

Structural members which carry compressive loads may be divided into two broad categories depending on their relative lengths and cross-sectional dimensions.

Columns: Short, thick members are generally termed columns and these usually fail by crushing when the yield stress of the material in compression is exceeded.



Struts: Long, slender columns are generally termed as struts, they fail by buckling some time before the yield stress in compression is reached. The buckling occurs owing to one the following reasons.

- the strut may not be perfectly straight initially.
 - the load may not be applied exactly along the axis of the Strut.
 - one part of the material may yield in compression more readily than others owing to some lack of uniformity in the material properties through out the strut.
-
- ❑ It is assumed that the deformation to be both progressive with increasing load and simple in form i.e. a member in simple tension/compression becomes progressively longer/shorter but remains straight. Under some circumstances, however, these assumptions may no longer hold good and the member become unstable. The term strut and column are widely used, often interchangeably in the context of buckling of slender members.
 - ❑ At values of load below the buckling load a strut will be in stable equilibrium where the displacement caused by any lateral disturbance will be totally recovered when the disturbance is removed. At the buckling load the strut is said to be in a state of neutral equilibrium, and theoretically it should than be possible to gently deflect the strut into a simple sine wave provided that the amplitude of wave is kept small.
 - ❑ Theoretically, it is possible for struts to achieve a condition of unstable equilibrium with loads exceeding the buckling load, any slight lateral disturbance then causing failure by buckling, this condition is never achieved in practice under static load conditions.

The resistance of any member to bending is determined by its flexural rigidity EI

where I = area of moment of inertia

A = area of the cross-section

k = radius of gyration.

The load per unit area which the member can withstand is therefore related to k .

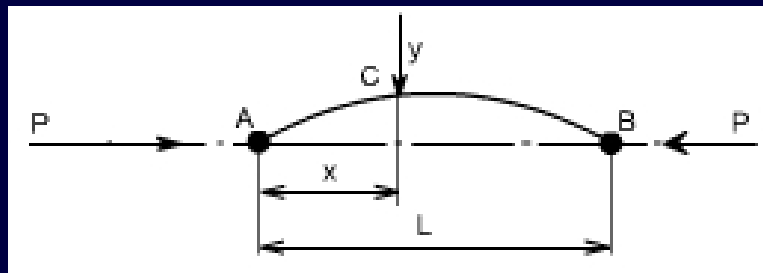
There will be two principal moments of inertia, if the least of these is taken then the ratio $\frac{l}{k}$ is called the slenderness ratio. Its numerical value indicates whether the member falls into the class of columns or struts.

$$\frac{l}{k} \quad \text{i.e.} \quad \frac{\text{length of member}}{\text{least radius of gyration}}$$

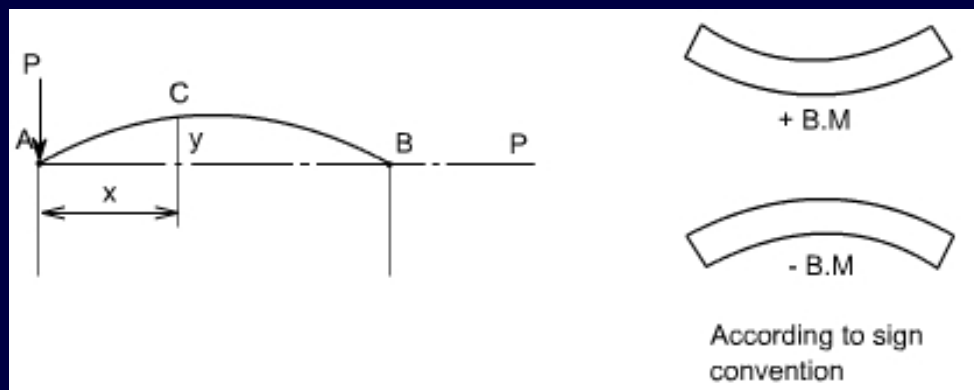
Euler's Theory

Case A: Strut with pinned ends: Consider an axially loaded strut, shown below, and is subjected to an axial load 'P' this load 'P' produces a deflection 'y' at a distance 'x' from one end.

Assume that the ends are either pin jointed or rounded so that there is no moment at either end.



Assumption: The strut is assumed to be initially straight, the end load being applied axially through centroid.



$$B. M|_C = -Py$$

Further, we know that

$$EI \frac{d^2 y}{dx^2} = M$$

$$EI \frac{d^2 y}{dx^2} = -P \cdot y = M$$

Thus,

$$EI \frac{d^2 y}{dx^2} + P y = 0$$

$$\frac{d^2 y}{dx^2} + \frac{Py}{EI} = 0$$

Let us define a operator $D = d/dx$
 $(D^2 + n^2) y = 0$ where $n^2 = P/EI$

$$y = A \cos \sqrt{\frac{P}{EI}} x + B \sin \sqrt{\frac{P}{EI}} x$$

In order to evaluate the constants A and B let us apply the boundary conditions,

- (i) at $x = 0$; $y = 0$
- (ii) at $x = L$; $y = 0$

Applying the first boundary condition yields $A = 0$.

$$B \sin \left(L \sqrt{\frac{P}{EI}} \right) = 0$$

$$\text{Thus either } B = 0, \text{ or } \sin \left(L \sqrt{\frac{P}{EI}} \right) = 0$$

if $B=0$, that $y=0$ for all values of x hence the strut has not buckled yet. Therefore, the solution required is

$$\sin \left(L \sqrt{\frac{P}{EI}} \right) = 0 \text{ or } \left(L \sqrt{\frac{P}{EI}} \right) = \pi \text{ or } nL = \pi$$

$$\text{or } \sqrt{\frac{P}{EI}} = \frac{\pi}{L} \text{ or } P = \frac{\pi^2 EI}{L^2}$$

From the above relationship the least value of P which will cause the strut to buckle, and it is called the **Euler Crippling Load** P_e from which we obtain.

$$P_e = \frac{\pi^2 EI}{L^2}$$

It may be noted that the value of I used in this expression is the least moment of inertia

It should be noted that the other solutions exist for the equation

$$\sin \left(\sqrt{\frac{P}{EI}} L \right) = 0 \quad \text{i.e. } \sin nL = 0$$

The interpretation of the above analysis is that for all the values of the load P, other than those which make $\sin nL = 0$; the strut will remain perfectly straight since $y = B \sin nL = 0$

For the particular value of

$$P_e = \frac{\pi^2 EI}{L^2}$$

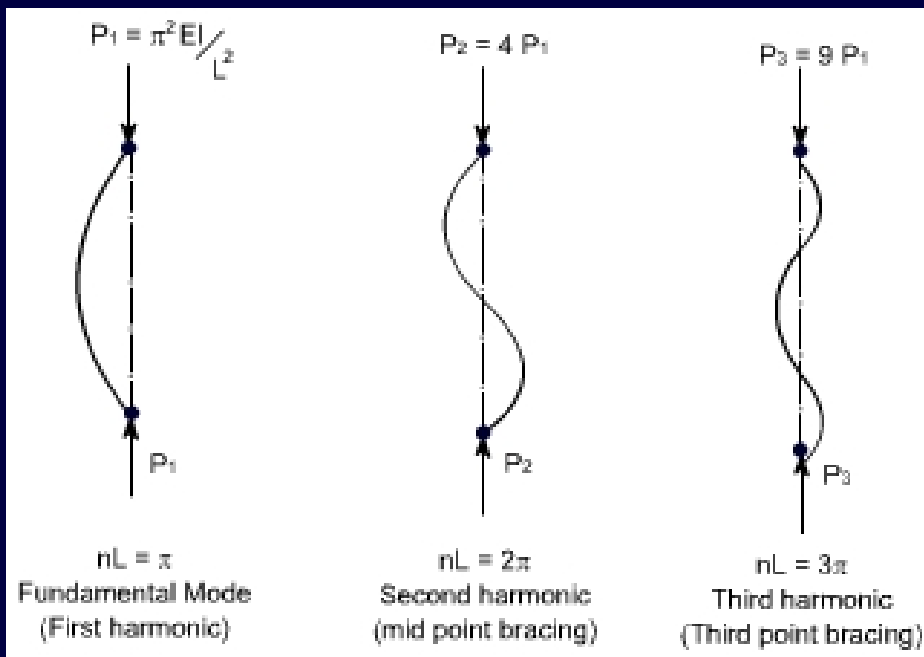
$$\sin nL = 0 \quad \text{or } nL = \pi$$

$$\text{Therefore } n = \frac{\pi}{L}$$

$$\text{Hence } y = B \sin nx = B \sin \frac{\pi x}{L}$$

Then we say that the strut is in a state of neutral equilibrium, and theoretically any deflection which it suffers will be maintained. This is subjected to the limitation that 'L' remains sensibly constant and in practice slight increase in load at the critical value will cause the deflection to increase appreciably until the material fails by yielding.

- ❖ Further it should be noted that the deflection is not proportional to load, and this applies to all strut problems; like wise it will be found that the maximum stress is not proportional to load.
- ❖ The solution chosen of $nL = \pi$ is just one particular solution; the solutions $nL = 2\pi, 3\pi, 5\pi$ etc are equally valid mathematically and they do, infact, produce values of ' P_e ' which are equally valid for modes of buckling of strut different from that of a simple bow. Theoretically therefore, there are an infinite number of values of P_e , each corresponding with a different mode of buckling.
- ❖ The value selected above is so called the fundamental mode value and is the lowest critical load producing the single bow buckling condition.
- ❖ The solution $nL = 2\pi$ produces buckling in two half waves, 3π in three half-waves etc.

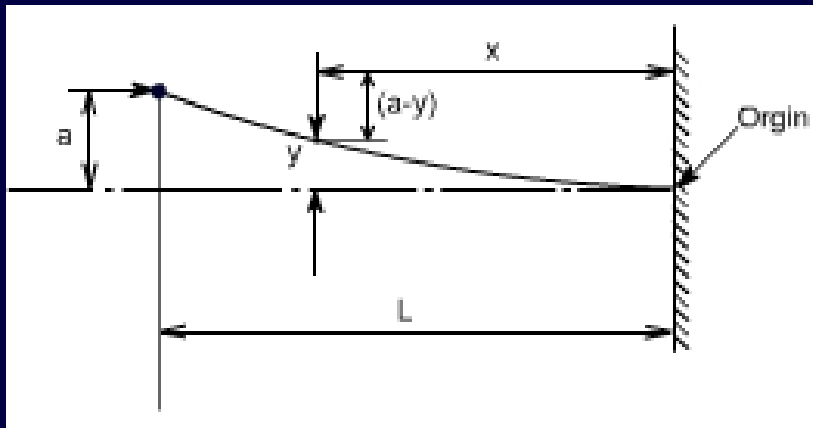


$$L\sqrt{\frac{P}{EI}} = \pi \text{ or } P_1 = \frac{\pi^2 EI}{L^2}$$

$$\text{If } L\sqrt{\frac{P}{EI}} = 2\pi \text{ or } P_2 = \frac{4\pi^2 EI}{L^2} = 4P_1$$

$$\text{If } L\sqrt{\frac{P}{EI}} = 3\pi \text{ or } P_3 = \frac{9\pi^2 EI}{L^2} = 9P_1$$

Case b: One end fixed and the other free:



$$B. M|_b = P(a - y)$$

Hence, the differential equation becomes,

$$EI \frac{d^2 y}{dx^2} = P(a - y)$$

On rearranging we get

$$\frac{d^2 y}{dx^2} + \frac{Py}{EI} = \frac{Pa}{EI}$$

$$\text{Let } \frac{P}{EI} = n^2$$

$$y_{\text{gen}} = A \cos(nx) + \sin(nx) + P.I$$

$P.I$ is a particular value of y which satisfies the differential equation, Hence $yP.I = a$

Therefore the complete solution becomes $Y = A \cos(nx) + B \sin(nx) + a$

Now imposing the boundary conditions to evaluate the constants A and B

(i) at $x = 0$; $y = 0$; This yields $A = -a$ and (ii) at $x = 0$; $dy/dx = 0$; This yields $B = 0$

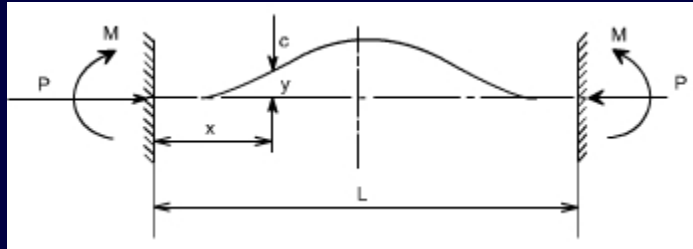
Hence, $y = -a \cos(nx) + a$; Further, at $x = L$; $y = a$, Therefore $a = -a \cos(nL) + a$ or $0 = \cos(nL)$

$$nL = \frac{\pi}{2}$$

$$\sqrt{\frac{P}{EI}} L = \frac{\pi}{2}, \text{ Therefore, the Euler's crippling load is given as}$$

$$P_e = \frac{\pi^2 EI}{4L^2}$$

Case 3: Strut with fixed ends



$$EI \frac{d^2 y}{dx^2} = M - Py$$

$$\text{or } \frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{M}{EI}$$

$$n^2 = \frac{P}{EI}, \text{ Therefore in the operator form, the equation reduces to}$$

$$(D^2 + n^2) y = \frac{M}{EI}$$

$$y_{\text{general}} = y_{\text{complementary}} + y_{\text{particular integral}}$$

$$y|_{p.i} = \frac{M}{n^2 EI} = \frac{M}{P}$$

Hence the general solution would be

$$y = B \cos nx + A \sin nx + \frac{M}{P}$$

Boundary conditions relevant to this case are at $x=0; y=0$

$$B = -\frac{M}{P}$$

Also at $x=0; \frac{dy}{dx} = 0$ hence

$$A=0$$

Therefore,

$$y = -\frac{M}{P} \cos nx + \frac{M}{P}$$

$$y = \frac{M}{P} (1 - \cos nx)$$

Further, it may be noted that at $x=L; y=0$

$$\text{Then } 0 = \frac{M}{P} (1 - \cos nL)$$

$$\text{Thus, either } \frac{M}{P} = 0 \text{ or } (1 - \cos nL) = 0$$

$$\text{obviously, } (1 - \cos nL) = 0$$

$$\cos nL = 1$$

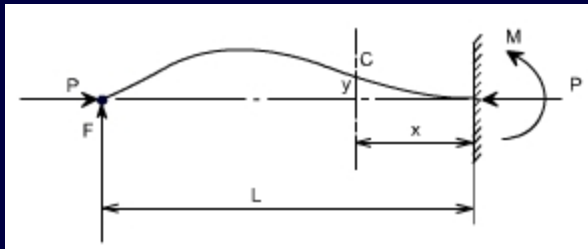
Hence the least solution would be

$$nL = 2\pi$$

$$\sqrt{\frac{P}{EI}} L = 2\pi, \text{ Thus, the buckling load or crippling load is}$$

$$P_e = \frac{4\pi^2 EI}{L^2}$$

Case 4 : One end fixed, the other pinned



$$EI \frac{d^2 y}{dx^2} = -Py + F(L-x)$$

$$EI \frac{d^2 y}{dx^2} + Py = F(L-x)$$

Hence

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{F}{EI} (L-x)$$

In the operator form the equation reduces to

$$(D^2 + n^2) y = \frac{F}{EI} (L-x)$$

$$y_{\text{particular}} = \frac{F}{n^2 EI} (L-x) \text{ or } y = \frac{F}{P} (L-x)$$

The full solution is therefore

$$y = A \cos nx + B \sin nx + \frac{F}{P} (L-x)$$

The boundary conditions relevant to the problem are at $x=0; y=0$

$$\text{Hence } A = -\frac{FL}{P}$$

$$\text{Also at } x=0; \frac{dy}{dx} = 0$$

$$\text{Hence } B = \frac{F}{nP}$$

$$\text{or } y = -\frac{FL}{P} \cos nx + \frac{F}{nP} \sin nx + \frac{F}{P} (L-x)$$

$$y = \frac{F}{nP} [\sin nx - nL \cos nx + n(L-x)]$$

Also when $x = L ; y = 0$

Therefore, $nL \cos nL = \sin nL$ or $\tan nL = nL$

The lowest value of nL (neglecting zero) which satisfies this condition and which therefore produces the fundamental buckling condition is $nL = 4.49 \text{radian}$

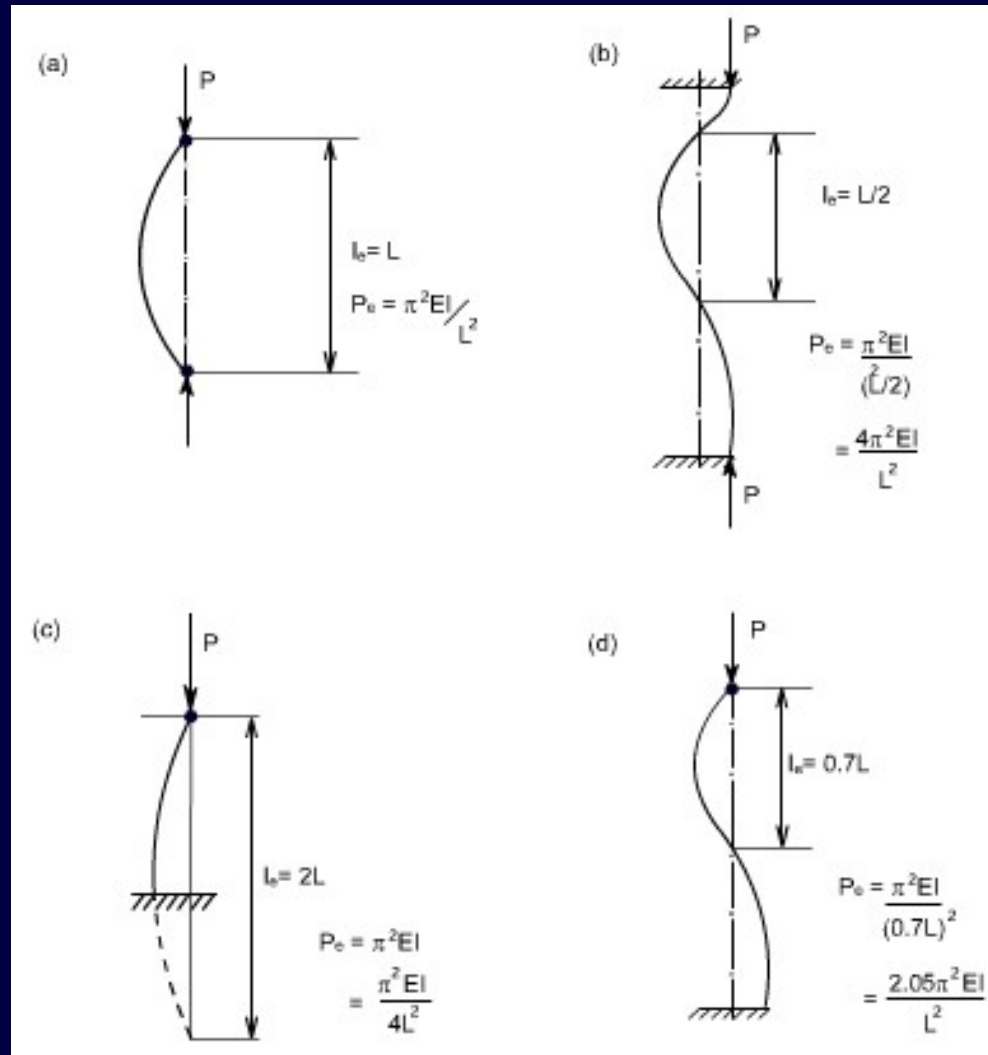
$$\text{or } \sqrt{\frac{P}{EI}} L = 4.49$$

$$\frac{P_e}{EI} L^2 = 20.2$$

$$P_e = \frac{2.05 \pi^2 EI}{L^2}$$

Equivalent Strut Length:

$$P_e = \frac{\pi^2 EI}{L_e^2}$$



$$\text{Euler's stress, } \sigma_e = \frac{P_e}{A} = \frac{\pi^2 EI}{A l^2}$$

$$\text{But, } I = A k^2$$

$$\sigma_e = \frac{\pi^2 E}{\left(\frac{l}{k}\right)^2}$$


$$P = \sigma_y A \left[1 - n \left(\frac{l}{k} \right) \right]$$

(b) Johnson parabolic formulae:

$$P = \sigma_y A \left[1 - b \left(\frac{l}{k} \right)^2 \right]$$

(c) Rankine Gordon Formulae :

$$\frac{1}{P_R} = \frac{1}{P_e} + \frac{1}{P_c}$$

where P_e = Euler crippling load

P_c = Crushing load or Yield point load in Compression

P_R = Actual load to cause failure or Rankine load

- Since the Rankine formulae is a combination of the Euler and crushing load for a strut.
- For a very short strut P_e is very large hence $1/P_e$ would be large so that $1/P_e$ can be neglected.
- Thus $P_R = P_c$, for very large struts, P_e is very small so $1/P_e$ would be large and $1/P_c$ can be neglected, hence $P_R = P_e$
- The Rankine formulae is therefore valid for extreme values of $1/k$. It is also found to be fairly accurate for the intermediate values in the range under consideration. Thus rewriting the formula in terms of stresses, we have

$$\frac{1}{\sigma A} = \frac{1}{\sigma_e A} + \frac{1}{\sigma_y A}$$

$$\frac{1}{\sigma} = \frac{1}{\sigma_e} + \frac{1}{\sigma_y}$$

$$\frac{1}{\sigma} = \frac{\sigma_e + \sigma_y}{\sigma_e \cdot \sigma_y}$$

$$\sigma = \frac{\sigma_e \cdot \sigma_y}{\sigma_e + \sigma_y} = \frac{\sigma_y}{1 + \frac{\sigma_y}{\sigma_e}}$$

For struts with both ends pinned

$$\sigma_e = \frac{\pi^2 E}{\left(\frac{l}{k}\right)^2}$$

$$\sigma = \frac{\sigma_y}{1 + \frac{\sigma_y}{\pi^2 E} \left(\frac{l}{k}\right)^2}$$

$$\sigma = \frac{\sigma_y}{1 + a \left(\frac{l}{k}\right)^2}$$

$$\text{Rankine load} = \frac{\sigma_y \cdot A}{1 + a \left(\frac{l}{k}\right)^2}$$

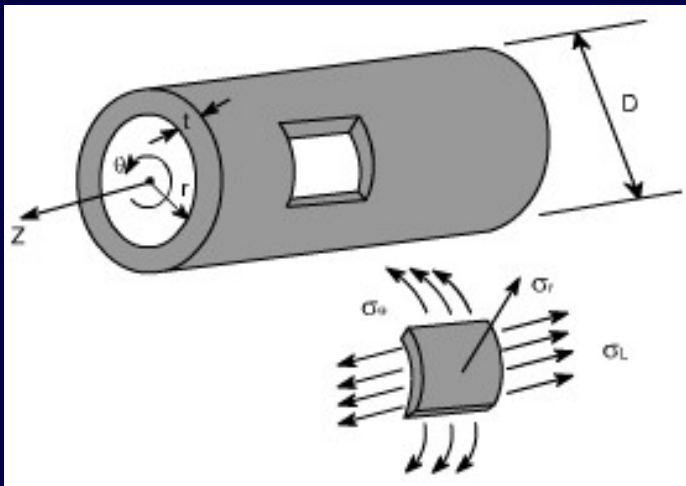
Material	σ_y or σ_c MN/m ²	Value of a	
		Pinned ends	Fixed ends
Low carbon steel	315	1/7500	1/30000
Cast Iron	540	1/1600	1/64000
Timber	35	1/3000	1/12000

note $a = 4 \times (a \text{ for fixed ends})$

Members Subjected to Axisymmetric Loads

Pressurized thin walled cylinder:

- ❖ **Preamble :** Pressure vessels are exceedingly important in industry. Normally two types of pressure vessel are used in common practice such as cylindrical pressure vessel and spherical pressure vessel.
- ❖ In the analysis of thin walled cylinders subjected to internal pressures it is assumed that the radial plane remains radial and the wall thickness does not change due to internal pressure. Although the internal pressure acting on the wall causes a local compressive stress (equal to pressure) but its value is negligibly small as compared to other stresses & hence the state of stress of an element of a thin walled pressure is considered a biaxial one.
- ❖ Further in the analysis of thin walled cylinders, the weight of the fluid is considered negligible.
- ❖ Let us consider a long cylinder of circular cross - section with an internal radius of R_2 and a constant wall thickness 't' as shown in fig.



- This cylinder is subjected to a difference of hydrostatic pressure of 'p' between its inner and outer surfaces. In many cases, 'p' between gauge pressure within the cylinder, taking outside pressure to be ambient.
- By thin walled cylinder we mean that the thickness 't' is very much smaller than the radius R_i and we may quantify this by stating that the ratio t / R_i of thickness of radius should be less than 0.1.
- An appropriate co-ordinate system to be used to describe such a system is the cylindrical polar one r, θ, z shown, where z axis lies along the axis of the cylinder, r is radial to it and θ is the angular co-ordinate about the axis.
- The small piece of the cylinder wall is shown in isolation, and stresses in respective direction have also been shown.

Type of failure: Such a component fails in since when subjected to an excessively high internal pressure. While it might fail by bursting along a path following the circumference of the cylinder. Under normal circumstances it fails by bursting along a path parallel to the axis. This suggests that the hoop stress is significantly higher than the axial stress.

Applications : Liquid storage tanks and containers, water pipes, boilers, submarine hulls, and certain air plane components are common examples of thin walled cylinders and spheres, roof domes.

ANALYSIS : In order to analyze the thin walled cylinders, let us make the following assumptions :

- There are no shear stresses acting in the wall.
- The longitudinal and hoop stresses do not vary through the wall.
- Radial stresses σ_r which acts normal to the curved plane of the isolated element are negligibly small as compared to other two stresses especially when

$$\left[\frac{t}{R_i} < \frac{1}{20} \right]$$

Thin Cylinders Subjected to Internal Pressure: When a thin walled cylinder is subjected to internal pressure, three mutually perpendicular principal stresses will be set up in the cylinder materials, namely

- Circumferential or hoop stress
- The radial stress
- Longitudinal stress

Hoop or circumferential stress: This is the stress which is set up in resisting the bursting effect of the applied pressure and can be most conveniently treated by considering the equilibrium of the cylinder.

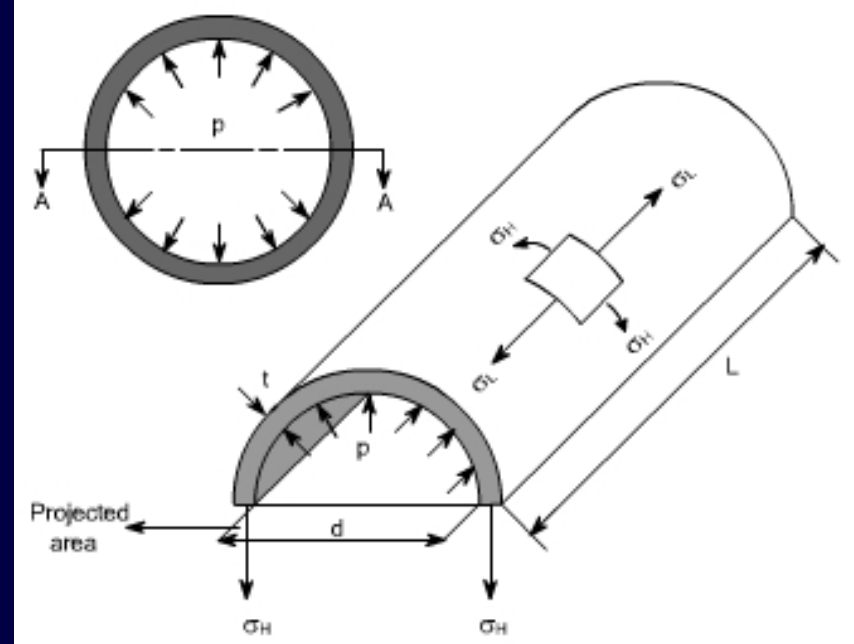
This cylinder is subjected to an internal pressure p .

i.e., p = internal pressure

d = inside diameter

L = Length of the cylinder

t = thickness of the wall



Total force on one half of the cylinder owing to the internal pressure ' p '

$$= p \times \text{Projected Area} = p \times d \times L$$

$$= p \cdot d \cdot L \quad \text{----- (1)}$$

The total resisting force owing to hoop stresses σ_H set up in the cylinder walls

$$= 2 \cdot \sigma_H \cdot L \cdot t \quad \text{-----(2)}$$

Because $\sigma_H \cdot L \cdot t$ is the force in the one wall of the half cylinder.

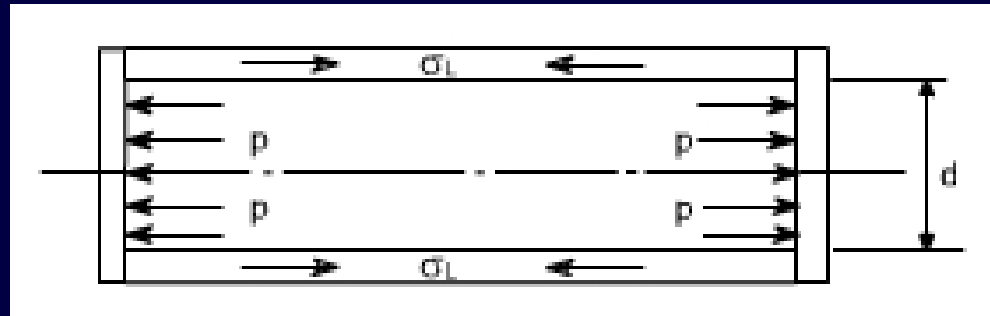
the equations (1) & (2) we get

$$2 \cdot \sigma_H \cdot L \cdot t = p \cdot d \cdot L$$

$$\Rightarrow \sigma_H = (p \cdot d) / 2t$$

$$\text{Circumferential or hoop Stress } (\sigma_H) = (p \cdot d) / 2t$$

Longitudinal Stress: Consider now again the same figure and the vessel could be considered to have closed ends and contains a fluid under a gauge pressure p . Then the walls of the cylinder will have a longitudinal stress as well as a circumferential stress.

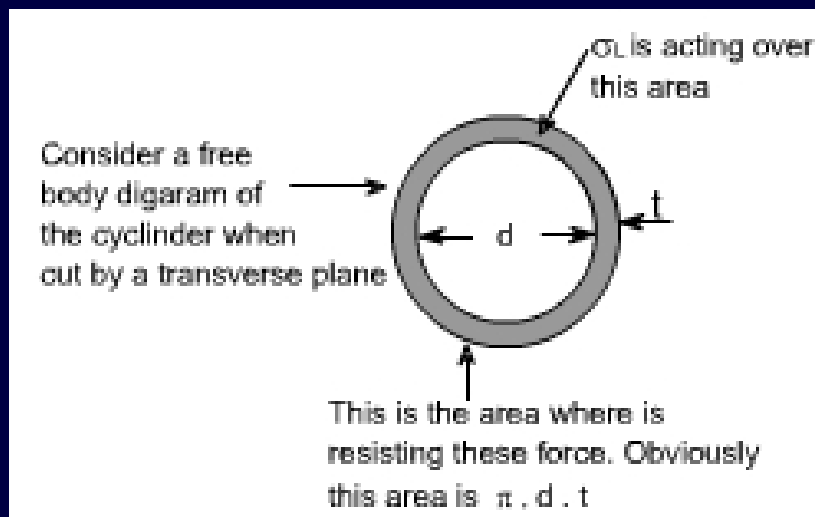


Total force on the end of the cylinder owing to internal pressure

$$= \text{pressure} \times \text{area} = p \times \pi d^2 / 4$$

Area of metal resisting this force $= \pi d \cdot t$. (approximately)

because πd is the circumference and this is multiplied by the wall thickness



Hence the longitudinal stresses

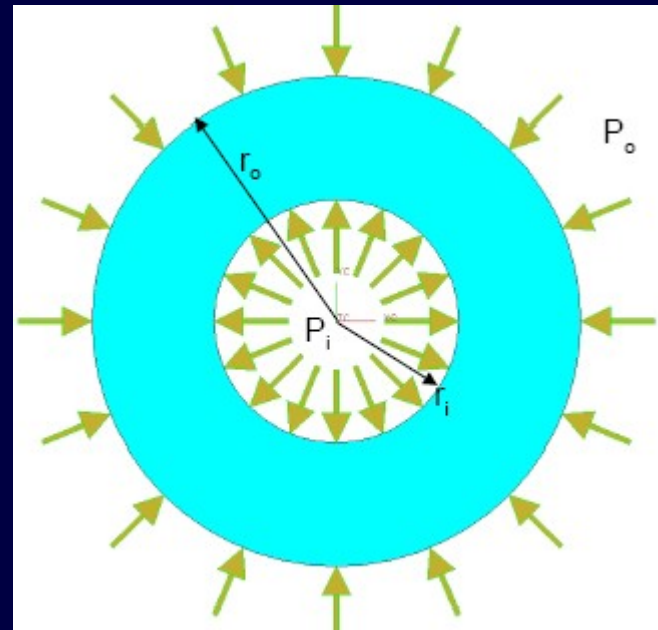
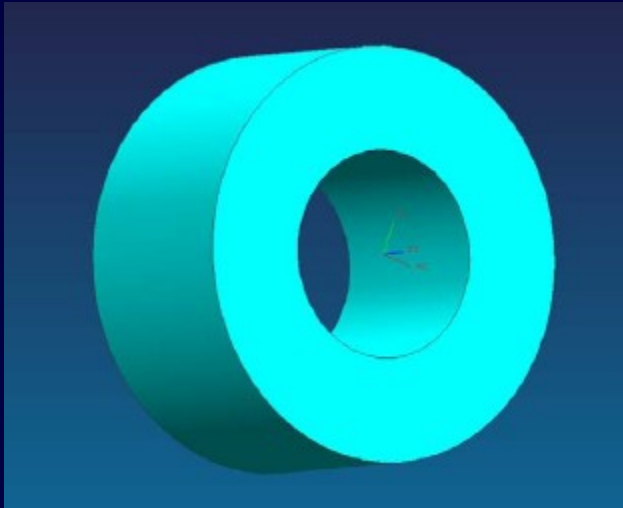
$$\begin{aligned} \text{Set up} &= \frac{\text{force}}{\text{area}} = \frac{[p \times \pi d^2 / 4]}{\pi d t} \\ &= \frac{pd}{4t} \quad \text{or} \quad \sigma_L = \frac{pd}{4t} \end{aligned}$$

or alternatively from equilibrium conditions

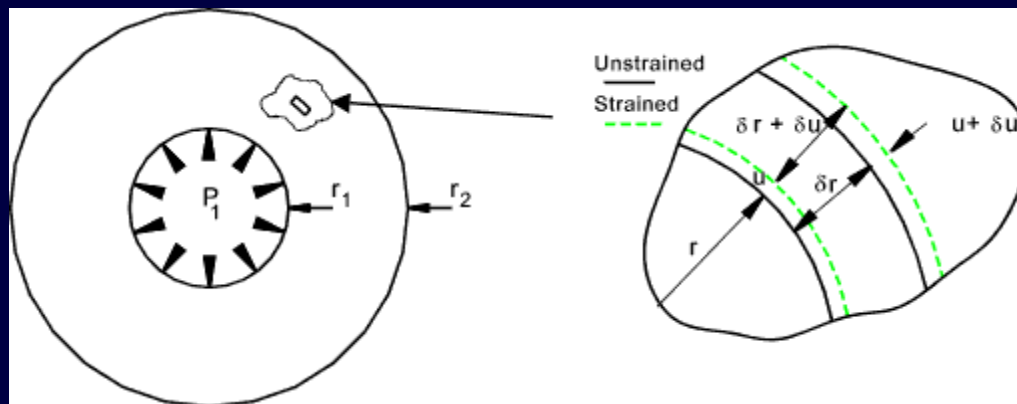
$$\sigma_L \cdot (\pi d t) = p \cdot \frac{\pi d^2}{4}$$

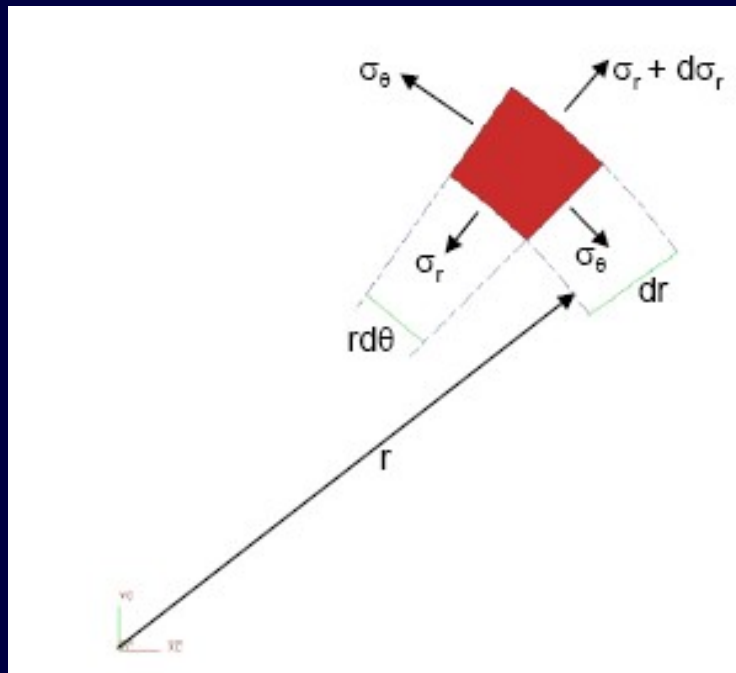
$$\text{Thus } \boxed{\sigma_L = \frac{pd}{4t}}$$

Thick Walled Cylinders



Consider a thick walled cylinder with open ends as shown above. It is loaded by internal pressure P_i and external pressure P_o as seen below. It has inner radius r_i and outer radius r_o .





Now consider an element at radius r and defined by an angle increment $d\theta$ and a radial increment dr . By circular symmetry, the stresses σ_θ and σ_r are functions of r only, not θ and the shear stress on the element must be zero. For an element of unit thickness, radial force equilibrium gives:

$$(\sigma_r + d\sigma_r)(r + dr)d\theta = \sigma_r r d\theta + \sigma_\theta d\theta dr$$

Ignoring second order terms gives:

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r + \sigma_\theta}{r} = 0 \quad \dots\dots\dots (1)$$

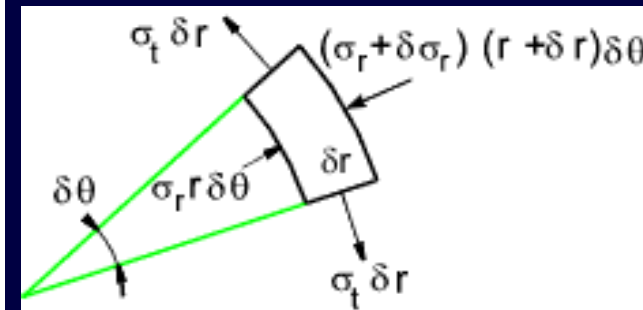
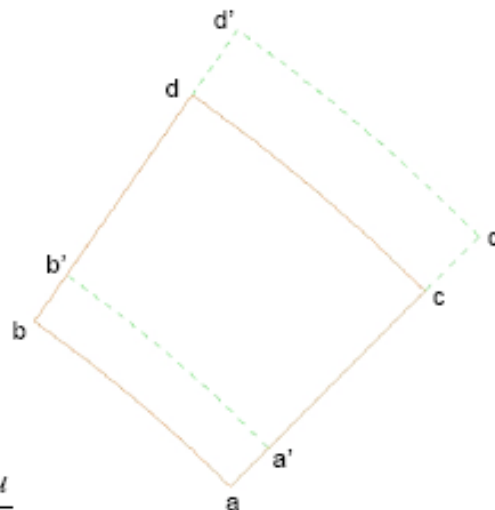
Assuming that there are no body forces.

Now consider strains in the element. By symmetry there is no θ displacement v . there is only a radial displacement u given by line aa' . Point c is displaced radially by $(u + du)$ given by line cc' . As the original radial length of the element is dr (line ac), the radial strain is:

$$\epsilon_r = \frac{u + du - u}{dr} = \frac{du}{dr}$$

Line ab has length $r d\theta$ and line $a'b'$ has length $(r + u) d\theta$. The tangential strain is thus:

$$\epsilon_\theta = \frac{(r + u)d\theta - r d\theta}{r d\theta} = \frac{u}{r}$$



As the ends are open, $\sigma_z = \sigma_3 = 0$ and we thus have plane stress conditions. From Hooke's law we get:

$$\varepsilon_r = \frac{du}{dr} = \frac{1}{E}(\sigma_r - \nu\sigma_\theta)$$

$$\varepsilon_\theta = \frac{u}{r} = \frac{1}{E}(\sigma_\theta - \nu\sigma_r)$$

Solving for the stresses gives:

$$\sigma_r = \frac{E}{1-\nu^2} \left(\frac{du}{dr} + \nu \frac{u}{r} \right) \quad \text{and} \quad \sigma_\theta = \frac{E}{1-\nu^2} \left(\frac{u}{r} + \nu \frac{du}{dr} \right)$$

Substituting into equation above yields:

$$\frac{d^2u}{dr^2} + \frac{1}{r} \frac{du}{dr} - \frac{u}{r^2} = 0$$

Which has solution : $u = C_1 r + \frac{C_2}{r}$

Giving the stresses as:

$$\sigma_r = \frac{E}{1-\nu^2} \left[C_1(1+\nu) - C_2 \left(\frac{1-\nu}{r^2} \right) \right] \dots\dots\dots (2)$$

$$\sigma_\theta = \frac{E}{1-\nu^2} \left[C_1(1+\nu) + C_2 \left(\frac{1-\nu}{r^2} \right) \right] \dots\dots\dots (3)$$

The boundary conditions are: $\sigma_r(r_i) = -P_i$ and $\sigma_r(r_o) = -P_o$

This yields the integration constants:

$$C_1 = \frac{1-\nu}{E} \left(\frac{r_i^2 P_i - r_o^2 P_o}{r_o^2 - r_i^2} \right) \quad \text{and} \quad C_2 = \frac{1-\nu}{E} \left(\frac{r_i^2 r_o^2 (P_i - P_o)}{r_o^2 - r_i^2} \right)$$

Giving the stresses as a function of radius:

$$\sigma_r = \frac{r_i^2 P_i - r_o^2 P_o}{(r_o^2 - r_i^2)} - \frac{(P_i - P_o) r_i^2 r_o^2}{(r_o^2 - r_i^2) r^2} \quad \sigma_\theta = \frac{r_i^2 P_i - r_o^2 P_o}{(r_o^2 - r_i^2)} + \frac{(P_i - P_o) r_i^2 r_o^2}{(r_o^2 - r_i^2) r^2}$$

These are known as Lamé's equations.

From equations 2 and 3 above we can see that the sum of the radial and tangential stresses is constant, regardless of radius: $\sigma_r + \sigma_\theta = 2EC_1/(1-\nu)$

Hence the longitudinal strain is also constant since:

$$\varepsilon_z = -\frac{\nu}{E} (\sigma_r + \sigma_\theta) = \text{constant.} \quad \text{Hence we get } \sigma_z = E\varepsilon_z = \text{constant} = c$$

If the ends of the cylinder are open and free we have $F_z = 0$, hence:

$$\int_{r_i}^{r_o} \sigma_z \cdot 2\pi r \, dr = \pi c (r_o^2 - r_i^2) = 0 \quad \text{or } c = \sigma_z = 0 \text{ as we assumed.}$$

If the cylinder has closed ends, the axial stress can be found separately using only force equilibrium considerations as was done for the thin walled cylinder. The result is then simply superimposed on the above equations.

The pressure P_i acts on area given by πr_i^2 .

The pressure P_o acts on area given by πr_o^2 .

The axial stress σ_z acts on an area given by $\pi(r_o^2 - r_i^2)$

Force equilibrium then gives:

$$\sigma_z = \left(\frac{P_i r_i^2 - P_o r_o^2}{r_o^2 - r_i^2} \right)$$

The following is a summary of the equations used to determine the stresses found in thick walled cylindrical pressure vessels. In the most general case the vessel is subject to both internal and external pressures. Most vessels also have closed ends - this results in an axial stress component.

Principal stresses at radius r :

$$\sigma_1 = \sigma_\theta = -K + C/r^2 \quad : \quad \sigma_2 = \sigma_r = -K + C/r^2$$

And, if the ends are closed, $\sigma_3 = \sigma_{axial} = -K$

Where:

$$C = (P_o - P_i) \left[\frac{r_i^2 r_o^2}{r_o^2 - r_i^2} \right] \quad : \quad K = \frac{(P_o r_o^2 - P_i r_i^2)}{r_o^2 - r_i^2}$$

(a) Internal Pressure only ($P_o = 0$):

$$\sigma_{\theta} = \frac{P_i r_i^2}{r_o^2 - r_i^2} \left[1 + \frac{r_o^2}{r^2} \right] \quad : \quad \sigma_r = \frac{P_i r_i^2}{r_o^2 - r_i^2} \left[1 - \frac{r_o^2}{r^2} \right] \quad : \quad \sigma_z = \frac{P_i r_i^2}{r_o^2 - r_i^2}$$

At inside surface, $r = r_i$:

$$\sigma_{\theta} = P_i \left[\frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} \right] \quad : \quad \sigma_r = -P_i \quad : \quad \sigma_z = \frac{P_i r_i^2}{r_o^2 - r_i^2}$$

At outside surface, $r = r_o$:

$$\sigma_{\theta} = \left[\frac{2P_i r_i^2}{r_o^2 - r_i^2} \right] \quad : \quad \sigma_r = 0 \quad : \quad \sigma_z = \frac{P_i r_i^2}{r_o^2 - r_i^2}$$

(b) External Pressure only ($P_i = 0$):

$$\sigma_{\theta} = \frac{-P_o r_o^2}{r_o^2 - r_i^2} \left[1 + \frac{r_i^2}{r^2} \right] \quad : \quad \sigma_r = \frac{-P_o r_o^2}{r_o^2 - r_i^2} \left[1 - \frac{r_i^2}{r^2} \right] \quad : \quad \sigma_z = \frac{-P_o r_o^2}{r_o^2 - r_i^2}$$

At inside surface, $r = r_i$:

$$\sigma_{\theta} = \left[\frac{-2P_o r_o^2}{r_o^2 - r_i^2} \right] \quad : \quad \sigma_r = 0 \quad : \quad \sigma_z = \frac{-P_o r_o^2}{r_o^2 - r_i^2}$$

At outside surface, $r = r_o$:

$$\sigma_{\theta} = -P_o \left[\frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} \right] \quad : \quad \sigma_r = -P_o \quad : \quad \sigma_z = \frac{-P_o r_o^2}{r_o^2 - r_i^2}$$

(c) Press and shrink fits

When a press or shrink fit is used between 2 cylinders of the same material, an interface pressure p_i is developed at the junction of the cylinders. If this pressure is calculated, the stresses in the cylinders can be found using the above equations. The pressure is:

$$p_i = \frac{E\delta}{b} \left[\frac{(c^2 - b^2)(b^2 - a^2)}{2b^2(c^2 - a^2)} \right]$$

Where:

E = Young's Modulus

δ = radial interference between the two cylinders

a = inner radius of the inner cylinder

b = outer radius of inner cylinder and inner radius of outer cylinder

c = outer radius of outer cylinder

It is assumed that δ is very small compared to the radius b and that there are no axial stresses. Thus we have $\delta = b_{\text{inner}} - b_{\text{outer}}$. Note that this small difference in the radii is ignored in the above equation.

1. The point of contraflexure occurs in

(a) Simply supported beams only, (b) Cantilever beams only, (c) Continuous beam only, (d) Overhanging beams only (e) All types of beams

2. A cantilever beam of length L , subjected to two loads W_1 and W_2 at distance L_1 and L_2 , respectively from the free end will have the greatest bending moment of

